

# Supplementary Material: Pathway-Centric Evaluation of Synthetic Educational Time-Series Data

## A How to Read This Supplement

In this supplement, a trajectory is one learner’s event sequence (one course enrollment in OULAD), while a pathway is a behavioral pattern identified by a rule that multiple trajectories can share. The main paper organizes evaluation around five learning-analytics questions: what occurs in the data, how learning unfolds, who is represented, what analysts can learn from the synthetic data, and what information might be exposed. This supplement provides the complete operational detail behind those questions. Sections C–D define every publication metric, including its intuition, direction, scope, and conditions under which it is not estimable. Section F then lists every numeric entry in the nine final `publication_metrics_v2` multi-seed summaries. Section H retains outcome-model collapse, a compact rare-pathway shape-support summary, and the complete dataset-by-metric applicability contract. The accompanying `rare-shape-estimability.csv` retains every seed-level support decision and its real/synthetic learner counts. Thus an absent numeric row is never left for the reader to interpret as either zero or an undocumented omission.

The final benchmark contains three datasets (ASSISTments, EdNet, and OULAD), three generators (Markov, sequence VAE, and TimeGAN), and the fixed seeds 20260703, 20260704, and 20260705. All reported standard deviations are population standard deviations across those generator seeds. Single-run entries are labeled “one available run” without an SD; the missing-run count remains explicit. Undefined values are never replaced by zero.

## B Implementation and Reproducibility Details

Trajectory frames and postprocessing. ASSISTments models skill, correctness, attempt, hint, response-time, and opportunity bins. EdNet models question, correctness, elapsed-time, timestamp-gap, attempt, and hint bins; deterministic question metadata are rebuilt from the generated question identifier. OULAD is an explicit weekly panel. Its generator-fitting frame contains course, dominant activity, registration, click, active-day, and profile attributes. Generated profile values, derived fields, and terminal outcomes are handled by the common train-fitted postprocessor detailed below. Learner identifiers and event order mark boundaries only and are excluded from the learned value representation. Table S1 states the complete implemented contract rather than only the fields retained in the final normalized output.

Table S1: Processed-data and generator column contract. Learner counts refer to trajectory identifiers; for OULAD these are course enrollments, while the split itself is by physical student. Structural fields establish sequence boundaries and are not learned values.

| Dataset     | Split unit          | Train/test trajectories | Train/test rows     | Generator-fitted value columns                                                                           | Rebuilt, replaced, or constructed after generation                                                                                                                                                           |
|-------------|---------------------|-------------------------|---------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASSISTments | Learner ID          | 3,429 / 788             | 282,118 / 64,742    | 6: skill ID, correctness, attempt bin, hint bin, response-time bin, opportunity bin                      | Learner ID and zero-based event order are newly assigned structural fields.                                                                                                                                  |
| EdNet       | Learner ID          | 7,703 / 1,963           | 1,021,565 / 264,798 | 6: question ID, correctness, elapsed-time bin, timestamp-gap bin, attempt bin, hint bin                  | Skill ID and part are deterministically rebuilt from question ID; learner ID and order are newly assigned.                                                                                                   |
| OULAD       | Physical student ID | 26,072 / 6,521          | 743,533 / 187,324   | 13: course ID, dominant activity, registration, click bin, active-days bin, and eight profile attributes | Module and presentation from course; engagement and gap from generated behavior; profiles independently resampled; dropout, failure, and final result constructed; learner ID and week order newly assigned. |

OULAD shared postprocessor. Normalization uses only the real-training split as reference and runs in a fixed order. First, course identifier is made learner-static and every retained enrollment-week is marked registered. The eight generated profile fields are then discarded. For each synthetic learner and each field independently, one value is sampled with replacement from real-training learners in the same course; the full real-training profile pool is the fallback for an unseen course. This targets, and preserves in expectation, course-conditional univariate marginals without copying one complete real profile as

a unit. Module and presentation are rebuilt from course; engagement is one exactly when click intensity is nonzero and the learner is registered. Non-engaged weeks receive activity none and zero active days, an engaged week decoded with zero active days is repaired to the 1–2 bin, and gap categories are recomputed from weeks since the most recent engaged week.

Dropout, failure, and final result are never generator inputs. A class-balanced logistic dropout ranker is fitted on real-training learners using course plus the mean and standard deviation of engagement after the first four observed weeks; missing post-prefix summaries are set to zero. The predictor features exclude terminal labels, total trajectory length, and behavior from the four-week prediction prefix; real dropout is used only as the fitting target. The frozen model ranks synthetic learners within course, with seeded random tie-breaking, and assigns the rounded product of the real-training course dropout rate and the number of synthetic learners in that course (using the global training rate if a course rate is unavailable). Dropouts receive Withdrawn and failure zero. Among completers, the lowest mean generated engagement values are assigned Fail to match the real-training course failure rate; among the remaining learners, the highest engagement values receive Distinction to match the real-training course distinction rate, and the remainder receive Pass. These labels are pipeline constructs, so OULAD outcome tasks evaluate the complete generation-and-postprocessing pipeline rather than direct outcome generation.

Generator settings. The Markov baseline is an order-2 tuple n-gram with minimum context count two and unigram backoff. The sequence VAE uses categorical windows of length 20, two 256-unit hidden layers, latent dimension 16, learning rate 0.001, KL weight 1.0, and sampling temperature 1.0. The external SynthCity TimeGAN uses boundary-safe length-20 categorical windows, one 50-unit generator layer, one 50-unit discriminator layer, RNN mode, and CPU execution. Both neural models receive all available training trajectories, use a batch-size cap of 200 and at most 100 observed categories per modeled column plus an <other> category when needed, and have a 50-epoch ceiling. The effective epoch count is chosen so fitting does not exceed approximately 4,000 minibatch updates. The effective settings, library version, source hashes, preprocessing and input hashes, and synthetic-output hash are recorded in each run report. A contiguous block-bootstrap generator remains in the software only as a copying-sensitive diagnostic; it is not one of the three ranked generators and is not proposed for privacy-safe release.

Targeting and output budgets. The targeted fit forms the union of eligible real-training pathway learners and assigns each learner three times the sampling weight of other learners; membership in multiple pathways does not compound the weight. This factor is fixed across datasets, generators, pathways, and seeds and is not tuned against evaluation outcomes. ASSISTments targets high hint use, late correctness decline, persistent low correctness, rapid low-accuracy responding, rare skill path, and recovery; EdNet uses the same set except high hint use; OULAD targets late disengagement, rare activity path, and reengagement. The selected unions contain 545 ASSISTments learners (15.89%), 1,790 EdNet learners (23.24%), and 2,968 OULAD enrollments (11.38%). Standard and targeted outputs have identical learner and row budgets. All generators use the same train-fitted, mean-calibrated truncated-lognormal length sampler; no individual real learner length is copied or empirically resampled. For neural outputs, the implementation allocates enough independently generated 20-observation windows to each learner, concatenates them in order, and truncates the final window to the sampled trajectory length. Cross-window continuity is not modeled by this assembly.

Evaluation sampling and uncertainty. Memory-heavy fidelity and privacy diagnostics use a deterministic whole-learner sample capped at 4,000 learners and 120,000 rows per split; source detection uses at most 100,000 rows per class. Rare-group prevalence and shape and the primary downstream tasks use the complete processed frames. Nearest-neighbor queries use at most 1,000 synthetic trajectories against the complete evaluated real-train reference; membership inference uses at most 1,000 members, nonmembers, and synthetic reference trajectories. The pipeline uses 200 learner-clustered percentile-bootstrap resamples for threshold-free downstream scores and paired differences. For OULAD, clusters are course-enrollment trajectory identifiers (module, presentation, and student ID), not physical students. All prediction rows within a sampled enrollment are retained together; different enrollments of the same student are not resampled jointly. The additional outcome tasks have one prediction row per enrollment. Rare-group prevalence uses an independent learner-membership bootstrap; conditional shape resamples qualifying learners, capped at 300 per side. Quantities below declared learner or class-count minima are reported as not estimable rather than replaced by zero.

Authoritative reports and schema. The final paired reports are under `experiments/reports/standard_vs_tail_targeted/<dataset>/<model>/`. Dataset keys are `assistments_2009_2010_skill_builder`, `ednet_kt1`, and `oulad_weekly_engagement`; model keys are `markov_ngram`, `sequence_vae_timevae`, and `timegan`. Each `seed_<seed>/generation_evaluation_report.json` contains the seed-level result, and `multi_seed_summary.json` contains the aggregate. The immutable `experiments/reports/standard/` directory is the RQ1 audit snapshot; it is neither averaged into the paired results nor counted as additional runs. Runtime manifests are under the corresponding `experiments/runs/` tree and are git-ignored.

Seed reports declare `publication.schema_version=publication_metrics_v2`, organize promoted results under `publication.metrics.rq1` and `publication.metrics.rq2`, retain exact exclusions and reasons under `publication.excluded_from_publication`; every `final_audit.status` must equal `passed` with an empty issues list. Aggregates declare `publication.schema_version=`

publication\_metrics\_v2 and metric\_summary\_scope=publication\_only. Their flattened numeric paths retain mean, population SD, minimum, maximum, estimable-seed count, and missing runs; publication\_status\_summary retains nonnumeric collapse and support states with their exact seeds. The deterministic learner split uses seed 20260703, and generator seeds are 20260703, 20260704, and 20260705. Seed-level bootstrap intervals remain in the seed reports and are not averaged.

## C Notation and Shared Comparison Measures

Let  $\mathcal{R}$  denote the real reference trajectories and  $\mathcal{S}$  the synthetic trajectories. Learner  $i$ 's ordered trajectory is  $x_i = ((z_{it}, c_{it}, \mathbf{v}_{it}, \theta_{it}))_{t=1}^{T_i}$ . The primary binary behavior is  $z$ : correctness in ASSISTments and EdNet and weekly engagement in OULAD. The categorical event type  $c$  is a skill, question, or activity. The vector  $\mathbf{v}$  contains the other modeled signals, and  $\theta$  is event time or week when a meaningful time signal is available. Jensen–Shannon divergence follows Lin (Lin, 1991); the geometric-mean normalization of mutual information follows Strehl and Ghosh (Strehl & Ghosh, 2002); source detectability follows classifier two-sample testing (Lopez-Paz & Oquab, 2017); precision–recall and ROC interpretation follows Davis and Goadrich (Davis & Goadrich, 2006), Saito and Rehmsmeier (Saito & Rehmsmeier, 2015), and Fawcett (Fawcett, 2006); train-on-synthetic, test-on-real evaluation follows prior time-series work (Esteban, Hyland, & Rätsch, 2017; Yoon, Jarrett, & van der Schaar, 2019); and the disclosure audit follows the membership-inference threat model of Shokri et al. (Shokri, Stronati, Song, & Shmatikov, 2017) and synthetic-data privacy cautions of Stadler et al. (Stadler, Oprisanu, & Troncoso, 2022).

Table S2: Shared notation and comparison functions.

| Quantity                       | Definition                                                                                                                           | Intuition                                                                                                                               |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Mean absolute error            | $\text{MAE}(\mathbf{a}, \mathbf{b}) = K^{-1} \sum_{k=1}^K  a_k - b_k .$                                                              | Average separation between matched summaries; zero means exact agreement.                                                               |
| Jensen–Shannon divergence      | $\text{JS}(P, Q) = \frac{1}{2} \text{KL}_2(P \  M) + \frac{1}{2} \text{KL}_2(Q \  M),$<br>where $M = (P + Q)/2.$                     | Symmetric difference between two discrete distributions. With base-2 logs it lies in $[0, 1]$ , and zero means identical distributions. |
| Normalized mutual information  | $\text{NMI}(X, Y) = I(X; Y) / \sqrt{H(X)H(Y)},$ with zero when either entropy is zero.                                               | Strength of a categorical relationship after normalizing for the variables' entropies.                                                  |
| Normalized trajectory position | Each learner's primary-behavior sequence is linearly interpolated onto ten positions from beginning to end before learner averaging. | Compares early-to-late development without allowing longer trajectories to contribute more learners' worth of evidence.                 |
| Population SD                  | $\sqrt{n^{-1} \sum_{s=1}^n (m_s - \bar{m})^2}$ over the estimable fixed generator seeds.                                             | Describes seed-to-seed generator variability; it is not a standard error over learners.                                                 |

## D Complete Metric Dictionary

The dictionary follows the five questions used in the main paper. Each subsection gives the metric definitions, learning-analytics intuition, preferred direction, and applicability conditions for one evaluation construct.

### D.1 Population Composition: What Occurs?

The population-composition metrics ask whether synthetic records reproduce the amount and mixture of observed behavior. Aggregate-behavior checks are necessary but insufficient because they ignore educational order and learner pathways.

Table S3: Population-composition metrics.

| Metric              | Formal or operational definition                                                                                                                                    | Learning-analytics intuition                                                        | Better |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------|
| Behavior-rate error | $ \bar{z}_{\mathcal{R}} - \bar{z}_{\mathcal{S}} $ , using event-weighted means. Because $z$ is binary, the negative-state error is identical and is not duplicated. | Does the synthetic dataset contain the same overall correctness or engagement rate? | Lower  |

| Metric                                    |  | Formal or operational definition                                                                                                                  | Learning-analytics intuition                                                                                                                                                                                | Better |
|-------------------------------------------|--|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Event-type frequency divergence           |  | $\text{JS}(P_{\mathcal{D}}(c), P_{\mathcal{S}}(c))$ .                                                                                             | Does it contain the same mixture of skills, questions, or weekly activity types?                                                                                                                            | Lower  |
| Average feature-distribution divergence   |  | $J^{-1} \sum_{j=1}^J \text{JS}(P_{\mathcal{D}}(v_j), P_{\mathcal{S}}(v_j))$ over prespecified generator-modeled attributes.                       | Are modeled attributes such as attempts, hints, timing bins, or activity intensities present in realistic proportions?                                                                                      | Lower  |
| Real-versus-synthetic detectability AUROC |  | $\max(A, 1 - A)$ , where $A$ is AUROC of a classifier distinguishing real from synthetic event records. Detector partitions are learner-disjoint. | Can a classifier easily tell which modeled records are artificial without seeing records from the same learner on both sides? A value of 0.5 is chance-level indistinguishability; 1 is perfect separation. | 0.5    |

## D.2 Learning Process: How Does Learning Unfold?

This group tests temporal order, progression, persistence, spacing, and relationships among educational signals. It distinguishes a realistic bag of events from a realistic learning trajectory.

Table S4: Learning-process metrics.

| Metric                                          |  | Formal or operational definition                                                                                                                                                                                           | Learning-analytics intuition                                                                                        | Better |
|-------------------------------------------------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------|
| Behavior-transition divergence                  |  | $\text{JS}$ between pooled within-learner adjacent pairs $(z_t, z_{t+1})$ .                                                                                                                                                | Are persistence, recovery, and decline transitions produced at realistic rates?                                     | Lower  |
| Three-event sequence divergence                 |  | $\text{JS}$ between pooled within-learner event-type triples $(c_t, c_{t+1}, c_{t+2})$ . Windows never cross learner boundaries.                                                                                           | Are short ordered skill or activity motifs realistic?                                                               | Lower  |
| Trajectory-shape error                          |  | $10^{-1} \sum_{b=1}^{10}  q_{\mathcal{D}}(b) - q_{\mathcal{S}}(b) $ , where $q_D(b)$ is the learner-averaged primary behavior at normalized position $b$ .                                                                 | Does the average trajectory rise, fall, or recover in the right way from beginning to end?                          | Lower  |
| Behavior-persistence error                      |  | $3^{-1} \sum_{\ell \in \{1,2,3\}}  \rho_{\mathcal{D}}(z_t, z_{t+\ell}) - \rho_{\mathcal{S}}(z_t, z_{t+\ell}) $ .                                                                                                           | Do correct/incorrect or engaged/disengaged states form similarly persistent streaks?                                | Lower  |
| Response/inactivity-gap distribution divergence |  | $\text{JS}(P_{\mathcal{D}}(g), P_{\mathcal{S}}(g))$ . In EdNet, $g$ is binned elapsed time since the prior response; in OULAD, it is binned weeks since the most recent engaged week. It is not applicable in ASSISTments. | Does the generator preserve response spacing or inactivity history, rather than response duration within one event? | Lower  |
| Behavior-hint lag error                         |  | Mean absolute difference in $\rho(z_t, h_{t+\ell})$ for $\ell \in \{0, 1\}$ . It is not estimated if hint use is absent or constant.                                                                                       | Is help-seeking related to current and next-step performance in the same way?                                       | Lower  |
| Feature-dependence error                        |  | Mean absolute real-synthetic NMI difference over prespecified scientifically relevant signal pairs.                                                                                                                        | Are relationships among modeled signals retained even if individual columns look realistic?                         | Lower  |

## D.3 Learner-Pathway Representation: Who Is Represented?

This group asks whether common learner strata and uncommon but intervention-relevant pathways appear in realistic proportions, and whether learners in those groups show realistic later behavior. Prefix-defined subgroups use a rule fitted on real training data and evaluate only later observations. Rare-group rules and thresholds are also fitted only on real training data and then applied unchanged to synthetic and held-out real learners.

Table S5: Learner-pathway representation metrics.

| Metric                                  | Formal or operational definition                                                                                                                  | Learning-analytics intuition                                                                                        | Better |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------|
| Subgroup behavior error                 | MAE across fixed prefix-defined groups in their later primary-behavior rates.                                                                     | Do learners with different early histories show realistic later correctness or engagement?                          | Lower  |
| Subgroup-size error                     | MAE across the same groups in learner share.                                                                                                      | Does the generator produce the same mixture of early-history learner types?                                         | Lower  |
| Subgroup hint-use error                 | MAE across eligible groups in later hint-request rates; not estimated without a varying hint signal.                                              | Do early-history learner groups retain their later help-seeking behavior?                                           | Lower  |
| Rare-group prevalence error             | For group $g$ , $ \pi_g(\mathcal{R}) - \pi_g(\mathcal{S}) $ , where $\pi_g(D)$ is learner prevalence.                                             | Does the generator include the right fraction of learners following a rare pathway?                                 | Lower  |
| Signed rare-group prevalence difference | For group $g$ , $\pi_g(\mathcal{S}) - \pi_g(\mathcal{R})$ . It is reported in percentage points alongside the absolute error.                     | Is the pathway underrepresented (negative), accurately represented (near zero), or overrepresented (positive)?      | Zero   |
| Rare-group trajectory-shape error       | Trajectory-shape error recomputed among learners satisfying $g$ . It requires at least ten qualifying real and ten qualifying synthetic learners. | Conditional on producing the group, does correctness or engagement among qualifying learners develop realistically? | Lower  |

Distribution-based rarity rules target the outer 5% of the real-training distribution. The 10% ceiling allows for ties in discretized variables without treating a common behavior as rare. Fixed change thresholds and minimum lengths are benchmark defaults for interpretable behavioral contrasts, not universal educational cut points. Prevalence remains reportable when no or too few synthetic learners qualify, because absence is itself a representation result. Shape is marked not estimable when either arm has fewer than ten qualifying learners. Publication rare groups have complete-real-training learner prevalence at most 10%. The published groups are high hint use, late correctness decline, persistent low correctness, rapid low-accuracy responding, rare skill path, and recovery where applicable; OULAD uses late disengagement, rare activity path, and reengagement. A coarse binned signal is excluded if no strict or inclusive boundary yields a nonempty group within the rarity ceiling.

Rare skill/activity paths are scored by the mean  $\log_2$  probability of successive skill/activity transitions under a real-training transition model with add-one smoothing. The bottom 5% defines the candidate rare-pathway group, subject to the rarity ceiling and at least one observed transition.

#### D.4 Analytic Usefulness: What Can Analysts Learn?

The main paper’s prediction-specification table lists the actual predictor inputs, fixed early-history lengths, and numerical eligibility requirements. Every downstream predictor is evaluated on the same held-out real learners. The primary task is next-response correctness for ASSISTments and EdNet and next-week engagement for OULAD. The five training arms are real only (TRTR), standard synthetic only (TSTR), real plus standard synthetic, pathway-targeted synthetic only, and real plus pathway-targeted synthetic. Learner-level outcome tasks use four arms because real plus targeted synthetic is not defined there. In synthetic-only arms, no real outcome rows fit the predictor, although all arms use the same real-training-fitted schema and transformations. AUPRC and AUROC are computed over eligible prediction rows, so longer trajectories contribute more instances; learner-clustered bootstrapping preserves within-learner dependence without changing that event-weighted point estimate.

Table S6: Analytic-usefulness metrics and contrasts.

| Metric | Formal or operational definition                                                                                   | Learning-analytics intuition                                                                                                        | Better |
|--------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------|
| AUPRC  | Area under the precision–recall curve on held-out real data. The observed positive rate is the no-skill reference. | Primary score for imbalanced educational outcomes and rare learner slices; rewards identifying positives without many false alarms. | Higher |

| Metric                                               |           | Formal or operational definition                                                                                                                            | Learning-analytics intuition                                                                                                 | Better |
|------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------|
| AUROC                                                |           | Probability that a randomly chosen positive receives a higher score than a randomly chosen negative.                                                        | Secondary threshold-free discrimination score; less sensitive than AUPRC to positive-class prevalence.                       | Higher |
| Standard utility loss                                | synthetic | $M_{\text{TRTR}} - M_{\text{standard}}$ , separately for AUPRC and AUROC.                                                                                   | How much held-out-real predictive performance is lost when analysts train only on ordinary synthetic data?                   | Lower  |
| Standard augmentation gain                           | augmenta- | $M_{\text{real+standard}} - M_{\text{TRTR}}$ .                                                                                                              | Does adding ordinary synthetic data improve a predictor already trained on real data?                                        | Higher |
| Targeted-vs-standard TSTR gain                       |           | $M_{\text{targeted}} - M_{\text{standard}}$ .                                                                                                               | Does emphasizing rare training learners make synthetic-only data more useful?                                                | Higher |
| Targeted augmentation gain                           | augmenta- | $M_{\text{real+targeted}} - M_{\text{TRTR}}$ .                                                                                                              | Does pathway-targeted synthetic augmentation improve over real-only training?                                                | Higher |
| Pathway-level predictive usefulness                  | predic-   | The same five arms and four contrasts, evaluated only on held-out real learners satisfying a fixed rare-group rule.                                         | Can a model trained on synthetic data learn the signal needed for uncommon, educationally consequential learner pathways?    | Varies |
| Rare-group prevalence-match change                   |           | $E_{\text{standard}}^{\text{prev}} - E_{\text{targeted}}^{\text{prev}}$ .                                                                                   | Did targeting move the learner proportion closer to its real-training reference, without treating overproduction as success? | Higher |
| Rare-group error reduction                           | shape-    | $E_{\text{standard}}^{\text{shape}} - E_{\text{targeted}}^{\text{shape}}$ , only when both errors are estimable.                                            | Did targeting improve the group’s within-trajectory primary-behavior pattern?                                                | Higher |
| Learner-level outcome-task AUPRC/AUROC and contrasts | outcome-  | Persistent-low-correctness and recovery prediction in ASSISTments/EdNet; dropout and course-failure prediction in OULAD, using only declared early history. | Does synthetic data support intervention-oriented outcomes beyond the primary next-step prediction task?                     | Varies |

OULAD terminal labels are excluded from generator inputs and constructed by the common leakage-safe post-generation pipeline. These pipeline-level outcome tasks evaluate the complete synthetic-data pipeline, not direct generation of future outcomes. The machine-readable reports retain the corresponding internal scope code for auditability.

## D.5 Disclosure Risk: What Might Be Exposed?

These measures are empirical memorization audits, not differential-privacy guarantees. A large nearest-neighbor distance should not be maximized without regard to fidelity: unrelated noise can be far from training data while being useless.

Trajectory representation and distance. Each event is represented by the tuple of all nonidentifier, non-order audit columns shared by the normalized real and synthetic frames. For trajectories  $x = (u_1, \dots, u_{T_x})$  and  $y = (v_1, \dots, v_{T_y})$ , the implemented positional distance is

$$d(x, y) = \frac{|T_x - T_y| + \sum_{t=1}^{\min(T_x, T_y)} \mathbf{1}[u_t \neq v_t]}{\max(T_x, T_y)}.$$

Distance is zero when both trajectories are empty. An aligned position counts as one mismatch when its complete event tuple differs, regardless of how many fields differ, and unmatched suffix positions count through the length term. Thus  $d \in [0, 1]$ , with no temporal realignment and  $d = 0$  indicating an exact normalized audit-trajectory match.

Near-copy and membership rules. The threshold  $\tau = 0.1$  is fixed for every dataset, generator, arm, and seed. Under this distance,  $d \leq 0.1$  means that at least 90% of positions in the longer trajectory match exactly after counting length mismatch. This is a prespecified benchmark flag, not a literature-derived or validated privacy boundary. The membership attack gives each real learner the fixed score  $-\min_{z \in \mathcal{S}} d(x, z)$ , so greater proximity always means “more likely a training member.” A below-chance AUROC is not inverted after seeing the labels; reported advantage is  $\max\{0, 2(A - 0.5)\}$ . For copying and

membership, targeting risk change is targeted minus standard; for mean distance it is standard minus targeted, so positive values consistently indicate greater empirical exposure after targeting. Sampling limits and bootstrap counts are given in Section B.

Table S7: Disclosure-risk metrics.

| Metric                                      | Formal or operational definition                                                                                                                            | Learning-analytics intuition                                                                                                 | Safer  |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------|
| Exact-duplicate rate                        | Fraction of synthetic learners whose full normalized audit trajectory exactly matches a real-training trajectory.                                           | How often does the generator literally reproduce a learner’s normalized sequence over the audit columns?                     | Lower  |
| Near-duplicate rate                         | Fraction of synthetic trajectories with $\min_{r \in \mathcal{R}} d(x, r) \leq \tau$ , where $\tau = 0.1$ .                                                 | How often does the generator produce a trajectory unusually close to one seen in training?                                   | Lower  |
| Mean nearest-training distance              | Mean $\min_{r \in \mathcal{R}} d(x, r)$ over evaluated synthetic trajectories.                                                                              | On average, how closely does each synthetic learner resemble its nearest training learner?                                   | Higher |
| Membership-inference advantage              | $\max(0, 2(A - 0.5))$ for the fixed proximity-oriented attack AUROC $A$ .                                                                                   | Can an attacker distinguish training learners from held-out learners using proximity to the synthetic data?                  | Lower  |
| Behavior-projection exact-duplicate rate    | Exact-copy rate after projecting OULAD to ordered dominant-activity/engagement pairs.                                                                       | How often are weekly behavioral patterns repeated when independently assigned profiles and constructed outcomes are removed? | Lower  |
| Targeted-minus-standard privacy-risk change | Targeted minus standard for copy and membership metrics; for distance, standard distance minus targeted distance so positive always means greater exposure. | Did pathway targeting make the empirical memorization signal stronger?                                                       | Lower  |

OULAD full-record nearest-neighbor and membership comparisons are excluded because demographic profiles are independently sampled and terminal outcomes are constructed after generation. Only exact copying on the ordered dominant-activity/engagement projection is promoted for comparison. Exploratory pathway-specific privacy blocks remain in raw reports, but the publication audit is deliberately dataset-level and does not estimate risk specifically for uncommon pathways.

## E Applicability, Uncertainty, and Exclusions

Metrics use the real training split as the fidelity reference because generators are trained on the real-training distribution. Preprocessing transforms, category maps, trajectory-length models, rare-group thresholds, subgroup cut points, postprocessors, and targeting membership are fit without held-out real test learners. Downstream evaluation uses held-out real learners only. OULAD is split by physical student, preventing different course enrollments from the same student from crossing the train/test boundary.

All numeric tables report means and population SD over fixed generator seeds. Seed-level bootstrap intervals over held-out learners remain in the JSON seed reports and are not averaged across seeds. Bootstrap intervals quantify learner-sampling uncertainty conditional on one fixed generated dataset and seed, whereas cross-seed SD describes generator stochasticity. With three generator seeds and one fixed learner-disjoint split, the paper treats cross-seed differences descriptively; neither summary captures split-to-split variability. A metric based on a constant or absent signal is not applicable. A rare-group conditional metric is not estimable below its declared support. Model-collapse outcomes are preserved as statuses. Table S8 lists every exclusion category present in the final seed reports. The seed-level source of truth is `publication.excluded_from_publication`; exclusion from comparison is distinct from non-estimability, whose support or class status remains a publication result.

Table S8: Complete publication-exclusion dictionary. Dataset-specific omitted-group reasons and fitted prevalences remain verbatim in each seed report.

| Report exclusion key                               | Excluded item                                                   | Reason                                                                                      |
|----------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <code>fixed_threshold_classification_scores</code> | Accuracy, precision, recall, and F1 at an untuned 0.5 threshold | Threshold-free AUPRC and AUROC are the pre-specified predictive comparisons.                |
| <code>threshold_metric_deltas</code>               | Targeting and augmentation contrasts for fixed-threshold scores | Only AUPRC/AUROC gaps are promoted.                                                         |
| <code>lightweight_downstream_utility</code>        | Skill-mean probability baseline                                 | It is a smoke-test diagnostic superseded by the declared histogram-gradient-boosting model. |

| Report exclusion key               | Excluded item                                                                                | Reason                                                                                                                                                |
|------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| sequence_length_model_ranking      | Length-distribution metrics                                                                  | All generators use the same train-fitted length sampler, so the result is not architecture-attributable.                                              |
| short_and_long_trajectory_tails    | Length-based uncommon pathways                                                               | The shared length sampler prevents generator attribution.                                                                                             |
| diversity_uniqueness_rates         | Raw diversity/coverage, uniqueness, entropy, and cardinality diagnostics                     | Novelty is not monotonic quality and remains a raw diagnostic.                                                                                        |
| raw_membership_auc_inversion       | Flipping a below-chance membership AU-ROC                                                    | Attack orientation is fixed before labels are inspected; the published advantage is not post-hoc inverted.                                            |
| raw_test_fidelity_duplicates       | Fidelity recomputed against held-out real test                                               | Real train is the generator’s fidelity reference; held-out real learners are reserved for generalization and nonmembership evaluation.                |
| raw_diagnostic_aggregates          | Unfiltered all-column marginals/dependence, detector accuracy, and minimum-distance extremes | These broad exploratory outputs do not belong to the prespecified construct metrics.                                                                  |
| omitted_standard_tail_groups       | Inapplicable or unsupported standard-arm pathway blocks                                      | Reasons include absent signals, postprocessed outcomes, and fewer than ten qualifying learners; prevalence is retained when defined.                  |
| omitted_targeted_tail_groups       | The corresponding targeted-arm pathway blocks                                                | The same frozen applicability and support rules apply to both generation arms.                                                                        |
| omitted_tail_learnability_groups   | Ineligible held-out-real pathway slices                                                      | Groups above the rarity ceiling, outside the dataset semantics, or below predictive support are not included in pathway-level prediction comparisons. |
| unavailable_percentile_tail_groups | Percentile pathways for which ties cannot yield a nonempty group at or below 10%             | A common discretized behavior is not relabeled as rare.                                                                                               |
| distinguishability                 | OULAD full-record source detection                                                           | Independently resampled profiles and constructed outcomes confound generator attribution.                                                             |

Dataset-specific scope rules add the following consequences. OULAD demographic subgroup fidelity, postprocessed dropout prevalence/shape, full-record source detection, and full-record nearest-neighbor/membership privacy are not used for generator comparison. The diagnostic persistent-misconception rule is omitted where the dataset does not support that pedagogical interpretation. Only the prespecified signal pairs, generator-attributable fields, and real-train fidelity reference enter publication comparisons.

The full applicability grid in Table S17 distinguishes three cases that would otherwise all appear as missing cells: a metric can be inapplicable because a required signal is absent or constant, deliberately excluded because postprocessing prevents a generator-comparable interpretation, or applicable but not estimable in a particular seed because learner or class support is insufficient.

## F Complete Numerical Publication Results

The following tables contain all 1,451 numeric entries in the nine final multi-seed publication summaries. The generation audit verifies equality between source and emitted row counts. Printed tables show mean, population SD, estimable-seed count, and missing-seed count. The accompanying supplement-generated/complete-publication-results.csv also retains the source minimum, maximum, exact JSON path, and all six summary fields for every row.

Table S9: Complete publication results for population composition. Every numeric metric-summary entry is shown; SD is the population SD across the fixed seeds and  $n$  is the number of estimable seeds.

| Dataset     | Generator | Scope / comparison           | Metric                                     | Mean $\pm$ SD ( $n$ )   | Direction |
|-------------|-----------|------------------------------|--------------------------------------------|-------------------------|-----------|
| ASSISTments | Markov    | All eligible learners/events | Average feature-distribution divergence    | 0.0014 $\pm$ 0.0001 (3) | Lower     |
| ASSISTments | Markov    | All eligible learners/events | Behavior-rate error                        | 0.0120 $\pm$ 0.0029 (3) | Lower     |
| ASSISTments | Markov    | All eligible learners/events | Event-type frequency divergence            | 0.0053 $\pm$ 0.0001 (3) | Lower     |
| ASSISTments | Markov    | All eligible learners/events | Real-versus-synthetic detectability AU-ROC | 0.5334 $\pm$ 0.0037 (3) | 0.5 best  |
| ASSISTments | SeqVAE    | All eligible learners/events | Average feature-distribution divergence    | 0.0003 $\pm$ 0.0000 (3) | Lower     |
| ASSISTments | SeqVAE    | All eligible learners/events | Behavior-rate error                        | 0.0012 $\pm$ 0.0008 (3) | Lower     |
| ASSISTments | SeqVAE    | All eligible learners/events | Event-type frequency divergence            | 0.0013 $\pm$ 0.0001 (3) | Lower     |
| ASSISTments | SeqVAE    | All eligible learners/events | Real-versus-synthetic detectability AU-ROC | 0.8500 $\pm$ 0.0008 (3) | 0.5 best  |
| ASSISTments | TimeGAN   | All eligible learners/events | Average feature-distribution divergence    | 0.1907 $\pm$ 0.0923 (3) | Lower     |
| ASSISTments | TimeGAN   | All eligible learners/events | Behavior-rate error                        | 0.3036 $\pm$ 0.0745 (3) | Lower     |
| ASSISTments | TimeGAN   | All eligible learners/events | Event-type frequency divergence            | 0.4101 $\pm$ 0.1875 (3) | Lower     |
| ASSISTments | TimeGAN   | All eligible learners/events | Real-versus-synthetic detectability AU-ROC | 0.9672 $\pm$ 0.0214 (3) | 0.5 best  |
| EdNet       | Markov    | All eligible learners/events | Average feature-distribution divergence    | 0.0152 $\pm$ 0.0000 (3) | Lower     |
| EdNet       | Markov    | All eligible learners/events | Behavior-rate error                        | 0.0163 $\pm$ 0.0013 (3) | Lower     |
| EdNet       | Markov    | All eligible learners/events | Event-type frequency divergence            | 0.0131 $\pm$ 0.0008 (3) | Lower     |
| EdNet       | Markov    | All eligible learners/events | Real-versus-synthetic detectability AU-ROC | 0.5794 $\pm$ 0.0009 (3) | 0.5 best  |
| EdNet       | SeqVAE    | All eligible learners/events | Average feature-distribution divergence    | 0.0070 $\pm$ 0.0002 (3) | Lower     |
| EdNet       | SeqVAE    | All eligible learners/events | Behavior-rate error                        | 0.0192 $\pm$ 0.0128 (3) | Lower     |
| EdNet       | SeqVAE    | All eligible learners/events | Event-type frequency divergence            | 0.0020 $\pm$ 0.0001 (3) | Lower     |
| EdNet       | SeqVAE    | All eligible learners/events | Real-versus-synthetic detectability AU-ROC | 0.6894 $\pm$ 0.0015 (3) | 0.5 best  |

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| Dataset | Generator | Scope / comparison           | Metric                                     | Mean $\pm$ SD ( $n$ )   | Direction |
|---------|-----------|------------------------------|--------------------------------------------|-------------------------|-----------|
| EdNet   | TimeGAN   | All eligible learners/events | Average feature-distribution divergence    | 0.1197 $\pm$ 0.0013 (3) | Lower     |
| EdNet   | TimeGAN   | All eligible learners/events | Behavior-rate error                        | 0.3345 $\pm$ 0.0004 (3) | Lower     |
| EdNet   | TimeGAN   | All eligible learners/events | Event-type frequency divergence            | 0.0101 $\pm$ 0.0001 (3) | Lower     |
| EdNet   | TimeGAN   | All eligible learners/events | Real-versus-synthetic detectability AU-ROC | 0.9092 $\pm$ 0.0008 (3) | 0.5 best  |
| OULAD   | Markov    | All eligible learners/events | Average feature-distribution divergence    | 0.0174 $\pm$ 0.0003 (3) | Lower     |
| OULAD   | Markov    | All eligible learners/events | Behavior-rate error                        | 0.1644 $\pm$ 0.0038 (3) | Lower     |
| OULAD   | Markov    | All eligible learners/events | Event-type frequency divergence            | 0.0203 $\pm$ 0.0009 (3) | Lower     |
| OULAD   | SeqVAE    | All eligible learners/events | Average feature-distribution divergence    | 0.0255 $\pm$ 0.0031 (3) | Lower     |
| OULAD   | SeqVAE    | All eligible learners/events | Behavior-rate error                        | 0.0100 $\pm$ 0.0051 (3) | Lower     |
| OULAD   | SeqVAE    | All eligible learners/events | Event-type frequency divergence            | 0.0231 $\pm$ 0.0032 (3) | Lower     |
| OULAD   | TimeGAN   | All eligible learners/events | Average feature-distribution divergence    | 0.1605 $\pm$ 0.1231 (3) | Lower     |
| OULAD   | TimeGAN   | All eligible learners/events | Behavior-rate error                        | 0.3494 $\pm$ 0.1877 (3) | Lower     |
| OULAD   | TimeGAN   | All eligible learners/events | Event-type frequency divergence            | 0.2222 $\pm$ 0.1331 (3) | Lower     |

Table S10: Complete publication results for learning process. Every numeric metric-summary entry is shown; SD is the population SD across the fixed seeds and  $n$  is the number of estimable seeds.

| Dataset     | Generator | Scope / comparison           | Metric                                          | Mean $\pm$ SD ( $n$ )   | Direction |
|-------------|-----------|------------------------------|-------------------------------------------------|-------------------------|-----------|
| ASSISTments | Markov    | All eligible learners/events | Feature-dependence error                        | 0.0017 $\pm$ 0.0004 (3) | Lower     |
| ASSISTments | Markov    | All eligible learners/events | Behavior-hint lag error                         | 0.0071 $\pm$ 0.0021 (3) | Lower     |
| ASSISTments | Markov    | All eligible learners/events | Behavior-persistence error                      | 0.0293 $\pm$ 0.0042 (3) | Lower     |
| ASSISTments | Markov    | All eligible learners/events | Trajectory-shape error                          | 0.0409 $\pm$ 0.0041 (3) | Lower     |
| ASSISTments | Markov    | All eligible learners/events | Behavior-transition divergence                  | 0.0002 $\pm$ 0.0001 (3) | Lower     |
| ASSISTments | Markov    | All eligible learners/events | Three-event sequence divergence                 | 0.0828 $\pm$ 0.0006 (3) | Lower     |
| ASSISTments | SeqVAE    | All eligible learners/events | Feature-dependence error                        | 0.0993 $\pm$ 0.0000 (3) | Lower     |
| ASSISTments | SeqVAE    | All eligible learners/events | Behavior-hint lag error                         | 0.4437 $\pm$ 0.0026 (3) | Lower     |
| ASSISTments | SeqVAE    | All eligible learners/events | Behavior-persistence error                      | 0.2017 $\pm$ 0.0009 (3) | Lower     |
| ASSISTments | SeqVAE    | All eligible learners/events | Trajectory-shape error                          | 0.0272 $\pm$ 0.0029 (3) | Lower     |
| ASSISTments | SeqVAE    | All eligible learners/events | Behavior-transition divergence                  | 0.0100 $\pm$ 0.0002 (3) | Lower     |
| ASSISTments | SeqVAE    | All eligible learners/events | Three-event sequence divergence                 | 0.8635 $\pm$ 0.0032 (3) | Lower     |
| ASSISTments | TimeGAN   | All eligible learners/events | Feature-dependence error                        | 0.1029 $\pm$ 0.0123 (3) | Lower     |
| ASSISTments | TimeGAN   | All eligible learners/events | Behavior-hint lag error                         | 0.3507 $\pm$ 0.1340 (3) | Lower     |
| ASSISTments | TimeGAN   | All eligible learners/events | Behavior-persistence error                      | 0.1881 $\pm$ 0.0192 (3) | Lower     |
| ASSISTments | TimeGAN   | All eligible learners/events | Trajectory-shape error                          | 0.3307 $\pm$ 0.0760 (3) | Lower     |
| ASSISTments | TimeGAN   | All eligible learners/events | Behavior-transition divergence                  | 0.2366 $\pm$ 0.1211 (3) | Lower     |
| ASSISTments | TimeGAN   | All eligible learners/events | Three-event sequence divergence                 | 0.7653 $\pm$ 0.0025 (3) | Lower     |
| EdNet       | Markov    | All eligible learners/events | Feature-dependence error                        | 0.0029 $\pm$ 0.0001 (3) | Lower     |
| EdNet       | Markov    | All eligible learners/events | Behavior-persistence error                      | 0.0230 $\pm$ 0.0014 (3) | Lower     |
| EdNet       | Markov    | All eligible learners/events | Trajectory-shape error                          | 0.1550 $\pm$ 0.0049 (3) | Lower     |
| EdNet       | Markov    | All eligible learners/events | Behavior-transition divergence                  | 0.0005 $\pm$ 0.0001 (3) | Lower     |
| EdNet       | Markov    | All eligible learners/events | Response/inactivity-gap distribution divergence | 0.0036 $\pm$ 0.0001 (3) | Lower     |
| EdNet       | Markov    | All eligible learners/events | Three-event sequence divergence                 | 0.4438 $\pm$ 0.0012 (3) | Lower     |
| EdNet       | SeqVAE    | All eligible learners/events | Feature-dependence error                        | 0.0088 $\pm$ 0.0000 (3) | Lower     |
| EdNet       | SeqVAE    | All eligible learners/events | Behavior-persistence error                      | 0.0941 $\pm$ 0.0006 (3) | Lower     |
| EdNet       | SeqVAE    | All eligible learners/events | Trajectory-shape error                          | 0.1460 $\pm$ 0.0126 (3) | Lower     |
| EdNet       | SeqVAE    | All eligible learners/events | Behavior-transition divergence                  | 0.0028 $\pm$ 0.0008 (3) | Lower     |
| EdNet       | SeqVAE    | All eligible learners/events | Response/inactivity-gap distribution divergence | 0.0008 $\pm$ 0.0002 (3) | Lower     |
| EdNet       | SeqVAE    | All eligible learners/events | Three-event sequence divergence                 | 0.8978 $\pm$ 0.0003 (3) | Lower     |
| EdNet       | TimeGAN   | All eligible learners/events | Feature-dependence error                        | 0.0087 $\pm$ 0.0001 (3) | Lower     |
| EdNet       | TimeGAN   | All eligible learners/events | Behavior-persistence error                      | 0.0932 $\pm$ 0.0015 (3) | Lower     |
| EdNet       | TimeGAN   | All eligible learners/events | Trajectory-shape error                          | 0.5047 $\pm$ 0.0002 (3) | Lower     |
| EdNet       | TimeGAN   | All eligible learners/events | Behavior-transition divergence                  | 0.3248 $\pm$ 0.0027 (3) | Lower     |
| EdNet       | TimeGAN   | All eligible learners/events | Response/inactivity-gap distribution divergence | 0.1745 $\pm$ 0.0008 (3) | Lower     |
| EdNet       | TimeGAN   | All eligible learners/events | Three-event sequence divergence                 | 0.8904 $\pm$ 0.0016 (3) | Lower     |
| OULAD       | Markov    | All eligible learners/events | Feature-dependence error                        | 0.0842 $\pm$ 0.0018 (3) | Lower     |
| OULAD       | Markov    | All eligible learners/events | Behavior-persistence error                      | 0.0269 $\pm$ 0.0017 (3) | Lower     |
| OULAD       | Markov    | All eligible learners/events | Trajectory-shape error                          | 0.1043 $\pm$ 0.0033 (3) | Lower     |
| OULAD       | Markov    | All eligible learners/events | Behavior-transition divergence                  | 0.0282 $\pm$ 0.0012 (3) | Lower     |
| OULAD       | Markov    | All eligible learners/events | Response/inactivity-gap distribution divergence | 0.0294 $\pm$ 0.0009 (3) | Lower     |
| OULAD       | Markov    | All eligible learners/events | Three-event sequence divergence                 | 0.0420 $\pm$ 0.0014 (3) | Lower     |
| OULAD       | SeqVAE    | All eligible learners/events | Feature-dependence error                        | 0.1971 $\pm$ 0.0036 (3) | Lower     |
| OULAD       | SeqVAE    | All eligible learners/events | Behavior-persistence error                      | 0.2506 $\pm$ 0.0033 (3) | Lower     |
| OULAD       | SeqVAE    | All eligible learners/events | Trajectory-shape error                          | 0.0841 $\pm$ 0.0090 (3) | Lower     |
| OULAD       | SeqVAE    | All eligible learners/events | Behavior-transition divergence                  | 0.0161 $\pm$ 0.0003 (3) | Lower     |
| OULAD       | SeqVAE    | All eligible learners/events | Response/inactivity-gap distribution divergence | 0.0091 $\pm$ 0.0006 (3) | Lower     |
| OULAD       | SeqVAE    | All eligible learners/events | Three-event sequence divergence                 | 0.1053 $\pm$ 0.0056 (3) | Lower     |
| OULAD       | TimeGAN   | All eligible learners/events | Feature-dependence error                        | 0.2012 $\pm$ 0.1637 (3) | Lower     |
| OULAD       | TimeGAN   | All eligible learners/events | Behavior-persistence error                      | 0.3582 $\pm$ 0.0930 (3) | Lower     |
| OULAD       | TimeGAN   | All eligible learners/events | Trajectory-shape error                          | 0.2853 $\pm$ 0.1825 (3) | Lower     |
| OULAD       | TimeGAN   | All eligible learners/events | Behavior-transition divergence                  | 0.2215 $\pm$ 0.2116 (3) | Lower     |
| OULAD       | TimeGAN   | All eligible learners/events | Response/inactivity-gap distribution divergence | 0.3152 $\pm$ 0.3144 (3) | Lower     |
| OULAD       | TimeGAN   | All eligible learners/events | Three-event sequence divergence                 | 0.4071 $\pm$ 0.1321 (3) | Lower     |

Table S11: Complete publication results for learner-pathway representation. Every numeric metric-summary entry is shown; SD is the population SD across the fixed seeds and  $n$  is the number of estimable seeds.

| Dataset     | Generator | Scope / comparison            | Metric                            | Mean $\pm$ SD ( $n$ )                   | Direction |
|-------------|-----------|-------------------------------|-----------------------------------|-----------------------------------------|-----------|
| ASSISTments | Markov    | All eligible learners/events  | Subgroup hint-use error           | $0.0509 \pm 0.0007$ (3)                 | Lower     |
| ASSISTments | Markov    | All eligible learners/events  | Subgroup behavior error           | $0.0646 \pm 0.0011$ (3)                 | Lower     |
| ASSISTments | Markov    | All eligible learners/events  | Subgroup-size error               | $0.0546 \pm 0.0030$ (3)                 | Lower     |
| ASSISTments | Markov    | High Hint Use                 | Rare-group prevalence error       | $0.0293 \pm 0.0015$ (3)                 | Lower     |
| ASSISTments | Markov    | High Hint Use                 | Rare-group trajectory-shape error | $0.0559 \pm 0.0099$ (3)                 | Lower     |
| ASSISTments | Markov    | Late Correctness Decline      | Rare-group prevalence error       | $0.0050 \pm 0.0013$ (3)                 | Lower     |
| ASSISTments | Markov    | Late Correctness Decline      | Rare-group trajectory-shape error | $0.0439 \pm 0.0153$ (3)                 | Lower     |
| ASSISTments | Markov    | Persistent Low Correctness    | Rare-group prevalence error       | $0.0197 \pm 0.0017$ (3)                 | Lower     |
| ASSISTments | Markov    | Persistent Low Correctness    | Rare-group trajectory-shape error | $0.0096 \pm 0.0029$ (2; 1 missing)      | Lower     |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding | Rare-group prevalence error       | $0.0140 \pm 0.0014$ (3)                 | Lower     |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding | Rare-group trajectory-shape error | $0.1054 \pm 0.0229$ (3)                 | Lower     |
| ASSISTments | Markov    | Rare Skill Path               | Rare-group prevalence error       | $0.0276 \pm 0.0076$ (3)                 | Lower     |
| ASSISTments | Markov    | Rare Skill Path               | Rare-group trajectory-shape error | $0.0540 \pm 0.0215$ (3)                 | Lower     |
| ASSISTments | Markov    | Recovery                      | Rare-group prevalence error       | $0.0031 \pm 0.0011$ (3)                 | Lower     |
| ASSISTments | Markov    | Recovery                      | Rare-group trajectory-shape error | $0.0659 \pm 0.0112$ (3)                 | Lower     |
| ASSISTments | SeqVAE    | All eligible learners/events  | Subgroup hint-use error           | $0.0515 \pm 0.0015$ (3)                 | Lower     |
| ASSISTments | SeqVAE    | All eligible learners/events  | Subgroup behavior error           | $0.0661 \pm 0.0014$ (3)                 | Lower     |
| ASSISTments | SeqVAE    | All eligible learners/events  | Subgroup-size error               | $0.0907 \pm 0.0060$ (3)                 | Lower     |
| ASSISTments | SeqVAE    | High Hint Use                 | Rare-group prevalence error       | $0.0390 \pm 0.0027$ (3)                 | Lower     |
| ASSISTments | SeqVAE    | High Hint Use                 | Rare-group trajectory-shape error | $0.5031$ (one available run; 2 missing) | Lower     |
| ASSISTments | SeqVAE    | Late Correctness Decline      | Rare-group prevalence error       | $0.0085 \pm 0.0070$ (3)                 | Lower     |
| ASSISTments | SeqVAE    | Late Correctness Decline      | Rare-group trajectory-shape error | $0.0730 \pm 0.0163$ (3)                 | Lower     |
| ASSISTments | SeqVAE    | Persistent Low Correctness    | Rare-group prevalence error       | $0.0237 \pm 0.0009$ (3)                 | Lower     |
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Responding | Rare-group prevalence error       | $0.0246 \pm 0.0006$ (3)                 | Lower     |
| ASSISTments | SeqVAE    | Rare Skill Path               | Rare-group prevalence error       | $0.9365 \pm 0.0073$ (3)                 | Lower     |
| ASSISTments | SeqVAE    | Rare Skill Path               | Rare-group trajectory-shape error | $0.0532 \pm 0.0042$ (3)                 | Lower     |
| ASSISTments | SeqVAE    | Recovery                      | Rare-group prevalence error       | $0.0103 \pm 0.0093$ (3)                 | Lower     |
| ASSISTments | SeqVAE    | Recovery                      | Rare-group trajectory-shape error | $0.0693 \pm 0.0160$ (3)                 | Lower     |
| ASSISTments | TimeGAN   | All eligible learners/events  | Subgroup hint-use error           | $0.1348 \pm 0.0357$ (3)                 | Lower     |
| ASSISTments | TimeGAN   | All eligible learners/events  | Subgroup behavior error           | $0.2930 \pm 0.0653$ (3)                 | Lower     |
| ASSISTments | TimeGAN   | All eligible learners/events  | Subgroup-size error               | $0.3320 \pm 0.0705$ (3)                 | Lower     |
| ASSISTments | TimeGAN   | High Hint Use                 | Rare-group prevalence error       | $0.0418 \pm 0.0003$ (3)                 | Lower     |
| ASSISTments | TimeGAN   | Late Correctness Decline      | Rare-group prevalence error       | $0.0237 \pm 0.0040$ (3)                 | Lower     |
| ASSISTments | TimeGAN   | Late Correctness Decline      | Rare-group trajectory-shape error | $0.0899$ (one available run; 2 missing) | Lower     |
| ASSISTments | TimeGAN   | Persistent Low Correctness    | Rare-group prevalence error       | $0.0245 \pm 0.0000$ (3)                 | Lower     |
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Responding | Rare-group prevalence error       | $0.0257 \pm 0.0000$ (3)                 | Lower     |
| ASSISTments | TimeGAN   | Rare Skill Path               | Rare-group prevalence error       | $0.1579 \pm 0.1623$ (3)                 | Lower     |
| ASSISTments | TimeGAN   | Rare Skill Path               | Rare-group trajectory-shape error | $0.1743 \pm 0.1345$ (2; 1 missing)      | Lower     |
| ASSISTments | TimeGAN   | Recovery                      | Rare-group prevalence error       | $0.0296 \pm 0.0051$ (3)                 | Lower     |
| ASSISTments | TimeGAN   | Recovery                      | Rare-group trajectory-shape error | $0.0804$ (one available run; 2 missing) | Lower     |
| EdNet       | Markov    | All eligible learners/events  | Subgroup behavior error           | $0.0194 \pm 0.0015$ (3)                 | Lower     |
| EdNet       | Markov    | All eligible learners/events  | Subgroup-size error               | $0.0755 \pm 0.0092$ (3)                 | Lower     |
| EdNet       | Markov    | Late Correctness Decline      | Rare-group prevalence error       | $0.0554 \pm 0.0014$ (3)                 | Lower     |
| EdNet       | Markov    | Late Correctness Decline      | Rare-group trajectory-shape error | $0.0705 \pm 0.0113$ (3)                 | Lower     |
| EdNet       | Markov    | Persistent Low Correctness    | Rare-group prevalence error       | $0.0441 \pm 0.0010$ (3)                 | Lower     |
| EdNet       | Markov    | Persistent Low Correctness    | Rare-group trajectory-shape error | $0.0343 \pm 0.0127$ (3)                 | Lower     |
| EdNet       | Markov    | Rapid Low-Accuracy Responding | Rare-group prevalence error       | $0.0353 \pm 0.0010$ (3)                 | Lower     |
| EdNet       | Markov    | Rapid Low-Accuracy Responding | Rare-group trajectory-shape error | $0.0719 \pm 0.0159$ (3)                 | Lower     |
| EdNet       | Markov    | Rare Skill Path               | Rare-group prevalence error       | $0.0325 \pm 0.0060$ (3)                 | Lower     |
| EdNet       | Markov    | Rare Skill Path               | Rare-group trajectory-shape error | $0.2191 \pm 0.0133$ (3)                 | Lower     |
| EdNet       | Markov    | Recovery                      | Rare-group prevalence error       | $0.0222 \pm 0.0007$ (3)                 | Lower     |
| EdNet       | Markov    | Recovery                      | Rare-group trajectory-shape error | $0.0422 \pm 0.0026$ (3)                 | Lower     |
| EdNet       | SeqVAE    | All eligible learners/events  | Subgroup behavior error           | $0.0234 \pm 0.0049$ (3)                 | Lower     |
| EdNet       | SeqVAE    | All eligible learners/events  | Subgroup-size error               | $0.0461 \pm 0.0116$ (3)                 | Lower     |
| EdNet       | SeqVAE    | Late Correctness Decline      | Rare-group prevalence error       | $0.0620 \pm 0.0035$ (3)                 | Lower     |
| EdNet       | SeqVAE    | Late Correctness Decline      | Rare-group trajectory-shape error | $0.0675 \pm 0.0098$ (3)                 | Lower     |
| EdNet       | SeqVAE    | Persistent Low Correctness    | Rare-group prevalence error       | $0.0459 \pm 0.0008$ (3)                 | Lower     |
| EdNet       | SeqVAE    | Persistent Low Correctness    | Rare-group trajectory-shape error | $0.0469 \pm 0.0091$ (2; 1 missing)      | Lower     |
| EdNet       | SeqVAE    | Rapid Low-Accuracy Responding | Rare-group prevalence error       | $0.0390 \pm 0.0002$ (3)                 | Lower     |
| EdNet       | SeqVAE    | Rare Skill Path               | Rare-group prevalence error       | $0.8528 \pm 0.0099$ (3)                 | Lower     |
| EdNet       | SeqVAE    | Rare Skill Path               | Rare-group trajectory-shape error | $0.2256 \pm 0.0140$ (3)                 | Lower     |
| EdNet       | SeqVAE    | Recovery                      | Rare-group prevalence error       | $0.0238 \pm 0.0040$ (3)                 | Lower     |
| EdNet       | SeqVAE    | Recovery                      | Rare-group trajectory-shape error | $0.0519 \pm 0.0054$ (3)                 | Lower     |
| EdNet       | TimeGAN   | All eligible learners/events  | Subgroup behavior error           | $0.3258 \pm 0.0052$ (3)                 | Lower     |
| EdNet       | TimeGAN   | All eligible learners/events  | Subgroup-size error               | $0.4389 \pm 0.0008$ (3)                 | Lower     |
| EdNet       | TimeGAN   | Late Correctness Decline      | Rare-group prevalence error       | $0.0761 \pm 0.0000$ (3)                 | Lower     |
| EdNet       | TimeGAN   | Persistent Low Correctness    | Rare-group prevalence error       | $0.0476 \pm 0.0000$ (3)                 | Lower     |
| EdNet       | TimeGAN   | Rapid Low-Accuracy Responding | Rare-group prevalence error       | $0.0392 \pm 0.0000$ (3)                 | Lower     |
| EdNet       | TimeGAN   | Rare Skill Path               | Rare-group prevalence error       | $0.8305 \pm 0.0119$ (3)                 | Lower     |
| EdNet       | TimeGAN   | Rare Skill Path               | Rare-group trajectory-shape error | $0.5816 \pm 0.0003$ (3)                 | Lower     |
| EdNet       | TimeGAN   | Recovery                      | Rare-group prevalence error       | $0.0435 \pm 0.0000$ (3)                 | Lower     |
| OULAD       | Markov    | All eligible learners/events  | Subgroup behavior error           | $0.0645 \pm 0.0054$ (3)                 | Lower     |
| OULAD       | Markov    | All eligible learners/events  | Subgroup-size error               | $0.0917 \pm 0.0017$ (3)                 | Lower     |
| OULAD       | Markov    | Late Disengagement            | Rare-group prevalence error       | $0.0256 \pm 0.0014$ (3)                 | Lower     |
| OULAD       | Markov    | Late Disengagement            | Rare-group trajectory-shape error | $0.0363 \pm 0.0044$ (3)                 | Lower     |
| OULAD       | Markov    | Rare Activity Path            | Rare-group prevalence error       | $0.0273 \pm 0.0007$ (3)                 | Lower     |
| OULAD       | Markov    | Rare Activity Path            | Rare-group trajectory-shape error | $0.0790 \pm 0.0057$ (3)                 | Lower     |

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| Dataset | Generator | Scope / comparison           | Metric                            | Mean $\pm$ SD ( $n$ )              | Direction |
|---------|-----------|------------------------------|-----------------------------------|------------------------------------|-----------|
| OULAD   | Markov    | Reengagement                 | Rare-group prevalence error       | $0.0321 \pm 0.0009$ (3)            | Lower     |
| OULAD   | Markov    | Reengagement                 | Rare-group trajectory-shape error | $0.0993 \pm 0.0053$ (3)            | Lower     |
| OULAD   | SeqVAE    | All eligible learners/events | Subgroup behavior error           | $0.1187 \pm 0.0069$ (3)            | Lower     |
| OULAD   | SeqVAE    | All eligible learners/events | Subgroup-size error               | $0.1072 \pm 0.0032$ (3)            | Lower     |
| OULAD   | SeqVAE    | Late Disengagement           | Rare-group prevalence error       | $0.0188 \pm 0.0033$ (3)            | Lower     |
| OULAD   | SeqVAE    | Late Disengagement           | Rare-group trajectory-shape error | $0.0768 \pm 0.0021$ (3)            | Lower     |
| OULAD   | SeqVAE    | Rare Activity Path           | Rare-group prevalence error       | $0.0209 \pm 0.0010$ (3)            | Lower     |
| OULAD   | SeqVAE    | Rare Activity Path           | Rare-group trajectory-shape error | $0.0652 \pm 0.0038$ (3)            | Lower     |
| OULAD   | SeqVAE    | Reengagement                 | Rare-group prevalence error       | $0.0219 \pm 0.0012$ (3)            | Lower     |
| OULAD   | SeqVAE    | Reengagement                 | Rare-group trajectory-shape error | $0.1045 \pm 0.0064$ (3)            | Lower     |
| OULAD   | TimeGAN   | All eligible learners/events | Subgroup behavior error           | $0.2205 \pm 0.0882$ (3)            | Lower     |
| OULAD   | TimeGAN   | All eligible learners/events | Subgroup-size error               | $0.2601 \pm 0.1935$ (3)            | Lower     |
| OULAD   | TimeGAN   | Late Disengagement           | Rare-group prevalence error       | $0.0558 \pm 0.0100$ (3)            | Lower     |
| OULAD   | TimeGAN   | Late Disengagement           | Rare-group trajectory-shape error | $0.0511 \pm 0.0030$ (2; 1 missing) | Lower     |
| OULAD   | TimeGAN   | Rare Activity Path           | Rare-group prevalence error       | $0.0426 \pm 0.0015$ (3)            | Lower     |
| OULAD   | TimeGAN   | Rare Activity Path           | Rare-group trajectory-shape error | $0.1263 \pm 0.0021$ (2; 1 missing) | Lower     |
| OULAD   | TimeGAN   | Reengagement                 | Rare-group prevalence error       | $0.0285 \pm 0.0186$ (3)            | Lower     |
| OULAD   | TimeGAN   | Reengagement                 | Rare-group trajectory-shape error | $0.1264 \pm 0.0013$ (2; 1 missing) | Lower     |

Table S12: Complete publication results for analytic usefulness. Every numeric metric-summary entry is shown; SD is the population SD across the fixed seeds and  $n$  is the number of estimable seeds.

| Dataset     | Generator | Scope / comparison                                                     | Metric | Mean $\pm$ SD ( $n$ )    | Direction |
|-------------|-----------|------------------------------------------------------------------------|--------|--------------------------|-----------|
| ASSISTments | Markov    | Real + standard synthetic                                              | AUPRC  | $0.8198 \pm 0.0002$ (3)  | Higher    |
| ASSISTments | Markov    | Real + standard synthetic                                              | AUROC  | $0.7340 \pm 0.0001$ (3)  | Higher    |
| ASSISTments | Markov    | Real + targeted synthetic                                              | AUPRC  | $0.8189 \pm 0.0004$ (3)  | Higher    |
| ASSISTments | Markov    | Real + targeted synthetic                                              | AUROC  | $0.7322 \pm 0.0000$ (3)  | Higher    |
| ASSISTments | Markov    | Real only (TRTR)                                                       | AUPRC  | $0.8232 \pm 0.0000$ (3)  | Higher    |
| ASSISTments | Markov    | Real only (TRTR)                                                       | AUROC  | $0.7384 \pm 0.0000$ (3)  | Higher    |
| ASSISTments | Markov    | Standard synthetic only (TSTR)                                         | AUPRC  | $0.7956 \pm 0.0002$ (3)  | Higher    |
| ASSISTments | Markov    | Standard synthetic only (TSTR)                                         | AUROC  | $0.7053 \pm 0.0007$ (3)  | Higher    |
| ASSISTments | Markov    | Targeted synthetic only                                                | AUPRC  | $0.7954 \pm 0.0002$ (3)  | Higher    |
| ASSISTments | Markov    | Targeted synthetic only                                                | AUROC  | $0.7048 \pm 0.0010$ (3)  | Higher    |
| ASSISTments | Markov    | Standard augmentation gain                                             | AUPRC  | $-0.0034 \pm 0.0002$ (3) | Higher    |
| ASSISTments | Markov    | Standard augmentation gain                                             | AUROC  | $-0.0043 \pm 0.0001$ (3) | Higher    |
| ASSISTments | Markov    | Standard synthetic utility loss                                        | AUPRC  | $0.0276 \pm 0.0002$ (3)  | Lower     |
| ASSISTments | Markov    | Standard synthetic utility loss                                        | AUROC  | $0.0331 \pm 0.0007$ (3)  | Lower     |
| ASSISTments | Markov    | Targeted-vs-standard TSTR gain                                         | AUPRC  | $-0.0002 \pm 0.0004$ (3) | Higher    |
| ASSISTments | Markov    | Targeted-vs-standard TSTR gain                                         | AUROC  | $-0.0005 \pm 0.0004$ (3) | Higher    |
| ASSISTments | Markov    | Targeted augmentation gain                                             | AUPRC  | $-0.0043 \pm 0.0004$ (3) | Higher    |
| ASSISTments | Markov    | Targeted augmentation gain                                             | AUROC  | $-0.0062 \pm 0.0000$ (3) | Higher    |
| ASSISTments | Markov    | Persistent Low-Correctness Prediction; Real + standard synthetic       | AUPRC  | $0.1593 \pm 0.0135$ (3)  | Higher    |
| ASSISTments | Markov    | Persistent Low-Correctness Prediction; Real + standard synthetic       | AUROC  | $0.7486 \pm 0.0176$ (3)  | Higher    |
| ASSISTments | Markov    | Persistent Low-Correctness Prediction; Real only (TRTR)                | AUPRC  | $0.1345 \pm 0.0000$ (3)  | Higher    |
| ASSISTments | Markov    | Persistent Low-Correctness Prediction; Real only (TRTR)                | AUROC  | $0.7011 \pm 0.0000$ (3)  | Higher    |
| ASSISTments | Markov    | Persistent Low-Correctness Prediction; Standard synthetic only (TSTR)  | AUPRC  | $0.0991 \pm 0.0287$ (3)  | Higher    |
| ASSISTments | Markov    | Persistent Low-Correctness Prediction; Standard synthetic only (TSTR)  | AUROC  | $0.5516 \pm 0.0840$ (3)  | Higher    |
| ASSISTments | Markov    | Persistent Low-Correctness Prediction; Targeted synthetic only         | AUPRC  | $0.1192 \pm 0.0164$ (3)  | Higher    |
| ASSISTments | Markov    | Persistent Low-Correctness Prediction; Targeted synthetic only         | AUROC  | $0.5745 \pm 0.0103$ (3)  | Higher    |
| ASSISTments | Markov    | Persistent Low-Correctness Prediction; Standard augmentation gain      | AUPRC  | $0.0248 \pm 0.0135$ (3)  | Higher    |
| ASSISTments | Markov    | Persistent Low-Correctness Prediction; Standard augmentation gain      | AUROC  | $0.0475 \pm 0.0176$ (3)  | Higher    |
| ASSISTments | Markov    | Persistent Low-Correctness Prediction; Standard synthetic utility loss | AUPRC  | $0.0354 \pm 0.0287$ (3)  | Lower     |
| ASSISTments | Markov    | Persistent Low-Correctness Prediction; Standard synthetic utility loss | AUROC  | $0.1495 \pm 0.0840$ (3)  | Lower     |
| ASSISTments | Markov    | Recovery Prediction; Real + standard synthetic                         | AUPRC  | $0.4226 \pm 0.0427$ (3)  | Higher    |
| ASSISTments | Markov    | Recovery Prediction; Real + standard synthetic                         | AUROC  | $0.6441 \pm 0.0229$ (3)  | Higher    |
| ASSISTments | Markov    | Recovery Prediction; Real only (TRTR)                                  | AUPRC  | $0.4387 \pm 0.0000$ (3)  | Higher    |
| ASSISTments | Markov    | Recovery Prediction; Real only (TRTR)                                  | AUROC  | $0.6897 \pm 0.0000$ (3)  | Higher    |
| ASSISTments | Markov    | Recovery Prediction; Standard synthetic only (TSTR)                    | AUPRC  | $0.3185 \pm 0.0339$ (3)  | Higher    |
| ASSISTments | Markov    | Recovery Prediction; Standard synthetic only (TSTR)                    | AUROC  | $0.5396 \pm 0.0316$ (3)  | Higher    |
| ASSISTments | Markov    | Recovery Prediction; Targeted synthetic only                           | AUPRC  | $0.3254 \pm 0.0197$ (3)  | Higher    |

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| Dataset     | Generator | Scope / comparison                                        | Metric     | Mean $\pm$ SD ( <i>n</i> ) | Direction |
|-------------|-----------|-----------------------------------------------------------|------------|----------------------------|-----------|
| ASSISTments | Markov    | Recovery Prediction; Targeted synthetic only              | syn- AUROC | 0.5303 $\pm$ 0.0383 (3)    | Higher    |
| ASSISTments | Markov    | Recovery Prediction; Standard augmentation gain           | aug- AUPRC | -0.0160 $\pm$ 0.0427 (3)   | Higher    |
| ASSISTments | Markov    | Recovery Prediction; Standard augmentation gain           | aug- AUROC | -0.0456 $\pm$ 0.0229 (3)   | Higher    |
| ASSISTments | Markov    | Recovery Prediction; Standard synthetic utility loss      | syn- AUPRC | 0.1201 $\pm$ 0.0339 (3)    | Lower     |
| ASSISTments | Markov    | Recovery Prediction; Standard synthetic utility loss      | syn- AUROC | 0.1502 $\pm$ 0.0316 (3)    | Lower     |
| ASSISTments | Markov    | High Hint Use; Real + standard synthetic                  | syn- AUPRC | 0.4179 $\pm$ 0.0047 (3)    | Higher    |
| ASSISTments | Markov    | High Hint Use; Real + standard synthetic                  | syn- AUROC | 0.8513 $\pm$ 0.0075 (3)    | Higher    |
| ASSISTments | Markov    | High Hint Use; Real + targeted synthetic                  | syn- AUPRC | 0.4100 $\pm$ 0.0058 (3)    | Higher    |
| ASSISTments | Markov    | High Hint Use; Real + targeted synthetic                  | syn- AUROC | 0.8518 $\pm$ 0.0063 (3)    | Higher    |
| ASSISTments | Markov    | High Hint Use; Real only (TRTR)                           | AUPRC      | 0.4399 $\pm$ 0.0000 (3)    | Higher    |
| ASSISTments | Markov    | High Hint Use; Real only (TRTR)                           | AUROC      | 0.8697 $\pm$ 0.0000 (3)    | Higher    |
| ASSISTments | Markov    | High Hint Use; Standard synthetic only (TSTR)             | AUPRC      | 0.3552 $\pm$ 0.0169 (3)    | Higher    |
| ASSISTments | Markov    | High Hint Use; Standard synthetic only (TSTR)             | AUROC      | 0.8225 $\pm$ 0.0073 (3)    | Higher    |
| ASSISTments | Markov    | High Hint Use; Targeted synthetic only                    | AUPRC      | 0.3774 $\pm$ 0.0077 (3)    | Higher    |
| ASSISTments | Markov    | High Hint Use; Targeted synthetic only                    | AUROC      | 0.8254 $\pm$ 0.0011 (3)    | Higher    |
| ASSISTments | Markov    | High Hint Use; Standard augmentation gain                 | AUPRC      | -0.0221 $\pm$ 0.0047 (3)   | Higher    |
| ASSISTments | Markov    | High Hint Use; Standard augmentation gain                 | AUROC      | -0.0184 $\pm$ 0.0075 (3)   | Higher    |
| ASSISTments | Markov    | High Hint Use; Standard synthetic utility loss            | AUPRC      | 0.0847 $\pm$ 0.0169 (3)    | Lower     |
| ASSISTments | Markov    | High Hint Use; Standard synthetic utility loss            | AUROC      | 0.0472 $\pm$ 0.0073 (3)    | Lower     |
| ASSISTments | Markov    | High Hint Use; Targeted-vs-standard TSTR gain             | AUPRC      | 0.0222 $\pm$ 0.0130 (3)    | Higher    |
| ASSISTments | Markov    | High Hint Use; Targeted-vs-standard TSTR gain             | AUROC      | 0.0030 $\pm$ 0.0074 (3)    | Higher    |
| ASSISTments | Markov    | High Hint Use; Targeted augmentation gain                 | AUPRC      | -0.0299 $\pm$ 0.0058 (3)   | Higher    |
| ASSISTments | Markov    | High Hint Use; Targeted augmentation gain                 | AUROC      | -0.0179 $\pm$ 0.0063 (3)   | Higher    |
| ASSISTments | Markov    | Late Correctness Decline; Real + standard synthetic       | AUPRC      | 0.7505 $\pm$ 0.0006 (3)    | Higher    |
| ASSISTments | Markov    | Late Correctness Decline; Real + standard synthetic       | AUROC      | 0.7374 $\pm$ 0.0013 (3)    | Higher    |
| ASSISTments | Markov    | Late Correctness Decline; Real + targeted synthetic       | AUPRC      | 0.7568 $\pm$ 0.0041 (3)    | Higher    |
| ASSISTments | Markov    | Late Correctness Decline; Real + targeted synthetic       | AUROC      | 0.7389 $\pm$ 0.0028 (3)    | Higher    |
| ASSISTments | Markov    | Late Correctness Decline; Real only (TRTR)                | AUPRC      | 0.7467 $\pm$ 0.0000 (3)    | Higher    |
| ASSISTments | Markov    | Late Correctness Decline; Real only (TRTR)                | AUROC      | 0.7451 $\pm$ 0.0000 (3)    | Higher    |
| ASSISTments | Markov    | Late Correctness Decline; Standard synthetic only (TSTR)  | AUPRC      | 0.7475 $\pm$ 0.0032 (3)    | Higher    |
| ASSISTments | Markov    | Late Correctness Decline; Standard synthetic only (TSTR)  | AUROC      | 0.7227 $\pm$ 0.0008 (3)    | Higher    |
| ASSISTments | Markov    | Late Correctness Decline; Targeted synthetic only         | AUPRC      | 0.7465 $\pm$ 0.0056 (3)    | Higher    |
| ASSISTments | Markov    | Late Correctness Decline; Targeted synthetic only         | AUROC      | 0.7200 $\pm$ 0.0086 (3)    | Higher    |
| ASSISTments | Markov    | Late Correctness Decline; Standard augmentation gain      | AUPRC      | 0.0037 $\pm$ 0.0006 (3)    | Higher    |
| ASSISTments | Markov    | Late Correctness Decline; Standard augmentation gain      | AUROC      | -0.0077 $\pm$ 0.0013 (3)   | Higher    |
| ASSISTments | Markov    | Late Correctness Decline; Standard synthetic utility loss | AUPRC      | -0.0008 $\pm$ 0.0032 (3)   | Lower     |
| ASSISTments | Markov    | Late Correctness Decline; Standard synthetic utility loss | AUROC      | 0.0224 $\pm$ 0.0008 (3)    | Lower     |
| ASSISTments | Markov    | Late Correctness Decline; Targeted-vs-standard TSTR gain  | AUPRC      | -0.0010 $\pm$ 0.0023 (3)   | Higher    |
| ASSISTments | Markov    | Late Correctness Decline; Targeted-vs-standard TSTR gain  | AUROC      | -0.0026 $\pm$ 0.0079 (3)   | Higher    |
| ASSISTments | Markov    | Late Correctness Decline; Targeted augmentation gain      | AUPRC      | 0.0100 $\pm$ 0.0041 (3)    | Higher    |
| ASSISTments | Markov    | Late Correctness Decline; Targeted augmentation gain      | AUROC      | -0.0062 $\pm$ 0.0028 (3)   | Higher    |
| ASSISTments | Markov    | Persistent Low Correctness; Real + standard synthetic     | AUPRC      | 0.1252 $\pm$ 0.0036 (3)    | Higher    |
| ASSISTments | Markov    | Persistent Low Correctness; Real + standard synthetic     | AUROC      | 0.7428 $\pm$ 0.0216 (3)    | Higher    |
| ASSISTments | Markov    | Persistent Low Correctness; Real + targeted synthetic     | AUPRC      | 0.1164 $\pm$ 0.0062 (3)    | Higher    |
| ASSISTments | Markov    | Persistent Low Correctness; Real + targeted synthetic     | AUROC      | 0.7761 $\pm$ 0.0253 (3)    | Higher    |

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| Dataset     | Generator | Scope / comparison                                             | Metric | Mean $\pm$ SD ( <i>n</i> ) | Direction |
|-------------|-----------|----------------------------------------------------------------|--------|----------------------------|-----------|
| ASSISTments | Markov    | Persistent Low Correctness; Real only (TRTR)                   | AUPRC  | 0.1436 $\pm$ 0.0000 (3)    | Higher    |
| ASSISTments | Markov    | Persistent Low Correctness; Real only (TRTR)                   | AUROC  | 0.8222 $\pm$ 0.0000 (3)    | Higher    |
| ASSISTments | Markov    | Persistent Low Correctness; Standard synthetic only (TSTR)     | AUPRC  | 0.0815 $\pm$ 0.0070 (3)    | Higher    |
| ASSISTments | Markov    | Persistent Low Correctness; Standard synthetic only (TSTR)     | AUROC  | 0.7503 $\pm$ 0.0138 (3)    | Higher    |
| ASSISTments | Markov    | Persistent Low Correctness; Targeted synthetic only            | AUPRC  | 0.0782 $\pm$ 0.0054 (3)    | Higher    |
| ASSISTments | Markov    | Persistent Low Correctness; Targeted synthetic only            | AUROC  | 0.7584 $\pm$ 0.0164 (3)    | Higher    |
| ASSISTments | Markov    | Persistent Low Correctness; Standard augmentation gain         | AUPRC  | -0.0184 $\pm$ 0.0036 (3)   | Higher    |
| ASSISTments | Markov    | Persistent Low Correctness; Standard augmentation gain         | AUROC  | -0.0794 $\pm$ 0.0216 (3)   | Higher    |
| ASSISTments | Markov    | Persistent Low Correctness; Standard synthetic utility loss    | AUPRC  | 0.0621 $\pm$ 0.0070 (3)    | Lower     |
| ASSISTments | Markov    | Persistent Low Correctness; Standard synthetic utility loss    | AUROC  | 0.0719 $\pm$ 0.0138 (3)    | Lower     |
| ASSISTments | Markov    | Persistent Low Correctness; Targeted-vs-standard TSTR gain     | AUPRC  | -0.0033 $\pm$ 0.0058 (3)   | Higher    |
| ASSISTments | Markov    | Persistent Low Correctness; Targeted-vs-standard TSTR gain     | AUROC  | 0.0081 $\pm$ 0.0302 (3)    | Higher    |
| ASSISTments | Markov    | Persistent Low Correctness; Targeted augmentation gain         | AUPRC  | -0.0273 $\pm$ 0.0062 (3)   | Higher    |
| ASSISTments | Markov    | Persistent Low Correctness; Targeted augmentation gain         | AUROC  | -0.0461 $\pm$ 0.0253 (3)   | Higher    |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding; Real + standard synthetic       | AUPRC  | 0.2812 $\pm$ 0.0013 (3)    | Higher    |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding; Real + standard synthetic       | AUROC  | 0.8515 $\pm$ 0.0086 (3)    | Higher    |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding; Real + targeted synthetic       | AUPRC  | 0.2591 $\pm$ 0.0101 (3)    | Higher    |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding; Real + targeted synthetic       | AUROC  | 0.8549 $\pm$ 0.0111 (3)    | Higher    |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding; Real only (TRTR)                | AUPRC  | 0.3409 $\pm$ 0.0000 (3)    | Higher    |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding; Real only (TRTR)                | AUROC  | 0.8810 $\pm$ 0.0000 (3)    | Higher    |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding; Standard synthetic only (TSTR)  | AUPRC  | 0.2165 $\pm$ 0.0181 (3)    | Higher    |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding; Standard synthetic only (TSTR)  | AUROC  | 0.8139 $\pm$ 0.0179 (3)    | Higher    |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding; Targeted synthetic only         | AUPRC  | 0.2197 $\pm$ 0.0137 (3)    | Higher    |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding; Targeted synthetic only         | AUROC  | 0.8159 $\pm$ 0.0062 (3)    | Higher    |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding; Standard augmentation gain      | AUPRC  | -0.0596 $\pm$ 0.0013 (3)   | Higher    |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding; Standard augmentation gain      | AUROC  | -0.0294 $\pm$ 0.0086 (3)   | Higher    |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding; Standard synthetic utility loss | AUPRC  | 0.1244 $\pm$ 0.0181 (3)    | Lower     |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding; Standard synthetic utility loss | AUROC  | 0.0671 $\pm$ 0.0179 (3)    | Lower     |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding; Targeted-vs-standard TSTR gain  | AUPRC  | 0.0033 $\pm$ 0.0177 (3)    | Higher    |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding; Targeted-vs-standard TSTR gain  | AUROC  | 0.0020 $\pm$ 0.0205 (3)    | Higher    |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding; Targeted augmentation gain      | AUPRC  | -0.0818 $\pm$ 0.0101 (3)   | Higher    |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding; Targeted augmentation gain      | AUROC  | -0.0261 $\pm$ 0.0111 (3)   | Higher    |
| ASSISTments | Markov    | Rare Skill Path; Real + standard synthetic                     | AUPRC  | 0.8973 $\pm$ 0.0072 (3)    | Higher    |
| ASSISTments | Markov    | Rare Skill Path; Real + standard synthetic                     | AUROC  | 0.7142 $\pm$ 0.0149 (3)    | Higher    |
| ASSISTments | Markov    | Rare Skill Path; Real + targeted synthetic                     | AUPRC  | 0.8881 $\pm$ 0.0036 (3)    | Higher    |
| ASSISTments | Markov    | Rare Skill Path; Real + targeted synthetic                     | AUROC  | 0.7093 $\pm$ 0.0045 (3)    | Higher    |
| ASSISTments | Markov    | Rare Skill Path; Real only (TRTR)                              | AUPRC  | 0.9009 $\pm$ 0.0000 (3)    | Higher    |
| ASSISTments | Markov    | Rare Skill Path; Real only (TRTR)                              | AUROC  | 0.7239 $\pm$ 0.0000 (3)    | Higher    |
| ASSISTments | Markov    | Rare Skill Path; Standard synthetic only (TSTR)                | AUPRC  | 0.8786 $\pm$ 0.0078 (3)    | Higher    |
| ASSISTments | Markov    | Rare Skill Path; Standard synthetic only (TSTR)                | AUROC  | 0.6843 $\pm$ 0.0163 (3)    | Higher    |
| ASSISTments | Markov    | Rare Skill Path; Targeted synthetic only                       | AUPRC  | 0.8784 $\pm$ 0.0063 (3)    | Higher    |
| ASSISTments | Markov    | Rare Skill Path; Targeted synthetic only                       | AUROC  | 0.7069 $\pm$ 0.0091 (3)    | Higher    |
| ASSISTments | Markov    | Rare Skill Path; Standard augmentation gain                    | AUPRC  | -0.0036 $\pm$ 0.0072 (3)   | Higher    |
| ASSISTments | Markov    | Rare Skill Path; Standard augmentation gain                    | AUROC  | -0.0097 $\pm$ 0.0149 (3)   | Higher    |

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| Dataset     | Generator | Scope / comparison                                                    | Metric                                      | Mean $\pm$ SD ( $n$ )               | Direction |
|-------------|-----------|-----------------------------------------------------------------------|---------------------------------------------|-------------------------------------|-----------|
| ASSISTments | Markov    | Rare Skill Path; Standard synthetic utility loss                      | AUPRC                                       | $0.0224 \pm 0.0078$ (3)             | Lower     |
| ASSISTments | Markov    | Rare Skill Path; Standard synthetic utility loss                      | AUROC                                       | $0.0396 \pm 0.0163$ (3)             | Lower     |
| ASSISTments | Markov    | Rare Skill Path; Targeted-vs-standard TSTR gain                       | AUPRC                                       | $-0.0002 \pm 0.0125$ (3)            | Higher    |
| ASSISTments | Markov    | Rare Skill Path; Targeted-vs-standard TSTR gain                       | AUROC                                       | $0.0226 \pm 0.0193$ (3)             | Higher    |
| ASSISTments | Markov    | Rare Skill Path; Targeted augmentation gain                           | AUPRC                                       | $-0.0129 \pm 0.0036$ (3)            | Higher    |
| ASSISTments | Markov    | Rare Skill Path; Targeted augmentation gain                           | AUROC                                       | $-0.0146 \pm 0.0045$ (3)            | Higher    |
| ASSISTments | Markov    | Recovery; Real + standard synthetic                                   | AUPRC                                       | $0.7534 \pm 0.0032$ (3)             | Higher    |
| ASSISTments | Markov    | Recovery; Real + standard synthetic                                   | AUROC                                       | $0.7322 \pm 0.0044$ (3)             | Higher    |
| ASSISTments | Markov    | Recovery; Real + targeted synthetic                                   | AUPRC                                       | $0.7437 \pm 0.0044$ (3)             | Higher    |
| ASSISTments | Markov    | Recovery; Real + targeted synthetic                                   | AUROC                                       | $0.7229 \pm 0.0065$ (3)             | Higher    |
| ASSISTments | Markov    | Recovery; Real only (TRTR)                                            | AUPRC                                       | $0.7445 \pm 0.0000$ (3)             | Higher    |
| ASSISTments | Markov    | Recovery; Real only (TRTR)                                            | AUROC                                       | $0.7329 \pm 0.0000$ (3)             | Higher    |
| ASSISTments | Markov    | Recovery; Standard synthetic only (TSTR)                              | AUPRC                                       | $0.7534 \pm 0.0096$ (3)             | Higher    |
| ASSISTments | Markov    | Recovery; Standard synthetic only (TSTR)                              | AUROC                                       | $0.7269 \pm 0.0086$ (3)             | Higher    |
| ASSISTments | Markov    | Recovery; Targeted synthetic only                                     | AUPRC                                       | $0.7310 \pm 0.0072$ (3)             | Higher    |
| ASSISTments | Markov    | Recovery; Targeted synthetic only                                     | AUROC                                       | $0.7060 \pm 0.0091$ (3)             | Higher    |
| ASSISTments | Markov    | Recovery; Standard augmentation gain                                  | AUPRC                                       | $0.0089 \pm 0.0032$ (3)             | Higher    |
| ASSISTments | Markov    | Recovery; Standard augmentation gain                                  | AUROC                                       | $-0.0006 \pm 0.0044$ (3)            | Higher    |
| ASSISTments | Markov    | Recovery; Standard synthetic utility loss                             | AUPRC                                       | $-0.0089 \pm 0.0096$ (3)            | Lower     |
| ASSISTments | Markov    | Recovery; Standard synthetic utility loss                             | AUROC                                       | $0.0060 \pm 0.0086$ (3)             | Lower     |
| ASSISTments | Markov    | Recovery; Targeted-vs-standard TSTR gain                              | AUPRC                                       | $-0.0224 \pm 0.0025$ (3)            | Higher    |
| ASSISTments | Markov    | Recovery; Targeted-vs-standard TSTR gain                              | AUROC                                       | $-0.0209 \pm 0.0047$ (3)            | Higher    |
| ASSISTments | Markov    | Recovery; Targeted augmentation gain                                  | AUPRC                                       | $-0.0008 \pm 0.0044$ (3)            | Higher    |
| ASSISTments | Markov    | Recovery; Targeted augmentation gain                                  | AUROC                                       | $-0.0099 \pm 0.0065$ (3)            | Higher    |
| ASSISTments | Markov    | High Hint Use                                                         | Rare-group prevalence-error reduction       | $0.0133 \pm 0.0043$ (3)             | Higher    |
| ASSISTments | Markov    | High Hint Use                                                         | Rare-group trajectory-shape-error reduction | $-0.0055 \pm 0.0034$ (3)            | Higher    |
| ASSISTments | Markov    | Late Correctness Decline                                              | Rare-group prevalence-error reduction       | $-0.0151 \pm 0.0037$ (3)            | Higher    |
| ASSISTments | Markov    | Late Correctness Decline                                              | Rare-group trajectory-shape-error reduction | $-0.0048 \pm 0.0103$ (3)            | Higher    |
| ASSISTments | Markov    | Persistent Low Correctness                                            | Rare-group prevalence-error reduction       | $0.0146 \pm 0.0013$ (3)             | Higher    |
| ASSISTments | Markov    | Persistent Low Correctness                                            | Rare-group trajectory-shape-error reduction | $-0.0044 \pm 0.0056$ (2; 1 missing) | Higher    |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding                                         | Rare-group prevalence-error reduction       | $-0.0016 \pm 0.0048$ (3)            | Higher    |
| ASSISTments | Markov    | Rapid Low-Accuracy Responding                                         | Rare-group trajectory-shape-error reduction | $0.0430 \pm 0.0110$ (3)             | Higher    |
| ASSISTments | Markov    | Rare Skill Path                                                       | Rare-group prevalence-error reduction       | $0.0066 \pm 0.0023$ (3)             | Higher    |
| ASSISTments | Markov    | Rare Skill Path                                                       | Rare-group trajectory-shape-error reduction | $-0.0080 \pm 0.0282$ (3)            | Higher    |
| ASSISTments | Markov    | Recovery                                                              | Rare-group prevalence-error reduction       | $-0.0067 \pm 0.0057$ (3)            | Higher    |
| ASSISTments | Markov    | Recovery                                                              | Rare-group trajectory-shape-error reduction | $-0.0079 \pm 0.0136$ (3)            | Higher    |
| ASSISTments | SeqVAE    | Real + standard synthetic                                             | AUPRC                                       | $0.8187 \pm 0.0003$ (3)             | Higher    |
| ASSISTments | SeqVAE    | Real + standard synthetic                                             | AUROC                                       | $0.7327 \pm 0.0006$ (3)             | Higher    |
| ASSISTments | SeqVAE    | Real + targeted synthetic                                             | AUPRC                                       | $0.8186 \pm 0.0004$ (3)             | Higher    |
| ASSISTments | SeqVAE    | Real + targeted synthetic                                             | AUROC                                       | $0.7326 \pm 0.0005$ (3)             | Higher    |
| ASSISTments | SeqVAE    | Real only (TRTR)                                                      | AUPRC                                       | $0.8232 \pm 0.0000$ (3)             | Higher    |
| ASSISTments | SeqVAE    | Real only (TRTR)                                                      | AUROC                                       | $0.7384 \pm 0.0000$ (3)             | Higher    |
| ASSISTments | SeqVAE    | Standard synthetic only (TSTR)                                        | AUPRC                                       | $0.6453 \pm 0.0056$ (3)             | Higher    |
| ASSISTments | SeqVAE    | Standard synthetic only (TSTR)                                        | AUROC                                       | $0.5056 \pm 0.0148$ (3)             | Higher    |
| ASSISTments | SeqVAE    | Targeted synthetic only                                               | AUPRC                                       | $0.6594 \pm 0.0137$ (3)             | Higher    |
| ASSISTments | SeqVAE    | Targeted synthetic only                                               | AUROC                                       | $0.5176 \pm 0.0220$ (3)             | Higher    |
| ASSISTments | SeqVAE    | Standard augmentation gain                                            | AUPRC                                       | $-0.0045 \pm 0.0003$ (3)            | Higher    |
| ASSISTments | SeqVAE    | Standard augmentation gain                                            | AUROC                                       | $-0.0056 \pm 0.0006$ (3)            | Higher    |
| ASSISTments | SeqVAE    | Standard synthetic utility loss                                       | AUPRC                                       | $0.1779 \pm 0.0056$ (3)             | Lower     |
| ASSISTments | SeqVAE    | Standard synthetic utility loss                                       | AUROC                                       | $0.2328 \pm 0.0148$ (3)             | Lower     |
| ASSISTments | SeqVAE    | Targeted-vs-standard TSTR gain                                        | AUPRC                                       | $0.0141 \pm 0.0081$ (3)             | Higher    |
| ASSISTments | SeqVAE    | Targeted-vs-standard TSTR gain                                        | AUROC                                       | $0.0120 \pm 0.0088$ (3)             | Higher    |
| ASSISTments | SeqVAE    | Targeted augmentation gain                                            | AUPRC                                       | $-0.0046 \pm 0.0004$ (3)            | Higher    |
| ASSISTments | SeqVAE    | Targeted augmentation gain                                            | AUROC                                       | $-0.0058 \pm 0.0005$ (3)            | Higher    |
| ASSISTments | SeqVAE    | Persistent Low-Correctness Prediction; Real + standard synthetic      | AUPRC                                       | $0.1964 \pm 0.0091$ (3)             | Higher    |
| ASSISTments | SeqVAE    | Persistent Low-Correctness Prediction; Real + standard synthetic      | AUROC                                       | $0.7731 \pm 0.0142$ (3)             | Higher    |
| ASSISTments | SeqVAE    | Persistent Low-Correctness Prediction; Real only (TRTR)               | AUPRC                                       | $0.1345 \pm 0.0000$ (3)             | Higher    |
| ASSISTments | SeqVAE    | Persistent Low-Correctness Prediction; Real only (TRTR)               | AUROC                                       | $0.7011 \pm 0.0000$ (3)             | Higher    |
| ASSISTments | SeqVAE    | Persistent Low-Correctness Prediction; Standard synthetic only (TSTR) | AUPRC                                       | $0.0681 \pm 0.0135$ (2; 1 missing)  | Higher    |
| ASSISTments | SeqVAE    | Persistent Low-Correctness Prediction; Standard synthetic only (TSTR) | AUROC                                       | $0.5538 \pm 0.1434$ (2; 1 missing)  | Higher    |
| ASSISTments | SeqVAE    | Persistent Low-Correctness Prediction; Targeted synthetic only        | AUPRC                                       | $0.0630 \pm 0.0225$ (3)             | Higher    |

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| Dataset     | Generator | Scope / comparison                                                     | Metric | Mean $\pm$ SD ( <i>n</i> )         | Direction |
|-------------|-----------|------------------------------------------------------------------------|--------|------------------------------------|-----------|
| ASSISTments | SeqVAE    | Persistent Low-Correctness Prediction; Targeted synthetic only         | AUROC  | 0.4811 $\pm$ 0.0780 (3)            | Higher    |
| ASSISTments | SeqVAE    | Persistent Low-Correctness Prediction; Standard augmentation gain      | AUPRC  | 0.0618 $\pm$ 0.0091 (3)            | Higher    |
| ASSISTments | SeqVAE    | Persistent Low-Correctness Prediction; Standard augmentation gain      | AUROC  | 0.0720 $\pm$ 0.0142 (3)            | Higher    |
| ASSISTments | SeqVAE    | Persistent Low-Correctness Prediction; Standard synthetic utility loss | AUPRC  | 0.0664 $\pm$ 0.0135 (2; 1 missing) | Lower     |
| ASSISTments | SeqVAE    | Persistent Low-Correctness Prediction; Standard synthetic utility loss | AUROC  | 0.1473 $\pm$ 0.1434 (2; 1 missing) | Lower     |
| ASSISTments | SeqVAE    | Recovery Prediction; Real + standard synthetic                         | AUPRC  | 0.3816 $\pm$ 0.0306 (3)            | Higher    |
| ASSISTments | SeqVAE    | Recovery Prediction; Real + standard synthetic                         | AUROC  | 0.6364 $\pm$ 0.0184 (3)            | Higher    |
| ASSISTments | SeqVAE    | Recovery Prediction; Real only (TRTR)                                  | AUPRC  | 0.4387 $\pm$ 0.0000 (3)            | Higher    |
| ASSISTments | SeqVAE    | Recovery Prediction; Real only (TRTR)                                  | AUROC  | 0.6897 $\pm$ 0.0000 (3)            | Higher    |
| ASSISTments | SeqVAE    | Recovery Prediction; Standard synthetic only (TSTR)                    | AUPRC  | 0.3080 $\pm$ 0.0356 (3)            | Higher    |
| ASSISTments | SeqVAE    | Recovery Prediction; Standard synthetic only (TSTR)                    | AUROC  | 0.5099 $\pm$ 0.0476 (3)            | Higher    |
| ASSISTments | SeqVAE    | Recovery Prediction; Targeted synthetic only                           | AUPRC  | 0.2984 $\pm$ 0.0227 (3)            | Higher    |
| ASSISTments | SeqVAE    | Recovery Prediction; Targeted synthetic only                           | AUROC  | 0.5012 $\pm$ 0.0487 (3)            | Higher    |
| ASSISTments | SeqVAE    | Recovery Prediction; Standard augmentation gain                        | AUPRC  | -0.0570 $\pm$ 0.0306 (3)           | Higher    |
| ASSISTments | SeqVAE    | Recovery Prediction; Standard augmentation gain                        | AUROC  | -0.0534 $\pm$ 0.0184 (3)           | Higher    |
| ASSISTments | SeqVAE    | Recovery Prediction; Standard synthetic utility loss                   | AUPRC  | 0.1307 $\pm$ 0.0356 (3)            | Lower     |
| ASSISTments | SeqVAE    | Recovery Prediction; Standard synthetic utility loss                   | AUROC  | 0.1798 $\pm$ 0.0476 (3)            | Lower     |
| ASSISTments | SeqVAE    | High Hint Use; Real + standard synthetic                               | AUPRC  | 0.4170 $\pm$ 0.0083 (3)            | Higher    |
| ASSISTments | SeqVAE    | High Hint Use; Real + standard synthetic                               | AUROC  | 0.8591 $\pm$ 0.0010 (3)            | Higher    |
| ASSISTments | SeqVAE    | High Hint Use; Real + targeted synthetic                               | AUPRC  | 0.3914 $\pm$ 0.0054 (3)            | Higher    |
| ASSISTments | SeqVAE    | High Hint Use; Real + targeted synthetic                               | AUROC  | 0.8494 $\pm$ 0.0052 (3)            | Higher    |
| ASSISTments | SeqVAE    | High Hint Use; Real only (TRTR)                                        | AUPRC  | 0.4399 $\pm$ 0.0000 (3)            | Higher    |
| ASSISTments | SeqVAE    | High Hint Use; Real only (TRTR)                                        | AUROC  | 0.8697 $\pm$ 0.0000 (3)            | Higher    |
| ASSISTments | SeqVAE    | High Hint Use; Standard synthetic only (TSTR)                          | AUPRC  | 0.1086 $\pm$ 0.0087 (3)            | Higher    |
| ASSISTments | SeqVAE    | High Hint Use; Standard synthetic only (TSTR)                          | AUROC  | 0.5344 $\pm$ 0.0237 (3)            | Higher    |
| ASSISTments | SeqVAE    | High Hint Use; Targeted synthetic only                                 | AUPRC  | 0.1042 $\pm$ 0.0184 (3)            | Higher    |
| ASSISTments | SeqVAE    | High Hint Use; Targeted synthetic only                                 | AUROC  | 0.4872 $\pm$ 0.0601 (3)            | Higher    |
| ASSISTments | SeqVAE    | High Hint Use; Standard augmentation gain                              | AUPRC  | -0.0229 $\pm$ 0.0083 (3)           | Higher    |
| ASSISTments | SeqVAE    | High Hint Use; Standard augmentation gain                              | AUROC  | -0.0106 $\pm$ 0.0010 (3)           | Higher    |
| ASSISTments | SeqVAE    | High Hint Use; Standard synthetic utility loss                         | AUPRC  | 0.3313 $\pm$ 0.0087 (3)            | Lower     |
| ASSISTments | SeqVAE    | High Hint Use; Standard synthetic utility loss                         | AUROC  | 0.3353 $\pm$ 0.0237 (3)            | Lower     |
| ASSISTments | SeqVAE    | High Hint Use; Targeted-vs-standard TSTR gain                          | AUPRC  | -0.0044 $\pm$ 0.0255 (3)           | Higher    |
| ASSISTments | SeqVAE    | High Hint Use; Targeted-vs-standard TSTR gain                          | AUROC  | -0.0472 $\pm$ 0.0765 (3)           | Higher    |
| ASSISTments | SeqVAE    | High Hint Use; Targeted augmentation gain                              | AUPRC  | -0.0485 $\pm$ 0.0054 (3)           | Higher    |
| ASSISTments | SeqVAE    | High Hint Use; Targeted augmentation gain                              | AUROC  | -0.0203 $\pm$ 0.0052 (3)           | Higher    |
| ASSISTments | SeqVAE    | Late Correctness Decline; Real + standard synthetic                    | AUPRC  | 0.7488 $\pm$ 0.0027 (3)            | Higher    |
| ASSISTments | SeqVAE    | Late Correctness Decline; Real + standard synthetic                    | AUROC  | 0.7420 $\pm$ 0.0024 (3)            | Higher    |
| ASSISTments | SeqVAE    | Late Correctness Decline; Real + targeted synthetic                    | AUPRC  | 0.7548 $\pm$ 0.0012 (3)            | Higher    |
| ASSISTments | SeqVAE    | Late Correctness Decline; Real + targeted synthetic                    | AUROC  | 0.7449 $\pm$ 0.0004 (3)            | Higher    |
| ASSISTments | SeqVAE    | Late Correctness Decline; Real only (TRTR)                             | AUPRC  | 0.7467 $\pm$ 0.0000 (3)            | Higher    |
| ASSISTments | SeqVAE    | Late Correctness Decline; Real only (TRTR)                             | AUROC  | 0.7451 $\pm$ 0.0000 (3)            | Higher    |
| ASSISTments | SeqVAE    | Late Correctness Decline; Standard synthetic only (TSTR)               | AUPRC  | 0.5608 $\pm$ 0.0143 (3)            | Higher    |
| ASSISTments | SeqVAE    | Late Correctness Decline; Standard synthetic only (TSTR)               | AUROC  | 0.5327 $\pm$ 0.0011 (3)            | Higher    |
| ASSISTments | SeqVAE    | Late Correctness Decline; Targeted synthetic only                      | AUPRC  | 0.5402 $\pm$ 0.0446 (3)            | Higher    |
| ASSISTments | SeqVAE    | Late Correctness Decline; Targeted synthetic only                      | AUROC  | 0.5150 $\pm$ 0.0468 (3)            | Higher    |

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| Dataset     | Generator | Scope / comparison                                             | Metric | Mean $\pm$ SD ( <i>n</i> ) | Direction |
|-------------|-----------|----------------------------------------------------------------|--------|----------------------------|-----------|
| ASSISTments | SeqVAE    | Late Correctness Decline; Standard augmentation gain           | AUPRC  | 0.0021 $\pm$ 0.0027 (3)    | Higher    |
| ASSISTments | SeqVAE    | Late Correctness Decline; Standard augmentation gain           | AUROC  | -0.0031 $\pm$ 0.0024 (3)   | Higher    |
| ASSISTments | SeqVAE    | Late Correctness Decline; Standard synthetic utility loss      | AUPRC  | 0.1860 $\pm$ 0.0143 (3)    | Lower     |
| ASSISTments | SeqVAE    | Late Correctness Decline; Standard synthetic utility loss      | AUROC  | 0.2124 $\pm$ 0.0011 (3)    | Lower     |
| ASSISTments | SeqVAE    | Late Correctness Decline; Targeted-vs-standard TSTR gain       | AUPRC  | -0.0206 $\pm$ 0.0419 (3)   | Higher    |
| ASSISTments | SeqVAE    | Late Correctness Decline; Targeted-vs-standard TSTR gain       | AUROC  | -0.0177 $\pm$ 0.0459 (3)   | Higher    |
| ASSISTments | SeqVAE    | Late Correctness Decline; Targeted augmentation gain           | AUPRC  | 0.0081 $\pm$ 0.0012 (3)    | Higher    |
| ASSISTments | SeqVAE    | Late Correctness Decline; Targeted augmentation gain           | AUROC  | -0.0002 $\pm$ 0.0004 (3)   | Higher    |
| ASSISTments | SeqVAE    | Persistent Low Correctness; Real + standard synthetic          | AUPRC  | 0.1212 $\pm$ 0.0073 (3)    | Higher    |
| ASSISTments | SeqVAE    | Persistent Low Correctness; Real + standard synthetic          | AUROC  | 0.7884 $\pm$ 0.0049 (3)    | Higher    |
| ASSISTments | SeqVAE    | Persistent Low Correctness; Real + targeted synthetic          | AUPRC  | 0.1084 $\pm$ 0.0056 (3)    | Higher    |
| ASSISTments | SeqVAE    | Persistent Low Correctness; Real + targeted synthetic          | AUROC  | 0.7558 $\pm$ 0.0178 (3)    | Higher    |
| ASSISTments | SeqVAE    | Persistent Low Correctness; Real only (TRTR)                   | AUPRC  | 0.1436 $\pm$ 0.0000 (3)    | Higher    |
| ASSISTments | SeqVAE    | Persistent Low Correctness; Real only (TRTR)                   | AUROC  | 0.8222 $\pm$ 0.0000 (3)    | Higher    |
| ASSISTments | SeqVAE    | Persistent Low Correctness; Standard synthetic only (TSTR)     | AUPRC  | 0.0361 $\pm$ 0.0071 (3)    | Higher    |
| ASSISTments | SeqVAE    | Persistent Low Correctness; Standard synthetic only (TSTR)     | AUROC  | 0.5410 $\pm$ 0.0544 (3)    | Higher    |
| ASSISTments | SeqVAE    | Persistent Low Correctness; Targeted synthetic only            | AUPRC  | 0.0302 $\pm$ 0.0059 (3)    | Higher    |
| ASSISTments | SeqVAE    | Persistent Low Correctness; Targeted synthetic only            | AUROC  | 0.5015 $\pm$ 0.0544 (3)    | Higher    |
| ASSISTments | SeqVAE    | Persistent Low Correctness; Standard augmentation gain         | AUPRC  | -0.0224 $\pm$ 0.0073 (3)   | Higher    |
| ASSISTments | SeqVAE    | Persistent Low Correctness; Standard augmentation gain         | AUROC  | -0.0338 $\pm$ 0.0049 (3)   | Higher    |
| ASSISTments | SeqVAE    | Persistent Low Correctness; Standard synthetic utility loss    | AUPRC  | 0.1076 $\pm$ 0.0071 (3)    | Lower     |
| ASSISTments | SeqVAE    | Persistent Low Correctness; Standard synthetic utility loss    | AUROC  | 0.2812 $\pm$ 0.0544 (3)    | Lower     |
| ASSISTments | SeqVAE    | Persistent Low Correctness; Targeted-vs-standard TSTR gain     | AUPRC  | -0.0058 $\pm$ 0.0130 (3)   | Higher    |
| ASSISTments | SeqVAE    | Persistent Low Correctness; Targeted-vs-standard TSTR gain     | AUROC  | -0.0395 $\pm$ 0.0964 (3)   | Higher    |
| ASSISTments | SeqVAE    | Persistent Low Correctness; Targeted augmentation gain         | AUPRC  | -0.0352 $\pm$ 0.0056 (3)   | Higher    |
| ASSISTments | SeqVAE    | Persistent Low Correctness; Targeted augmentation gain         | AUROC  | -0.0664 $\pm$ 0.0178 (3)   | Higher    |
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Responding; Real + standard synthetic       | AUPRC  | 0.2959 $\pm$ 0.0053 (3)    | Higher    |
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Responding; Real + standard synthetic       | AUROC  | 0.8593 $\pm$ 0.0013 (3)    | Higher    |
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Responding; Real + targeted synthetic       | AUPRC  | 0.2828 $\pm$ 0.0080 (3)    | Higher    |
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Responding; Real + targeted synthetic       | AUROC  | 0.8448 $\pm$ 0.0088 (3)    | Higher    |
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Responding; Real only (TRTR)                | AUPRC  | 0.3409 $\pm$ 0.0000 (3)    | Higher    |
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Responding; Real only (TRTR)                | AUROC  | 0.8810 $\pm$ 0.0000 (3)    | Higher    |
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Responding; Standard synthetic only (TSTR)  | AUPRC  | 0.0870 $\pm$ 0.0060 (3)    | Higher    |
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Responding; Standard synthetic only (TSTR)  | AUROC  | 0.5216 $\pm$ 0.0335 (3)    | Higher    |
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Responding; Targeted synthetic only         | AUPRC  | 0.0743 $\pm$ 0.0129 (3)    | Higher    |
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Responding; Targeted synthetic only         | AUROC  | 0.4541 $\pm$ 0.0738 (3)    | Higher    |
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Responding; Standard augmentation gain      | AUPRC  | -0.0450 $\pm$ 0.0053 (3)   | Higher    |
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Responding; Standard augmentation gain      | AUROC  | -0.0216 $\pm$ 0.0013 (3)   | Higher    |
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Responding; Standard synthetic utility loss | AUPRC  | 0.2539 $\pm$ 0.0060 (3)    | Lower     |
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Responding; Standard synthetic utility loss | AUROC  | 0.3594 $\pm$ 0.0335 (3)    | Lower     |
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Responding; Targeted-vs-standard TSTR gain  | AUPRC  | -0.0127 $\pm$ 0.0189 (3)   | Higher    |
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Responding; Targeted-vs-standard TSTR gain  | AUROC  | -0.0675 $\pm$ 0.1069 (3)   | Higher    |
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Responding; Targeted augmentation gain      | AUPRC  | -0.0580 $\pm$ 0.0080 (3)   | Higher    |

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| Dataset     | Generator | Scope / comparison                                        | Metric                                      | Mean $\pm$ SD ( <i>n</i> )              | Direction |
|-------------|-----------|-----------------------------------------------------------|---------------------------------------------|-----------------------------------------|-----------|
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Responding; Targeted augmentation gain | AUROC                                       | $-0.0362 \pm 0.0088$ (3)                | Higher    |
| ASSISTments | SeqVAE    | Rare Skill Path; Real + standard synthetic                | AUPRC                                       | $0.8805 \pm 0.0061$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Rare Skill Path; Real + standard synthetic                | AUROC                                       | $0.7039 \pm 0.0032$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Rare Skill Path; Real + targeted synthetic                | AUPRC                                       | $0.8953 \pm 0.0005$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Rare Skill Path; Real + targeted synthetic                | AUROC                                       | $0.7309 \pm 0.0064$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Rare Skill Path; Real only (TRTR)                         | AUPRC                                       | $0.9009 \pm 0.0000$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Rare Skill Path; Real only (TRTR)                         | AUROC                                       | $0.7239 \pm 0.0000$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Rare Skill Path; Standard synthetic only (TSTR)           | AUPRC                                       | $0.7866 \pm 0.0348$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Rare Skill Path; Standard synthetic only (TSTR)           | AUROC                                       | $0.4846 \pm 0.0572$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Rare Skill Path; Targeted synthetic only                  | AUPRC                                       | $0.8085 \pm 0.0168$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Rare Skill Path; Targeted synthetic only                  | AUROC                                       | $0.5413 \pm 0.0480$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Rare Skill Path; Standard augmentation gain               | AUPRC                                       | $-0.0205 \pm 0.0061$ (3)                | Higher    |
| ASSISTments | SeqVAE    | Rare Skill Path; Standard augmentation gain               | AUROC                                       | $-0.0201 \pm 0.0032$ (3)                | Higher    |
| ASSISTments | SeqVAE    | Rare Skill Path; Standard synthetic utility loss          | AUPRC                                       | $0.1143 \pm 0.0348$ (3)                 | Lower     |
| ASSISTments | SeqVAE    | Rare Skill Path; Standard synthetic utility loss          | AUROC                                       | $0.2393 \pm 0.0572$ (3)                 | Lower     |
| ASSISTments | SeqVAE    | Rare Skill Path; Targeted-vs-standard TSTR gain           | AUPRC                                       | $0.0219 \pm 0.0489$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Rare Skill Path; Targeted-vs-standard TSTR gain           | AUROC                                       | $0.0567 \pm 0.1009$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Rare Skill Path; Targeted augmentation gain               | AUPRC                                       | $-0.0056 \pm 0.0005$ (3)                | Higher    |
| ASSISTments | SeqVAE    | Rare Skill Path; Targeted augmentation gain               | AUROC                                       | $0.0070 \pm 0.0064$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Recovery; Real + standard synthetic                       | AUPRC                                       | $0.7538 \pm 0.0056$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Recovery; Real + standard synthetic                       | AUROC                                       | $0.7353 \pm 0.0039$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Recovery; Real + targeted synthetic                       | AUPRC                                       | $0.7379 \pm 0.0088$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Recovery; Real + targeted synthetic                       | AUROC                                       | $0.7246 \pm 0.0064$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Recovery; Real only (TRTR)                                | AUPRC                                       | $0.7445 \pm 0.0000$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Recovery; Real only (TRTR)                                | AUROC                                       | $0.7329 \pm 0.0000$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Recovery; Standard synthetic only (TSTR)                  | AUPRC                                       | $0.5886 \pm 0.0867$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Recovery; Standard synthetic only (TSTR)                  | AUROC                                       | $0.5290 \pm 0.0933$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Recovery; Targeted synthetic only                         | AUPRC                                       | $0.5215 \pm 0.0252$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Recovery; Targeted synthetic only                         | AUROC                                       | $0.4260 \pm 0.0431$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Recovery; Standard augmentation gain                      | AUPRC                                       | $0.0093 \pm 0.0056$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Recovery; Standard augmentation gain                      | AUROC                                       | $0.0024 \pm 0.0039$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Recovery; Standard synthetic utility loss                 | AUPRC                                       | $0.1559 \pm 0.0867$ (3)                 | Lower     |
| ASSISTments | SeqVAE    | Recovery; Standard synthetic utility loss                 | AUROC                                       | $0.2038 \pm 0.0933$ (3)                 | Lower     |
| ASSISTments | SeqVAE    | Recovery; Targeted-vs-standard TSTR gain                  | AUPRC                                       | $-0.0671 \pm 0.0764$ (3)                | Higher    |
| ASSISTments | SeqVAE    | Recovery; Targeted-vs-standard TSTR gain                  | AUROC                                       | $-0.1030 \pm 0.0921$ (3)                | Higher    |
| ASSISTments | SeqVAE    | Recovery; Targeted augmentation gain                      | AUPRC                                       | $-0.0065 \pm 0.0088$ (3)                | Higher    |
| ASSISTments | SeqVAE    | Recovery; Targeted augmentation gain                      | AUROC                                       | $-0.0083 \pm 0.0064$ (3)                | Higher    |
| ASSISTments | SeqVAE    | High Hint Use                                             | Rare-group prevalence-error reduction       | $0.0020 \pm 0.0017$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | High Hint Use                                             | Rare-group trajectory-shape-error reduction | $0.0411$ (one available run; 2 missing) | Higher    |
| ASSISTments | SeqVAE    | Late Correctness Decline                                  | Rare-group prevalence-error reduction       | $0.0039 \pm 0.0048$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Late Correctness Decline                                  | Rare-group trajectory-shape-error reduction | $0.0143 \pm 0.0197$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Persistent Low Correctness                                | Rare-group prevalence-error reduction       | $0.0001 \pm 0.0001$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Responding                             | Rare-group prevalence-error reduction       | $0.0001 \pm 0.0010$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Rare Skill Path                                           | Rare-group prevalence-error reduction       | $-0.0046 \pm 0.0014$ (3)                | Higher    |
| ASSISTments | SeqVAE    | Rare Skill Path                                           | Rare-group trajectory-shape-error reduction | $-0.0282 \pm 0.0038$ (3)                | Higher    |
| ASSISTments | SeqVAE    | Recovery                                                  | Rare-group prevalence-error reduction       | $0.0022 \pm 0.0037$ (3)                 | Higher    |
| ASSISTments | SeqVAE    | Recovery                                                  | Rare-group trajectory-shape-error reduction | $-0.0126 \pm 0.0142$ (3)                | Higher    |
| ASSISTments | TimeGAN   | Real + standard synthetic                                 | AUPRC                                       | $0.8146 \pm 0.0094$ (3)                 | Higher    |
| ASSISTments | TimeGAN   | Real + standard synthetic                                 | AUROC                                       | $0.7335 \pm 0.0062$ (3)                 | Higher    |
| ASSISTments | TimeGAN   | Real + targeted synthetic                                 | AUPRC                                       | $0.8176 \pm 0.0048$ (3)                 | Higher    |
| ASSISTments | TimeGAN   | Real + targeted synthetic                                 | AUROC                                       | $0.7358 \pm 0.0026$ (3)                 | Higher    |
| ASSISTments | TimeGAN   | Real only (TRTR)                                          | AUPRC                                       | $0.8232 \pm 0.0000$ (3)                 | Higher    |
| ASSISTments | TimeGAN   | Real only (TRTR)                                          | AUROC                                       | $0.7384 \pm 0.0000$ (3)                 | Higher    |
| ASSISTments | TimeGAN   | Standard synthetic only (TSTR)                            | AUPRC                                       | $0.6630 \pm 0.0381$ (3)                 | Higher    |
| ASSISTments | TimeGAN   | Standard synthetic only (TSTR)                            | AUROC                                       | $0.5259 \pm 0.0671$ (3)                 | Higher    |
| ASSISTments | TimeGAN   | Targeted synthetic only                                   | AUPRC                                       | $0.6612 \pm 0.0293$ (3)                 | Higher    |
| ASSISTments | TimeGAN   | Targeted synthetic only                                   | AUROC                                       | $0.5210 \pm 0.0567$ (3)                 | Higher    |
| ASSISTments | TimeGAN   | Standard augmentation gain                                | AUPRC                                       | $-0.0086 \pm 0.0094$ (3)                | Higher    |
| ASSISTments | TimeGAN   | Standard augmentation gain                                | AUROC                                       | $-0.0049 \pm 0.0062$ (3)                | Higher    |

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| Dataset     | Generator | Scope / comparison                                                | Metric | Mean $\pm$ SD ( <i>n</i> )            | Direction |
|-------------|-----------|-------------------------------------------------------------------|--------|---------------------------------------|-----------|
| ASSISTments | TimeGAN   | Standard synthetic utility loss                                   | AUPRC  | 0.1602 $\pm$ 0.0381 (3)               | Lower     |
| ASSISTments | TimeGAN   | Standard synthetic utility loss                                   | AUROC  | 0.2125 $\pm$ 0.0671 (3)               | Lower     |
| ASSISTments | TimeGAN   | Targeted-vs-standard TSTR gain                                    | AUPRC  | -0.0017 $\pm$ 0.0187 (3)              | Higher    |
| ASSISTments | TimeGAN   | Targeted-vs-standard TSTR gain                                    | AUROC  | -0.0050 $\pm$ 0.0336 (3)              | Higher    |
| ASSISTments | TimeGAN   | Targeted augmentation gain                                        | AUPRC  | -0.0056 $\pm$ 0.0048 (3)              | Higher    |
| ASSISTments | TimeGAN   | Targeted augmentation gain                                        | AUROC  | -0.0025 $\pm$ 0.0026 (3)              | Higher    |
| ASSISTments | TimeGAN   | Persistent Low-Correctness Prediction; Real + standard synthetic  | AUPRC  | 0.1349 $\pm$ 0.0154 (3)               | Higher    |
| ASSISTments | TimeGAN   | Persistent Low-Correctness Prediction; Real + standard synthetic  | AUROC  | 0.7280 $\pm$ 0.0011 (3)               | Higher    |
| ASSISTments | TimeGAN   | Persistent Low-Correctness Prediction; Real only (TRTR)           | AUPRC  | 0.1345 $\pm$ 0.0000 (3)               | Higher    |
| ASSISTments | TimeGAN   | Persistent Low-Correctness Prediction; Real only (TRTR)           | AUROC  | 0.7011 $\pm$ 0.0000 (3)               | Higher    |
| ASSISTments | TimeGAN   | Persistent Low-Correctness Prediction; Standard augmentation gain | AUPRC  | 0.0004 $\pm$ 0.0154 (3)               | Higher    |
| ASSISTments | TimeGAN   | Persistent Low-Correctness Prediction; Standard augmentation gain | AUROC  | 0.0269 $\pm$ 0.0011 (3)               | Higher    |
| ASSISTments | TimeGAN   | Recovery Prediction; Real + standard synthetic                    | AUPRC  | 0.4304 $\pm$ 0.0116 (3)               | Higher    |
| ASSISTments | TimeGAN   | Recovery Prediction; Real + standard synthetic                    | AUROC  | 0.6759 $\pm$ 0.0196 (3)               | Higher    |
| ASSISTments | TimeGAN   | Recovery Prediction; Real only (TRTR)                             | AUPRC  | 0.4387 $\pm$ 0.0000 (3)               | Higher    |
| ASSISTments | TimeGAN   | Recovery Prediction; Real only (TRTR)                             | AUROC  | 0.6897 $\pm$ 0.0000 (3)               | Higher    |
| ASSISTments | TimeGAN   | Recovery Prediction; Standard synthetic only (TSTR)               | AUPRC  | 0.3738 (one available run; 2 missing) | Higher    |
| ASSISTments | TimeGAN   | Recovery Prediction; Standard synthetic only (TSTR)               | AUROC  | 0.5787 (one available run; 2 missing) | Higher    |
| ASSISTments | TimeGAN   | Recovery Prediction; Targeted synthetic only                      | AUPRC  | 0.2593 (one available run; 2 missing) | Higher    |
| ASSISTments | TimeGAN   | Recovery Prediction; Targeted synthetic only                      | AUROC  | 0.4649 (one available run; 2 missing) | Higher    |
| ASSISTments | TimeGAN   | Recovery Prediction; Standard augmentation gain                   | AUPRC  | -0.0082 $\pm$ 0.0116 (3)              | Higher    |
| ASSISTments | TimeGAN   | Recovery Prediction; Standard augmentation gain                   | AUROC  | -0.0138 $\pm$ 0.0196 (3)              | Higher    |
| ASSISTments | TimeGAN   | Recovery Prediction; Standard synthetic utility loss              | AUPRC  | 0.0649 (one available run; 2 missing) | Lower     |
| ASSISTments | TimeGAN   | Recovery Prediction; Standard synthetic utility loss              | AUROC  | 0.1111 (one available run; 2 missing) | Lower     |
| ASSISTments | TimeGAN   | High Hint Use; Real + standard synthetic                          | AUPRC  | 0.4302 $\pm$ 0.0068 (3)               | Higher    |
| ASSISTments | TimeGAN   | High Hint Use; Real + standard synthetic                          | AUROC  | 0.8583 $\pm$ 0.0064 (3)               | Higher    |
| ASSISTments | TimeGAN   | High Hint Use; Real + targeted synthetic                          | AUPRC  | 0.4327 $\pm$ 0.0063 (3)               | Higher    |
| ASSISTments | TimeGAN   | High Hint Use; Real + targeted synthetic                          | AUROC  | 0.8578 $\pm$ 0.0073 (3)               | Higher    |
| ASSISTments | TimeGAN   | High Hint Use; Real only (TRTR)                                   | AUPRC  | 0.4399 $\pm$ 0.0000 (3)               | Higher    |
| ASSISTments | TimeGAN   | High Hint Use; Real only (TRTR)                                   | AUROC  | 0.8697 $\pm$ 0.0000 (3)               | Higher    |
| ASSISTments | TimeGAN   | High Hint Use; Standard synthetic only (TSTR)                     | AUPRC  | 0.1550 $\pm$ 0.0671 (3)               | Higher    |
| ASSISTments | TimeGAN   | High Hint Use; Standard synthetic only (TSTR)                     | AUROC  | 0.5438 $\pm$ 0.1026 (3)               | Higher    |
| ASSISTments | TimeGAN   | High Hint Use; Targeted synthetic only                            | AUPRC  | 0.1250 $\pm$ 0.0487 (3)               | Higher    |
| ASSISTments | TimeGAN   | High Hint Use; Targeted synthetic only                            | AUROC  | 0.4969 $\pm$ 0.0691 (3)               | Higher    |
| ASSISTments | TimeGAN   | High Hint Use; Standard augmentation gain                         | AUPRC  | -0.0098 $\pm$ 0.0068 (3)              | Higher    |
| ASSISTments | TimeGAN   | High Hint Use; Standard augmentation gain                         | AUROC  | -0.0113 $\pm$ 0.0064 (3)              | Higher    |
| ASSISTments | TimeGAN   | High Hint Use; Standard synthetic utility loss                    | AUPRC  | 0.2850 $\pm$ 0.0671 (3)               | Lower     |
| ASSISTments | TimeGAN   | High Hint Use; Standard synthetic utility loss                    | AUROC  | 0.3258 $\pm$ 0.1026 (3)               | Lower     |
| ASSISTments | TimeGAN   | High Hint Use; Targeted-vs-standard TSTR gain                     | AUPRC  | -0.0300 $\pm$ 0.0219 (3)              | Higher    |
| ASSISTments | TimeGAN   | High Hint Use; Targeted-vs-standard TSTR gain                     | AUROC  | -0.0469 $\pm$ 0.0421 (3)              | Higher    |
| ASSISTments | TimeGAN   | High Hint Use; Targeted augmentation gain                         | AUPRC  | -0.0072 $\pm$ 0.0063 (3)              | Higher    |
| ASSISTments | TimeGAN   | High Hint Use; Targeted augmentation gain                         | AUROC  | -0.0119 $\pm$ 0.0073 (3)              | Higher    |
| ASSISTments | TimeGAN   | Late Correctness Decline; Real + standard synthetic               | AUPRC  | 0.7269 $\pm$ 0.0145 (3)               | Higher    |
| ASSISTments | TimeGAN   | Late Correctness Decline; Real + standard synthetic               | AUROC  | 0.7344 $\pm$ 0.0101 (3)               | Higher    |
| ASSISTments | TimeGAN   | Late Correctness Decline; Real + targeted synthetic               | AUPRC  | 0.7239 $\pm$ 0.0118 (3)               | Higher    |
| ASSISTments | TimeGAN   | Late Correctness Decline; Real + targeted synthetic               | AUROC  | 0.7355 $\pm$ 0.0064 (3)               | Higher    |
| ASSISTments | TimeGAN   | Late Correctness Decline; Real only (TRTR)                        | AUPRC  | 0.7467 $\pm$ 0.0000 (3)               | Higher    |
| ASSISTments | TimeGAN   | Late Correctness Decline; Real only (TRTR)                        | AUROC  | 0.7451 $\pm$ 0.0000 (3)               | Higher    |

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| Dataset     | Generator | Scope / comparison                                             | Metric | Mean $\pm$ SD ( <i>n</i> ) | Direction |
|-------------|-----------|----------------------------------------------------------------|--------|----------------------------|-----------|
| ASSISTments | TimeGAN   | Late Correctness Decline; Standard synthetic only (TSTR)       | AUPRC  | 0.5569 $\pm$ 0.0491 (3)    | Higher    |
| ASSISTments | TimeGAN   | Late Correctness Decline; Standard synthetic only (TSTR)       | AUROC  | 0.5803 $\pm$ 0.0548 (3)    | Higher    |
| ASSISTments | TimeGAN   | Late Correctness Decline; Targeted synthetic only              | AUPRC  | 0.5160 $\pm$ 0.0229 (3)    | Higher    |
| ASSISTments | TimeGAN   | Late Correctness Decline; Targeted synthetic only              | AUROC  | 0.5377 $\pm$ 0.0273 (3)    | Higher    |
| ASSISTments | TimeGAN   | Late Correctness Decline; Standard augmentation gain           | AUPRC  | -0.0198 $\pm$ 0.0145 (3)   | Higher    |
| ASSISTments | TimeGAN   | Late Correctness Decline; Standard augmentation gain           | AUROC  | -0.0107 $\pm$ 0.0101 (3)   | Higher    |
| ASSISTments | TimeGAN   | Late Correctness Decline; Standard synthetic utility loss      | AUPRC  | 0.1898 $\pm$ 0.0491 (3)    | Lower     |
| ASSISTments | TimeGAN   | Late Correctness Decline; Standard synthetic utility loss      | AUROC  | 0.1648 $\pm$ 0.0548 (3)    | Lower     |
| ASSISTments | TimeGAN   | Late Correctness Decline; Targeted-vs-standard TSTR gain       | AUPRC  | -0.0409 $\pm$ 0.0718 (3)   | Higher    |
| ASSISTments | TimeGAN   | Late Correctness Decline; Targeted-vs-standard TSTR gain       | AUROC  | -0.0426 $\pm$ 0.0791 (3)   | Higher    |
| ASSISTments | TimeGAN   | Late Correctness Decline; Targeted augmentation gain           | AUPRC  | -0.0228 $\pm$ 0.0118 (3)   | Higher    |
| ASSISTments | TimeGAN   | Late Correctness Decline; Targeted augmentation gain           | AUROC  | -0.0096 $\pm$ 0.0064 (3)   | Higher    |
| ASSISTments | TimeGAN   | Persistent Low Correctness; Real + standard synthetic          | AUPRC  | 0.1541 $\pm$ 0.0212 (3)    | Higher    |
| ASSISTments | TimeGAN   | Persistent Low Correctness; Real + standard synthetic          | AUROC  | 0.7800 $\pm$ 0.0231 (3)    | Higher    |
| ASSISTments | TimeGAN   | Persistent Low Correctness; Real + targeted synthetic          | AUPRC  | 0.1439 $\pm$ 0.0032 (3)    | Higher    |
| ASSISTments | TimeGAN   | Persistent Low Correctness; Real + targeted synthetic          | AUROC  | 0.7862 $\pm$ 0.0269 (3)    | Higher    |
| ASSISTments | TimeGAN   | Persistent Low Correctness; Real only (TRTR)                   | AUPRC  | 0.1436 $\pm$ 0.0000 (3)    | Higher    |
| ASSISTments | TimeGAN   | Persistent Low Correctness; Real only (TRTR)                   | AUROC  | 0.8222 $\pm$ 0.0000 (3)    | Higher    |
| ASSISTments | TimeGAN   | Persistent Low Correctness; Standard synthetic only (TSTR)     | AUPRC  | 0.0833 $\pm$ 0.0670 (3)    | Higher    |
| ASSISTments | TimeGAN   | Persistent Low Correctness; Standard synthetic only (TSTR)     | AUROC  | 0.5595 $\pm$ 0.1198 (3)    | Higher    |
| ASSISTments | TimeGAN   | Persistent Low Correctness; Targeted synthetic only            | AUPRC  | 0.0631 $\pm$ 0.0576 (3)    | Higher    |
| ASSISTments | TimeGAN   | Persistent Low Correctness; Targeted synthetic only            | AUROC  | 0.4593 $\pm$ 0.1080 (3)    | Higher    |
| ASSISTments | TimeGAN   | Persistent Low Correctness; Standard augmentation gain         | AUPRC  | 0.0105 $\pm$ 0.0212 (3)    | Higher    |
| ASSISTments | TimeGAN   | Persistent Low Correctness; Standard augmentation gain         | AUROC  | -0.0422 $\pm$ 0.0231 (3)   | Higher    |
| ASSISTments | TimeGAN   | Persistent Low Correctness; Standard synthetic utility loss    | AUPRC  | 0.0603 $\pm$ 0.0670 (3)    | Lower     |
| ASSISTments | TimeGAN   | Persistent Low Correctness; Standard synthetic utility loss    | AUROC  | 0.2627 $\pm$ 0.1198 (3)    | Lower     |
| ASSISTments | TimeGAN   | Persistent Low Correctness; Targeted-vs-standard TSTR gain     | AUPRC  | -0.0203 $\pm$ 0.0106 (3)   | Higher    |
| ASSISTments | TimeGAN   | Persistent Low Correctness; Targeted-vs-standard TSTR gain     | AUROC  | -0.1001 $\pm$ 0.0382 (3)   | Higher    |
| ASSISTments | TimeGAN   | Persistent Low Correctness; Targeted augmentation gain         | AUPRC  | 0.0003 $\pm$ 0.0032 (3)    | Higher    |
| ASSISTments | TimeGAN   | Persistent Low Correctness; Targeted augmentation gain         | AUROC  | -0.0360 $\pm$ 0.0269 (3)   | Higher    |
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Responding; Real + standard synthetic       | AUPRC  | 0.3171 $\pm$ 0.0103 (3)    | Higher    |
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Responding; Real + standard synthetic       | AUROC  | 0.8622 $\pm$ 0.0152 (3)    | Higher    |
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Responding; Real + targeted synthetic       | AUPRC  | 0.3161 $\pm$ 0.0073 (3)    | Higher    |
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Responding; Real + targeted synthetic       | AUROC  | 0.8652 $\pm$ 0.0099 (3)    | Higher    |
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Responding; Real only (TRTR)                | AUPRC  | 0.3409 $\pm$ 0.0000 (3)    | Higher    |
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Responding; Real only (TRTR)                | AUROC  | 0.8810 $\pm$ 0.0000 (3)    | Higher    |
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Responding; Standard synthetic only (TSTR)  | AUPRC  | 0.1296 $\pm$ 0.0627 (3)    | Higher    |
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Responding; Standard synthetic only (TSTR)  | AUROC  | 0.5301 $\pm$ 0.1321 (3)    | Higher    |
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Responding; Targeted synthetic only         | AUPRC  | 0.1017 $\pm$ 0.0397 (3)    | Higher    |
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Responding; Targeted synthetic only         | AUROC  | 0.4677 $\pm$ 0.1231 (3)    | Higher    |
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Responding; Standard augmentation gain      | AUPRC  | -0.0238 $\pm$ 0.0103 (3)   | Higher    |
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Responding; Standard augmentation gain      | AUROC  | -0.0188 $\pm$ 0.0152 (3)   | Higher    |
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Responding; Standard synthetic utility loss | AUPRC  | 0.2112 $\pm$ 0.0627 (3)    | Lower     |

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| Dataset     | Generator | Scope / comparison                                             | Metric                                      | Mean $\pm$ SD ( <i>n</i> )          | Direction |
|-------------|-----------|----------------------------------------------------------------|---------------------------------------------|-------------------------------------|-----------|
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Responding; Standard synthetic utility loss | AUROC                                       | $0.3509 \pm 0.1321$ (3)             | Lower     |
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Responding; Targeted-vs-standard TSTR gain  | AUPRC                                       | $-0.0279 \pm 0.0307$ (3)            | Higher    |
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Responding; Targeted-vs-standard TSTR gain  | AUROC                                       | $-0.0624 \pm 0.1196$ (3)            | Higher    |
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Responding; Targeted augmentation gain      | AUPRC                                       | $-0.0248 \pm 0.0073$ (3)            | Higher    |
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Responding; Targeted augmentation gain      | AUROC                                       | $-0.0157 \pm 0.0099$ (3)            | Higher    |
| ASSISTments | TimeGAN   | Rare Skill Path; Real + standard synthetic                     | AUPRC                                       | $0.8878 \pm 0.0169$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Rare Skill Path; Real + standard synthetic                     | AUROC                                       | $0.7094 \pm 0.0032$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Rare Skill Path; Real + targeted synthetic                     | AUPRC                                       | $0.8950 \pm 0.0136$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Rare Skill Path; Real + targeted synthetic                     | AUROC                                       | $0.7159 \pm 0.0101$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Rare Skill Path; Real only (TRTR)                              | AUPRC                                       | $0.9009 \pm 0.0000$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Rare Skill Path; Real only (TRTR)                              | AUROC                                       | $0.7239 \pm 0.0000$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Rare Skill Path; Standard synthetic only (TSTR)                | AUPRC                                       | $0.8103 \pm 0.0480$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Rare Skill Path; Standard synthetic only (TSTR)                | AUROC                                       | $0.5168 \pm 0.1208$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Rare Skill Path; Targeted synthetic only                       | AUPRC                                       | $0.8049 \pm 0.0345$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Rare Skill Path; Targeted synthetic only                       | AUROC                                       | $0.5161 \pm 0.0898$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Rare Skill Path; Standard augmentation gain                    | AUPRC                                       | $-0.0131 \pm 0.0169$ (3)            | Higher    |
| ASSISTments | TimeGAN   | Rare Skill Path; Standard augmentation gain                    | AUROC                                       | $-0.0145 \pm 0.0032$ (3)            | Higher    |
| ASSISTments | TimeGAN   | Rare Skill Path; Standard synthetic utility loss               | AUPRC                                       | $0.0907 \pm 0.0480$ (3)             | Lower     |
| ASSISTments | TimeGAN   | Rare Skill Path; Standard synthetic utility loss               | AUROC                                       | $0.2072 \pm 0.1208$ (3)             | Lower     |
| ASSISTments | TimeGAN   | Rare Skill Path; Targeted-vs-standard TSTR gain                | AUPRC                                       | $-0.0054 \pm 0.0286$ (3)            | Higher    |
| ASSISTments | TimeGAN   | Rare Skill Path; Targeted-vs-standard TSTR gain                | AUROC                                       | $-0.0006 \pm 0.0762$ (3)            | Higher    |
| ASSISTments | TimeGAN   | Rare Skill Path; Targeted augmentation gain                    | AUPRC                                       | $-0.0059 \pm 0.0136$ (3)            | Higher    |
| ASSISTments | TimeGAN   | Rare Skill Path; Targeted augmentation gain                    | AUROC                                       | $-0.0080 \pm 0.0101$ (3)            | Higher    |
| ASSISTments | TimeGAN   | Recovery; Real + standard synthetic                            | AUPRC                                       | $0.7340 \pm 0.0033$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Recovery; Real + standard synthetic                            | AUROC                                       | $0.7231 \pm 0.0021$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Recovery; Real + targeted synthetic                            | AUPRC                                       | $0.7330 \pm 0.0067$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Recovery; Real + targeted synthetic                            | AUROC                                       | $0.7243 \pm 0.0046$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Recovery; Real only (TRTR)                                     | AUPRC                                       | $0.7445 \pm 0.0000$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Recovery; Real only (TRTR)                                     | AUROC                                       | $0.7329 \pm 0.0000$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Recovery; Standard synthetic only (TSTR)                       | AUPRC                                       | $0.5913 \pm 0.0318$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Recovery; Standard synthetic only (TSTR)                       | AUROC                                       | $0.5224 \pm 0.0568$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Recovery; Targeted synthetic only                              | AUPRC                                       | $0.6155 \pm 0.0619$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Recovery; Targeted synthetic only                              | AUROC                                       | $0.5607 \pm 0.0721$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Recovery; Standard augmentation gain                           | AUPRC                                       | $-0.0105 \pm 0.0033$ (3)            | Higher    |
| ASSISTments | TimeGAN   | Recovery; Standard augmentation gain                           | AUROC                                       | $-0.0098 \pm 0.0021$ (3)            | Higher    |
| ASSISTments | TimeGAN   | Recovery; Standard synthetic utility loss                      | AUPRC                                       | $0.1532 \pm 0.0318$ (3)             | Lower     |
| ASSISTments | TimeGAN   | Recovery; Standard synthetic utility loss                      | AUROC                                       | $0.2104 \pm 0.0568$ (3)             | Lower     |
| ASSISTments | TimeGAN   | Recovery; Targeted-vs-standard TSTR gain                       | AUPRC                                       | $0.0242 \pm 0.0479$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Recovery; Targeted-vs-standard TSTR gain                       | AUROC                                       | $0.0383 \pm 0.0707$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Recovery; Targeted augmentation gain                           | AUPRC                                       | $-0.0115 \pm 0.0067$ (3)            | Higher    |
| ASSISTments | TimeGAN   | Recovery; Targeted augmentation gain                           | AUROC                                       | $-0.0086 \pm 0.0046$ (3)            | Higher    |
| ASSISTments | TimeGAN   | High Hint Use                                                  | Rare-group prevalence-error reduction       | $-0.0001 \pm 0.0001$ (3)            | Higher    |
| ASSISTments | TimeGAN   | Late Correctness Decline                                       | Rare-group prevalence-error reduction       | $-0.0023 \pm 0.0033$ (3)            | Higher    |
| ASSISTments | TimeGAN   | Persistent Low Correctness                                     | Rare-group prevalence-error reduction       | $0.0000 \pm 0.0000$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Responding                                  | Rare-group prevalence-error reduction       | $0.0000 \pm 0.0000$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Rare Skill Path                                                | Rare-group prevalence-error reduction       | $0.0617 \pm 0.0904$ (3)             | Higher    |
| ASSISTments | TimeGAN   | Rare Skill Path                                                | Rare-group trajectory-shape-error reduction | $-0.0380 \pm 0.0380$ (2; 1 missing) | Higher    |
| ASSISTments | TimeGAN   | Recovery                                                       | Rare-group prevalence-error reduction       | $-0.0028 \pm 0.0040$ (3)            | Higher    |
| EdNet       | Markov    | Real + standard synthetic                                      | AUPRC                                       | $0.7664 \pm 0.0003$ (3)             | Higher    |
| EdNet       | Markov    | Real + standard synthetic                                      | AUROC                                       | $0.6684 \pm 0.0003$ (3)             | Higher    |
| EdNet       | Markov    | Real + targeted synthetic                                      | AUPRC                                       | $0.7607 \pm 0.0005$ (3)             | Higher    |
| EdNet       | Markov    | Real + targeted synthetic                                      | AUROC                                       | $0.6609 \pm 0.0002$ (3)             | Higher    |
| EdNet       | Markov    | Real only (TRTR)                                               | AUPRC                                       | $0.7687 \pm 0.0000$ (3)             | Higher    |
| EdNet       | Markov    | Real only (TRTR)                                               | AUROC                                       | $0.6722 \pm 0.0000$ (3)             | Higher    |
| EdNet       | Markov    | Standard synthetic only (TSTR)                                 | AUPRC                                       | $0.7380 \pm 0.0012$ (3)             | Higher    |
| EdNet       | Markov    | Standard synthetic only (TSTR)                                 | AUROC                                       | $0.6359 \pm 0.0017$ (3)             | Higher    |
| EdNet       | Markov    | Targeted synthetic only                                        | AUPRC                                       | $0.7313 \pm 0.0016$ (3)             | Higher    |
| EdNet       | Markov    | Targeted synthetic only                                        | AUROC                                       | $0.6245 \pm 0.0011$ (3)             | Higher    |

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| Dataset | Generator | Scope / comparison                                                     | Metric | Mean $\pm$ SD ( <i>n</i> ) | Direction |
|---------|-----------|------------------------------------------------------------------------|--------|----------------------------|-----------|
| EdNet   | Markov    | Standard augmentation gain                                             | AUPRC  | $-0.0024 \pm 0.0003$ (3)   | Higher    |
| EdNet   | Markov    | Standard augmentation gain                                             | AUROC  | $-0.0037 \pm 0.0003$ (3)   | Higher    |
| EdNet   | Markov    | Standard synthetic utility loss                                        | AUPRC  | $0.0308 \pm 0.0012$ (3)    | Lower     |
| EdNet   | Markov    | Standard synthetic utility loss                                        | AUROC  | $0.0362 \pm 0.0017$ (3)    | Lower     |
| EdNet   | Markov    | Targeted-vs-standard TSTR gain                                         | AUPRC  | $-0.0066 \pm 0.0020$ (3)   | Higher    |
| EdNet   | Markov    | Targeted-vs-standard TSTR gain                                         | AUROC  | $-0.0114 \pm 0.0024$ (3)   | Higher    |
| EdNet   | Markov    | Targeted augmentation gain                                             | AUPRC  | $-0.0080 \pm 0.0005$ (3)   | Higher    |
| EdNet   | Markov    | Targeted augmentation gain                                             | AUROC  | $-0.0112 \pm 0.0002$ (3)   | Higher    |
| EdNet   | Markov    | Persistent Low-Correctness Prediction; Real + standard synthetic       | AUPRC  | $0.1130 \pm 0.0014$ (3)    | Higher    |
| EdNet   | Markov    | Persistent Low-Correctness Prediction; Real + standard synthetic       | AUROC  | $0.5596 \pm 0.0057$ (3)    | Higher    |
| EdNet   | Markov    | Persistent Low-Correctness Prediction; Real only (TRTR)                | AUPRC  | $0.1310 \pm 0.0000$ (3)    | Higher    |
| EdNet   | Markov    | Persistent Low-Correctness Prediction; Real only (TRTR)                | AUROC  | $0.5859 \pm 0.0000$ (3)    | Higher    |
| EdNet   | Markov    | Persistent Low-Correctness Prediction; Standard synthetic only (TSTR)  | AUPRC  | $0.0877 \pm 0.0044$ (3)    | Higher    |
| EdNet   | Markov    | Persistent Low-Correctness Prediction; Standard synthetic only (TSTR)  | AUROC  | $0.4889 \pm 0.0139$ (3)    | Higher    |
| EdNet   | Markov    | Persistent Low-Correctness Prediction; Targeted synthetic only         | AUPRC  | $0.0943 \pm 0.0055$ (3)    | Higher    |
| EdNet   | Markov    | Persistent Low-Correctness Prediction; Targeted synthetic only         | AUROC  | $0.4945 \pm 0.0166$ (3)    | Higher    |
| EdNet   | Markov    | Persistent Low-Correctness Prediction; Standard augmentation gain      | AUPRC  | $-0.0179 \pm 0.0014$ (3)   | Higher    |
| EdNet   | Markov    | Persistent Low-Correctness Prediction; Standard augmentation gain      | AUROC  | $-0.0263 \pm 0.0057$ (3)   | Higher    |
| EdNet   | Markov    | Persistent Low-Correctness Prediction; Standard synthetic utility loss | AUPRC  | $0.0432 \pm 0.0044$ (3)    | Lower     |
| EdNet   | Markov    | Persistent Low-Correctness Prediction; Standard synthetic utility loss | AUROC  | $0.0969 \pm 0.0139$ (3)    | Lower     |
| EdNet   | Markov    | Recovery Prediction; Real + standard synthetic                         | AUPRC  | $0.2130 \pm 0.0116$ (3)    | Higher    |
| EdNet   | Markov    | Recovery Prediction; Real + standard synthetic                         | AUROC  | $0.4764 \pm 0.0151$ (3)    | Higher    |
| EdNet   | Markov    | Recovery Prediction; Real only (TRTR)                                  | AUPRC  | $0.2194 \pm 0.0000$ (3)    | Higher    |
| EdNet   | Markov    | Recovery Prediction; Real only (TRTR)                                  | AUROC  | $0.4910 \pm 0.0000$ (3)    | Higher    |
| EdNet   | Markov    | Recovery Prediction; Standard synthetic only (TSTR)                    | AUPRC  | $0.2239 \pm 0.0052$ (3)    | Higher    |
| EdNet   | Markov    | Recovery Prediction; Standard synthetic only (TSTR)                    | AUROC  | $0.5052 \pm 0.0194$ (3)    | Higher    |
| EdNet   | Markov    | Recovery Prediction; Targeted synthetic only                           | AUPRC  | $0.2142 \pm 0.0125$ (3)    | Higher    |
| EdNet   | Markov    | Recovery Prediction; Targeted synthetic only                           | AUROC  | $0.4924 \pm 0.0199$ (3)    | Higher    |
| EdNet   | Markov    | Recovery Prediction; Standard augmentation gain                        | AUPRC  | $-0.0064 \pm 0.0116$ (3)   | Higher    |
| EdNet   | Markov    | Recovery Prediction; Standard augmentation gain                        | AUROC  | $-0.0146 \pm 0.0151$ (3)   | Higher    |
| EdNet   | Markov    | Recovery Prediction; Standard synthetic utility loss                   | AUPRC  | $-0.0045 \pm 0.0052$ (3)   | Lower     |
| EdNet   | Markov    | Recovery Prediction; Standard synthetic utility loss                   | AUROC  | $-0.0142 \pm 0.0194$ (3)   | Lower     |
| EdNet   | Markov    | Late Correctness Decline; Real + standard synthetic                    | AUPRC  | $0.6295 \pm 0.0015$ (3)    | Higher    |
| EdNet   | Markov    | Late Correctness Decline; Real + standard synthetic                    | AUROC  | $0.6826 \pm 0.0019$ (3)    | Higher    |
| EdNet   | Markov    | Late Correctness Decline; Real + targeted synthetic                    | AUPRC  | $0.6280 \pm 0.0049$ (3)    | Higher    |
| EdNet   | Markov    | Late Correctness Decline; Real + targeted synthetic                    | AUROC  | $0.6813 \pm 0.0021$ (3)    | Higher    |
| EdNet   | Markov    | Late Correctness Decline; Real only (TRTR)                             | AUPRC  | $0.6426 \pm 0.0000$ (3)    | Higher    |
| EdNet   | Markov    | Late Correctness Decline; Real only (TRTR)                             | AUROC  | $0.6902 \pm 0.0000$ (3)    | Higher    |
| EdNet   | Markov    | Late Correctness Decline; Standard synthetic only (TSTR)               | AUPRC  | $0.6286 \pm 0.0103$ (3)    | Higher    |
| EdNet   | Markov    | Late Correctness Decline; Standard synthetic only (TSTR)               | AUROC  | $0.6819 \pm 0.0092$ (3)    | Higher    |
| EdNet   | Markov    | Late Correctness Decline; Targeted synthetic only                      | AUPRC  | $0.6197 \pm 0.0051$ (3)    | Higher    |
| EdNet   | Markov    | Late Correctness Decline; Targeted synthetic only                      | AUROC  | $0.6619 \pm 0.0043$ (3)    | Higher    |
| EdNet   | Markov    | Late Correctness Decline; Standard augmentation gain                   | AUPRC  | $-0.0131 \pm 0.0015$ (3)   | Higher    |
| EdNet   | Markov    | Late Correctness Decline; Standard augmentation gain                   | AUROC  | $-0.0076 \pm 0.0019$ (3)   | Higher    |
| EdNet   | Markov    | Late Correctness Decline; Standard synthetic utility loss              | AUPRC  | $0.0140 \pm 0.0103$ (3)    | Lower     |
| EdNet   | Markov    | Late Correctness Decline; Standard synthetic utility loss              | AUROC  | $0.0083 \pm 0.0092$ (3)    | Lower     |
| EdNet   | Markov    | Late Correctness Decline; Targeted-vs-standard TSTR gain               | AUPRC  | $-0.0088 \pm 0.0057$ (3)   | Higher    |

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| Dataset | Generator | Scope / comparison                                             | Metric | Mean $\pm$ SD ( <i>n</i> ) | Direction |
|---------|-----------|----------------------------------------------------------------|--------|----------------------------|-----------|
| EdNet   | Markov    | Late Correctness Decline; Targeted-vs-standard TSTR gain       | AUROC  | $-0.0200 \pm 0.0071$ (3)   | Higher    |
| EdNet   | Markov    | Late Correctness Decline; Targeted augmentation gain           | AUPRC  | $-0.0146 \pm 0.0049$ (3)   | Higher    |
| EdNet   | Markov    | Late Correctness Decline; Targeted augmentation gain           | AUROC  | $-0.0089 \pm 0.0021$ (3)   | Higher    |
| EdNet   | Markov    | Persistent Low Correctness; Real + standard synthetic          | AUPRC  | $0.1065 \pm 0.0004$ (3)    | Higher    |
| EdNet   | Markov    | Persistent Low Correctness; Real + standard synthetic          | AUROC  | $0.5436 \pm 0.0006$ (3)    | Higher    |
| EdNet   | Markov    | Persistent Low Correctness; Real + targeted synthetic          | AUPRC  | $0.0947 \pm 0.0005$ (3)    | Higher    |
| EdNet   | Markov    | Persistent Low Correctness; Real + targeted synthetic          | AUROC  | $0.5238 \pm 0.0027$ (3)    | Higher    |
| EdNet   | Markov    | Persistent Low Correctness; Real only (TRTR)                   | AUPRC  | $0.1132 \pm 0.0000$ (3)    | Higher    |
| EdNet   | Markov    | Persistent Low Correctness; Real only (TRTR)                   | AUROC  | $0.5416 \pm 0.0000$ (3)    | Higher    |
| EdNet   | Markov    | Persistent Low Correctness; Standard synthetic only (TSTR)     | AUPRC  | $0.1009 \pm 0.0023$ (3)    | Higher    |
| EdNet   | Markov    | Persistent Low Correctness; Standard synthetic only (TSTR)     | AUROC  | $0.5186 \pm 0.0048$ (3)    | Higher    |
| EdNet   | Markov    | Persistent Low Correctness; Targeted synthetic only            | AUPRC  | $0.0887 \pm 0.0042$ (3)    | Higher    |
| EdNet   | Markov    | Persistent Low Correctness; Targeted synthetic only            | AUROC  | $0.4886 \pm 0.0130$ (3)    | Higher    |
| EdNet   | Markov    | Persistent Low Correctness; Standard augmentation gain         | AUPRC  | $-0.0067 \pm 0.0004$ (3)   | Higher    |
| EdNet   | Markov    | Persistent Low Correctness; Standard augmentation gain         | AUROC  | $0.0020 \pm 0.0006$ (3)    | Higher    |
| EdNet   | Markov    | Persistent Low Correctness; Standard synthetic utility loss    | AUPRC  | $0.0123 \pm 0.0023$ (3)    | Lower     |
| EdNet   | Markov    | Persistent Low Correctness; Standard synthetic utility loss    | AUROC  | $0.0230 \pm 0.0048$ (3)    | Lower     |
| EdNet   | Markov    | Persistent Low Correctness; Targeted-vs-standard TSTR gain     | AUPRC  | $-0.0123 \pm 0.0063$ (3)   | Higher    |
| EdNet   | Markov    | Persistent Low Correctness; Targeted-vs-standard TSTR gain     | AUROC  | $-0.0301 \pm 0.0178$ (3)   | Higher    |
| EdNet   | Markov    | Persistent Low Correctness; Targeted augmentation gain         | AUPRC  | $-0.0186 \pm 0.0005$ (3)   | Higher    |
| EdNet   | Markov    | Persistent Low Correctness; Targeted augmentation gain         | AUROC  | $-0.0178 \pm 0.0027$ (3)   | Higher    |
| EdNet   | Markov    | Rapid Low-Accuracy Responding; Real + standard synthetic       | AUPRC  | $0.3019 \pm 0.0036$ (3)    | Higher    |
| EdNet   | Markov    | Rapid Low-Accuracy Responding; Real + standard synthetic       | AUROC  | $0.5622 \pm 0.0009$ (3)    | Higher    |
| EdNet   | Markov    | Rapid Low-Accuracy Responding; Real + targeted synthetic       | AUPRC  | $0.3096 \pm 0.0011$ (3)    | Higher    |
| EdNet   | Markov    | Rapid Low-Accuracy Responding; Real + targeted synthetic       | AUROC  | $0.5849 \pm 0.0038$ (3)    | Higher    |
| EdNet   | Markov    | Rapid Low-Accuracy Responding; Real only (TRTR)                | AUPRC  | $0.3092 \pm 0.0000$ (3)    | Higher    |
| EdNet   | Markov    | Rapid Low-Accuracy Responding; Real only (TRTR)                | AUROC  | $0.5619 \pm 0.0000$ (3)    | Higher    |
| EdNet   | Markov    | Rapid Low-Accuracy Responding; Standard synthetic only (TSTR)  | AUPRC  | $0.2954 \pm 0.0035$ (3)    | Higher    |
| EdNet   | Markov    | Rapid Low-Accuracy Responding; Standard synthetic only (TSTR)  | AUROC  | $0.5631 \pm 0.0030$ (3)    | Higher    |
| EdNet   | Markov    | Rapid Low-Accuracy Responding; Targeted synthetic only         | AUPRC  | $0.3110 \pm 0.0037$ (3)    | Higher    |
| EdNet   | Markov    | Rapid Low-Accuracy Responding; Targeted synthetic only         | AUROC  | $0.5904 \pm 0.0043$ (3)    | Higher    |
| EdNet   | Markov    | Rapid Low-Accuracy Responding; Standard augmentation gain      | AUPRC  | $-0.0073 \pm 0.0036$ (3)   | Higher    |
| EdNet   | Markov    | Rapid Low-Accuracy Responding; Standard augmentation gain      | AUROC  | $0.0003 \pm 0.0009$ (3)    | Higher    |
| EdNet   | Markov    | Rapid Low-Accuracy Responding; Standard synthetic utility loss | AUPRC  | $0.0137 \pm 0.0035$ (3)    | Lower     |
| EdNet   | Markov    | Rapid Low-Accuracy Responding; Standard synthetic utility loss | AUROC  | $-0.0012 \pm 0.0030$ (3)   | Lower     |
| EdNet   | Markov    | Rapid Low-Accuracy Responding; Targeted-vs-standard TSTR gain  | AUPRC  | $0.0155 \pm 0.0023$ (3)    | Higher    |
| EdNet   | Markov    | Rapid Low-Accuracy Responding; Targeted-vs-standard TSTR gain  | AUROC  | $0.0273 \pm 0.0020$ (3)    | Higher    |
| EdNet   | Markov    | Rapid Low-Accuracy Responding; Targeted augmentation gain      | AUPRC  | $0.0004 \pm 0.0011$ (3)    | Higher    |
| EdNet   | Markov    | Rapid Low-Accuracy Responding; Targeted augmentation gain      | AUROC  | $0.0230 \pm 0.0038$ (3)    | Higher    |
| EdNet   | Markov    | Rare Skill Path; Real + standard synthetic                     | AUPRC  | $0.5333 \pm 0.0052$ (3)    | Higher    |
| EdNet   | Markov    | Rare Skill Path; Real + standard synthetic                     | AUROC  | $0.5799 \pm 0.0056$ (3)    | Higher    |
| EdNet   | Markov    | Rare Skill Path; Real + targeted synthetic                     | AUPRC  | $0.5212 \pm 0.0100$ (3)    | Higher    |
| EdNet   | Markov    | Rare Skill Path; Real + targeted synthetic                     | AUROC  | $0.5504 \pm 0.0025$ (3)    | Higher    |

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| Dataset | Generator | Scope / comparison                                               | Metric                                      | Mean $\pm$ SD ( <i>n</i> ) | Direction |
|---------|-----------|------------------------------------------------------------------|---------------------------------------------|----------------------------|-----------|
| EdNet   | Markov    | Rare Skill Path; Real only (TRTR)                                | AUPRC                                       | 0.5203 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | Markov    | Rare Skill Path; Real only (TRTR)                                | AUROC                                       | 0.5892 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | Markov    | Rare Skill Path; Standard synthetic only (TSTR)                  | AUPRC                                       | 0.5230 $\pm$ 0.0030 (3)    | Higher    |
| EdNet   | Markov    | Rare Skill Path; Standard synthetic only (TSTR)                  | AUROC                                       | 0.5556 $\pm$ 0.0059 (3)    | Higher    |
| EdNet   | Markov    | Rare Skill Path; Targeted synthetic only                         | AUPRC                                       | 0.5315 $\pm$ 0.0170 (3)    | Higher    |
| EdNet   | Markov    | Rare Skill Path; Targeted synthetic only                         | AUROC                                       | 0.5611 $\pm$ 0.0196 (3)    | Higher    |
| EdNet   | Markov    | Rare Skill Path; Standard augmentation gain                      | AUPRC                                       | 0.0130 $\pm$ 0.0052 (3)    | Higher    |
| EdNet   | Markov    | Rare Skill Path; Standard augmentation gain                      | AUROC                                       | -0.0093 $\pm$ 0.0056 (3)   | Higher    |
| EdNet   | Markov    | Rare Skill Path; Standard synthetic utility loss                 | AUPRC                                       | -0.0027 $\pm$ 0.0030 (3)   | Lower     |
| EdNet   | Markov    | Rare Skill Path; Standard synthetic utility loss                 | AUROC                                       | 0.0337 $\pm$ 0.0059 (3)    | Lower     |
| EdNet   | Markov    | Rare Skill Path; Targeted-vs-standard TSTR gain                  | AUPRC                                       | 0.0085 $\pm$ 0.0150 (3)    | Higher    |
| EdNet   | Markov    | Rare Skill Path; Targeted-vs-standard TSTR gain                  | AUROC                                       | 0.0056 $\pm$ 0.0199 (3)    | Higher    |
| EdNet   | Markov    | Rare Skill Path; Targeted augmentation gain                      | AUPRC                                       | 0.0009 $\pm$ 0.0100 (3)    | Higher    |
| EdNet   | Markov    | Rare Skill Path; Targeted augmentation gain                      | AUROC                                       | -0.0388 $\pm$ 0.0025 (3)   | Higher    |
| EdNet   | Markov    | Recovery; Real + standard synthetic                              | AUPRC                                       | 0.7186 $\pm$ 0.0009 (3)    | Higher    |
| EdNet   | Markov    | Recovery; Real + standard synthetic                              | AUROC                                       | 0.7010 $\pm$ 0.0014 (3)    | Higher    |
| EdNet   | Markov    | Recovery; Real + targeted synthetic                              | AUPRC                                       | 0.6883 $\pm$ 0.0015 (3)    | Higher    |
| EdNet   | Markov    | Recovery; Real + targeted synthetic                              | AUROC                                       | 0.6689 $\pm$ 0.0037 (3)    | Higher    |
| EdNet   | Markov    | Recovery; Real only (TRTR)                                       | AUPRC                                       | 0.7117 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | Markov    | Recovery; Real only (TRTR)                                       | AUROC                                       | 0.6912 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | Markov    | Recovery; Standard synthetic only (TSTR)                         | AUPRC                                       | 0.7122 $\pm$ 0.0043 (3)    | Higher    |
| EdNet   | Markov    | Recovery; Standard synthetic only (TSTR)                         | AUROC                                       | 0.6953 $\pm$ 0.0019 (3)    | Higher    |
| EdNet   | Markov    | Recovery; Targeted synthetic only                                | AUPRC                                       | 0.6670 $\pm$ 0.0089 (3)    | Higher    |
| EdNet   | Markov    | Recovery; Targeted synthetic only                                | AUROC                                       | 0.6456 $\pm$ 0.0160 (3)    | Higher    |
| EdNet   | Markov    | Recovery; Standard augmentation gain                             | AUPRC                                       | 0.0069 $\pm$ 0.0009 (3)    | Higher    |
| EdNet   | Markov    | Recovery; Standard augmentation gain                             | AUROC                                       | 0.0098 $\pm$ 0.0014 (3)    | Higher    |
| EdNet   | Markov    | Recovery; Standard synthetic utility loss                        | AUPRC                                       | -0.0005 $\pm$ 0.0043 (3)   | Lower     |
| EdNet   | Markov    | Recovery; Standard synthetic utility loss                        | AUROC                                       | -0.0041 $\pm$ 0.0019 (3)   | Lower     |
| EdNet   | Markov    | Recovery; Targeted-vs-standard TSTR gain                         | AUPRC                                       | -0.0452 $\pm$ 0.0125 (3)   | Higher    |
| EdNet   | Markov    | Recovery; Targeted-vs-standard TSTR gain                         | AUROC                                       | -0.0497 $\pm$ 0.0179 (3)   | Higher    |
| EdNet   | Markov    | Recovery; Targeted augmentation gain                             | AUPRC                                       | -0.0234 $\pm$ 0.0015 (3)   | Higher    |
| EdNet   | Markov    | Recovery; Targeted augmentation gain                             | AUROC                                       | -0.0224 $\pm$ 0.0037 (3)   | Higher    |
| EdNet   | Markov    | Late Correctness Decline                                         | Rare-group prevalence-error reduction       | 0.0117 $\pm$ 0.0020 (3)    | Higher    |
| EdNet   | Markov    | Late Correctness Decline                                         | Rare-group trajectory-shape-error reduction | -0.0078 $\pm$ 0.0036 (3)   | Higher    |
| EdNet   | Markov    | Persistent Low Correctness                                       | Rare-group prevalence-error reduction       | 0.0043 $\pm$ 0.0002 (3)    | Higher    |
| EdNet   | Markov    | Persistent Low Correctness                                       | Rare-group trajectory-shape-error reduction | -0.0002 $\pm$ 0.0033 (3)   | Higher    |
| EdNet   | Markov    | Rapid Low-Accuracy Responding                                    | Rare-group prevalence-error reduction       | -0.0018 $\pm$ 0.0048 (3)   | Higher    |
| EdNet   | Markov    | Rapid Low-Accuracy Responding                                    | Rare-group trajectory-shape-error reduction | 0.0035 $\pm$ 0.0170 (3)    | Higher    |
| EdNet   | Markov    | Rare Skill Path                                                  | Rare-group prevalence-error reduction       | 0.0031 $\pm$ 0.0043 (3)    | Higher    |
| EdNet   | Markov    | Rare Skill Path                                                  | Rare-group trajectory-shape-error reduction | 0.1097 $\pm$ 0.0027 (3)    | Higher    |
| EdNet   | Markov    | Recovery                                                         | Rare-group prevalence-error reduction       | -0.0009 $\pm$ 0.0012 (3)   | Higher    |
| EdNet   | Markov    | Recovery                                                         | Rare-group trajectory-shape-error reduction | -0.0009 $\pm$ 0.0049 (3)   | Higher    |
| EdNet   | SeqVAE    | Real + standard synthetic                                        | AUPRC                                       | 0.7642 $\pm$ 0.0005 (3)    | Higher    |
| EdNet   | SeqVAE    | Real + standard synthetic                                        | AUROC                                       | 0.6670 $\pm$ 0.0005 (3)    | Higher    |
| EdNet   | SeqVAE    | Real + targeted synthetic                                        | AUPRC                                       | 0.7635 $\pm$ 0.0001 (3)    | Higher    |
| EdNet   | SeqVAE    | Real + targeted synthetic                                        | AUROC                                       | 0.6663 $\pm$ 0.0001 (3)    | Higher    |
| EdNet   | SeqVAE    | Real only (TRTR)                                                 | AUPRC                                       | 0.7687 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | SeqVAE    | Real only (TRTR)                                                 | AUROC                                       | 0.6722 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | SeqVAE    | Standard synthetic only (TSTR)                                   | AUPRC                                       | 0.6257 $\pm$ 0.0159 (3)    | Higher    |
| EdNet   | SeqVAE    | Standard synthetic only (TSTR)                                   | AUROC                                       | 0.4906 $\pm$ 0.0212 (3)    | Higher    |
| EdNet   | SeqVAE    | Targeted synthetic only                                          | AUPRC                                       | 0.6326 $\pm$ 0.0130 (3)    | Higher    |
| EdNet   | SeqVAE    | Targeted synthetic only                                          | AUROC                                       | 0.5114 $\pm$ 0.0264 (3)    | Higher    |
| EdNet   | SeqVAE    | Standard augmentation gain                                       | AUPRC                                       | -0.0046 $\pm$ 0.0005 (3)   | Higher    |
| EdNet   | SeqVAE    | Standard augmentation gain                                       | AUROC                                       | -0.0051 $\pm$ 0.0005 (3)   | Higher    |
| EdNet   | SeqVAE    | Standard synthetic utility loss                                  | AUPRC                                       | 0.1430 $\pm$ 0.0159 (3)    | Lower     |
| EdNet   | SeqVAE    | Standard synthetic utility loss                                  | AUROC                                       | 0.1816 $\pm$ 0.0212 (3)    | Lower     |
| EdNet   | SeqVAE    | Targeted-vs-standard TSTR gain                                   | AUPRC                                       | 0.0069 $\pm$ 0.0056 (3)    | Higher    |
| EdNet   | SeqVAE    | Targeted-vs-standard TSTR gain                                   | AUROC                                       | 0.0208 $\pm$ 0.0073 (3)    | Higher    |
| EdNet   | SeqVAE    | Targeted augmentation gain                                       | AUPRC                                       | -0.0052 $\pm$ 0.0001 (3)   | Higher    |
| EdNet   | SeqVAE    | Targeted augmentation gain                                       | AUROC                                       | -0.0059 $\pm$ 0.0001 (3)   | Higher    |
| EdNet   | SeqVAE    | Persistent Low-Correctness Prediction; Real + standard synthetic | AUPRC                                       | 0.1227 $\pm$ 0.0049 (3)    | Higher    |

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| Dataset | Generator | Scope / comparison                                                     | Metric | Mean $\pm$ SD ( <i>n</i> ) | Direction |
|---------|-----------|------------------------------------------------------------------------|--------|----------------------------|-----------|
| EdNet   | SeqVAE    | Persistent Low-Correctness Prediction; Real + standard synthetic       | AUROC  | 0.5725 $\pm$ 0.0124 (3)    | Higher    |
| EdNet   | SeqVAE    | Persistent Low-Correctness Prediction; Real only (TRTR)                | AUPRC  | 0.1310 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | SeqVAE    | Persistent Low-Correctness Prediction; Real only (TRTR)                | AUROC  | 0.5859 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | SeqVAE    | Persistent Low-Correctness Prediction; Standard synthetic only (TSTR)  | AUPRC  | 0.0810 $\pm$ 0.0007 (3)    | Higher    |
| EdNet   | SeqVAE    | Persistent Low-Correctness Prediction; Standard synthetic only (TSTR)  | AUROC  | 0.4758 $\pm$ 0.0071 (3)    | Higher    |
| EdNet   | SeqVAE    | Persistent Low-Correctness Prediction; Targeted synthetic only         | AUPRC  | 0.0799 $\pm$ 0.0034 (3)    | Higher    |
| EdNet   | SeqVAE    | Persistent Low-Correctness Prediction; Targeted synthetic only         | AUROC  | 0.4500 $\pm$ 0.0185 (3)    | Higher    |
| EdNet   | SeqVAE    | Persistent Low-Correctness Prediction; Standard augmentation gain      | AUPRC  | -0.0082 $\pm$ 0.0049 (3)   | Higher    |
| EdNet   | SeqVAE    | Persistent Low-Correctness Prediction; Standard augmentation gain      | AUROC  | -0.0133 $\pm$ 0.0124 (3)   | Higher    |
| EdNet   | SeqVAE    | Persistent Low-Correctness Prediction; Standard synthetic utility loss | AUPRC  | 0.0500 $\pm$ 0.0007 (3)    | Lower     |
| EdNet   | SeqVAE    | Persistent Low-Correctness Prediction; Standard synthetic utility loss | AUROC  | 0.1101 $\pm$ 0.0071 (3)    | Lower     |
| EdNet   | SeqVAE    | Recovery Prediction; Real + standard synthetic                         | AUPRC  | 0.2222 $\pm$ 0.0175 (3)    | Higher    |
| EdNet   | SeqVAE    | Recovery Prediction; Real + standard synthetic                         | AUROC  | 0.4739 $\pm$ 0.0140 (3)    | Higher    |
| EdNet   | SeqVAE    | Recovery Prediction; Real only (TRTR)                                  | AUPRC  | 0.2194 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | SeqVAE    | Recovery Prediction; Real only (TRTR)                                  | AUROC  | 0.4910 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | SeqVAE    | Recovery Prediction; Standard synthetic only (TSTR)                    | AUPRC  | 0.2026 $\pm$ 0.0040 (3)    | Higher    |
| EdNet   | SeqVAE    | Recovery Prediction; Standard synthetic only (TSTR)                    | AUROC  | 0.4699 $\pm$ 0.0190 (3)    | Higher    |
| EdNet   | SeqVAE    | Recovery Prediction; Targeted synthetic only                           | AUPRC  | 0.2206 $\pm$ 0.0101 (3)    | Higher    |
| EdNet   | SeqVAE    | Recovery Prediction; Targeted synthetic only                           | AUROC  | 0.5181 $\pm$ 0.0140 (3)    | Higher    |
| EdNet   | SeqVAE    | Recovery Prediction; Standard augmentation gain                        | AUPRC  | 0.0027 $\pm$ 0.0175 (3)    | Higher    |
| EdNet   | SeqVAE    | Recovery Prediction; Standard augmentation gain                        | AUROC  | -0.0171 $\pm$ 0.0140 (3)   | Higher    |
| EdNet   | SeqVAE    | Recovery Prediction; Standard synthetic utility loss                   | AUPRC  | 0.0168 $\pm$ 0.0040 (3)    | Lower     |
| EdNet   | SeqVAE    | Recovery Prediction; Standard synthetic utility loss                   | AUROC  | 0.0211 $\pm$ 0.0190 (3)    | Lower     |
| EdNet   | SeqVAE    | Late Correctness Decline; Real + standard synthetic                    | AUPRC  | 0.6061 $\pm$ 0.0043 (3)    | Higher    |
| EdNet   | SeqVAE    | Late Correctness Decline; Real + standard synthetic                    | AUROC  | 0.6556 $\pm$ 0.0030 (3)    | Higher    |
| EdNet   | SeqVAE    | Late Correctness Decline; Real + targeted synthetic                    | AUPRC  | 0.6107 $\pm$ 0.0052 (3)    | Higher    |
| EdNet   | SeqVAE    | Late Correctness Decline; Real + targeted synthetic                    | AUROC  | 0.6621 $\pm$ 0.0044 (3)    | Higher    |
| EdNet   | SeqVAE    | Late Correctness Decline; Real only (TRTR)                             | AUPRC  | 0.6426 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | SeqVAE    | Late Correctness Decline; Real only (TRTR)                             | AUROC  | 0.6902 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | SeqVAE    | Late Correctness Decline; Standard synthetic only (TSTR)               | AUPRC  | 0.3975 $\pm$ 0.0139 (3)    | Higher    |
| EdNet   | SeqVAE    | Late Correctness Decline; Standard synthetic only (TSTR)               | AUROC  | 0.4341 $\pm$ 0.0209 (3)    | Higher    |
| EdNet   | SeqVAE    | Late Correctness Decline; Targeted synthetic only                      | AUPRC  | 0.4080 $\pm$ 0.0160 (3)    | Higher    |
| EdNet   | SeqVAE    | Late Correctness Decline; Targeted synthetic only                      | AUROC  | 0.4711 $\pm$ 0.0428 (3)    | Higher    |
| EdNet   | SeqVAE    | Late Correctness Decline; Standard augmentation gain                   | AUPRC  | -0.0365 $\pm$ 0.0043 (3)   | Higher    |
| EdNet   | SeqVAE    | Late Correctness Decline; Standard augmentation gain                   | AUROC  | -0.0346 $\pm$ 0.0030 (3)   | Higher    |
| EdNet   | SeqVAE    | Late Correctness Decline; Standard synthetic utility loss              | AUPRC  | 0.2451 $\pm$ 0.0139 (3)    | Lower     |
| EdNet   | SeqVAE    | Late Correctness Decline; Standard synthetic utility loss              | AUROC  | 0.2561 $\pm$ 0.0209 (3)    | Lower     |
| EdNet   | SeqVAE    | Late Correctness Decline; Targeted-vs-standard TSTR gain               | AUPRC  | 0.0105 $\pm$ 0.0152 (3)    | Higher    |
| EdNet   | SeqVAE    | Late Correctness Decline; Targeted-vs-standard TSTR gain               | AUROC  | 0.0369 $\pm$ 0.0278 (3)    | Higher    |
| EdNet   | SeqVAE    | Late Correctness Decline; Targeted augmentation gain                   | AUPRC  | -0.0319 $\pm$ 0.0052 (3)   | Higher    |
| EdNet   | SeqVAE    | Late Correctness Decline; Targeted augmentation gain                   | AUROC  | -0.0281 $\pm$ 0.0044 (3)   | Higher    |
| EdNet   | SeqVAE    | Persistent Low Correctness; Real + standard synthetic                  | AUPRC  | 0.1083 $\pm$ 0.0090 (3)    | Higher    |
| EdNet   | SeqVAE    | Persistent Low Correctness; Real + standard synthetic                  | AUROC  | 0.5288 $\pm$ 0.0017 (3)    | Higher    |

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| Dataset | Generator | Scope / comparison                                             | Metric | Mean $\pm$ SD ( $n$ )    | Direction |
|---------|-----------|----------------------------------------------------------------|--------|--------------------------|-----------|
| EdNet   | SeqVAE    | Persistent Low Correctness; Real + targeted synthetic          | AUPRC  | $0.1068 \pm 0.0030$ (3)  | Higher    |
| EdNet   | SeqVAE    | Persistent Low Correctness; Real + targeted synthetic          | AUROC  | $0.5390 \pm 0.0055$ (3)  | Higher    |
| EdNet   | SeqVAE    | Persistent Low Correctness; Real only (TRTR)                   | AUPRC  | $0.1132 \pm 0.0000$ (3)  | Higher    |
| EdNet   | SeqVAE    | Persistent Low Correctness; Real only (TRTR)                   | AUROC  | $0.5416 \pm 0.0000$ (3)  | Higher    |
| EdNet   | SeqVAE    | Persistent Low Correctness; Standard synthetic only (TSTR)     | AUPRC  | $0.0992 \pm 0.0021$ (3)  | Higher    |
| EdNet   | SeqVAE    | Persistent Low Correctness; Standard synthetic only (TSTR)     | AUROC  | $0.4726 \pm 0.0111$ (3)  | Higher    |
| EdNet   | SeqVAE    | Persistent Low Correctness; Targeted synthetic only            | AUPRC  | $0.0918 \pm 0.0049$ (3)  | Higher    |
| EdNet   | SeqVAE    | Persistent Low Correctness; Targeted synthetic only            | AUROC  | $0.4718 \pm 0.0242$ (3)  | Higher    |
| EdNet   | SeqVAE    | Persistent Low Correctness; Standard augmentation gain         | AUPRC  | $-0.0049 \pm 0.0090$ (3) | Higher    |
| EdNet   | SeqVAE    | Persistent Low Correctness; Standard augmentation gain         | AUROC  | $-0.0129 \pm 0.0017$ (3) | Higher    |
| EdNet   | SeqVAE    | Persistent Low Correctness; Standard synthetic utility loss    | AUPRC  | $0.0141 \pm 0.0021$ (3)  | Lower     |
| EdNet   | SeqVAE    | Persistent Low Correctness; Standard synthetic utility loss    | AUROC  | $0.0691 \pm 0.0111$ (3)  | Lower     |
| EdNet   | SeqVAE    | Persistent Low Correctness; Targeted-vs-standard TSTR gain     | AUPRC  | $-0.0073 \pm 0.0063$ (3) | Higher    |
| EdNet   | SeqVAE    | Persistent Low Correctness; Targeted-vs-standard TSTR gain     | AUROC  | $-0.0008 \pm 0.0214$ (3) | Higher    |
| EdNet   | SeqVAE    | Persistent Low Correctness; Targeted augmentation gain         | AUPRC  | $-0.0064 \pm 0.0030$ (3) | Higher    |
| EdNet   | SeqVAE    | Persistent Low Correctness; Targeted augmentation gain         | AUROC  | $-0.0027 \pm 0.0055$ (3) | Higher    |
| EdNet   | SeqVAE    | Rapid Low-Accuracy Responding; Real + standard synthetic       | AUPRC  | $0.3009 \pm 0.0020$ (3)  | Higher    |
| EdNet   | SeqVAE    | Rapid Low-Accuracy Responding; Real + standard synthetic       | AUROC  | $0.5487 \pm 0.0005$ (3)  | Higher    |
| EdNet   | SeqVAE    | Rapid Low-Accuracy Responding; Real + targeted synthetic       | AUPRC  | $0.3010 \pm 0.0021$ (3)  | Higher    |
| EdNet   | SeqVAE    | Rapid Low-Accuracy Responding; Real + targeted synthetic       | AUROC  | $0.5487 \pm 0.0024$ (3)  | Higher    |
| EdNet   | SeqVAE    | Rapid Low-Accuracy Responding; Real only (TRTR)                | AUPRC  | $0.3092 \pm 0.0000$ (3)  | Higher    |
| EdNet   | SeqVAE    | Rapid Low-Accuracy Responding; Real only (TRTR)                | AUROC  | $0.5619 \pm 0.0000$ (3)  | Higher    |
| EdNet   | SeqVAE    | Rapid Low-Accuracy Responding; Standard synthetic only (TSTR)  | AUPRC  | $0.2962 \pm 0.0156$ (3)  | Higher    |
| EdNet   | SeqVAE    | Rapid Low-Accuracy Responding; Standard synthetic only (TSTR)  | AUROC  | $0.5348 \pm 0.0233$ (3)  | Higher    |
| EdNet   | SeqVAE    | Rapid Low-Accuracy Responding; Targeted synthetic only         | AUPRC  | $0.2575 \pm 0.0058$ (3)  | Higher    |
| EdNet   | SeqVAE    | Rapid Low-Accuracy Responding; Targeted synthetic only         | AUROC  | $0.4839 \pm 0.0125$ (3)  | Higher    |
| EdNet   | SeqVAE    | Rapid Low-Accuracy Responding; Standard augmentation gain      | AUPRC  | $-0.0083 \pm 0.0020$ (3) | Higher    |
| EdNet   | SeqVAE    | Rapid Low-Accuracy Responding; Standard augmentation gain      | AUROC  | $-0.0132 \pm 0.0005$ (3) | Higher    |
| EdNet   | SeqVAE    | Rapid Low-Accuracy Responding; Standard synthetic utility loss | AUPRC  | $0.0130 \pm 0.0156$ (3)  | Lower     |
| EdNet   | SeqVAE    | Rapid Low-Accuracy Responding; Standard synthetic utility loss | AUROC  | $0.0271 \pm 0.0233$ (3)  | Lower     |
| EdNet   | SeqVAE    | Rapid Low-Accuracy Responding; Targeted-vs-standard TSTR gain  | AUPRC  | $-0.0387 \pm 0.0098$ (3) | Higher    |
| EdNet   | SeqVAE    | Rapid Low-Accuracy Responding; Targeted-vs-standard TSTR gain  | AUROC  | $-0.0508 \pm 0.0120$ (3) | Higher    |
| EdNet   | SeqVAE    | Rapid Low-Accuracy Responding; Targeted augmentation gain      | AUPRC  | $-0.0082 \pm 0.0021$ (3) | Higher    |
| EdNet   | SeqVAE    | Rapid Low-Accuracy Responding; Targeted augmentation gain      | AUROC  | $-0.0132 \pm 0.0024$ (3) | Higher    |
| EdNet   | SeqVAE    | Rare Skill Path; Real + standard synthetic                     | AUPRC  | $0.5408 \pm 0.0099$ (3)  | Higher    |
| EdNet   | SeqVAE    | Rare Skill Path; Real + standard synthetic                     | AUROC  | $0.5871 \pm 0.0025$ (3)  | Higher    |
| EdNet   | SeqVAE    | Rare Skill Path; Real + targeted synthetic                     | AUPRC  | $0.5349 \pm 0.0082$ (3)  | Higher    |
| EdNet   | SeqVAE    | Rare Skill Path; Real + targeted synthetic                     | AUROC  | $0.5921 \pm 0.0032$ (3)  | Higher    |
| EdNet   | SeqVAE    | Rare Skill Path; Real only (TRTR)                              | AUPRC  | $0.5203 \pm 0.0000$ (3)  | Higher    |
| EdNet   | SeqVAE    | Rare Skill Path; Real only (TRTR)                              | AUROC  | $0.5892 \pm 0.0000$ (3)  | Higher    |
| EdNet   | SeqVAE    | Rare Skill Path; Standard synthetic only (TSTR)                | AUPRC  | $0.4843 \pm 0.0154$ (3)  | Higher    |
| EdNet   | SeqVAE    | Rare Skill Path; Standard synthetic only (TSTR)                | AUROC  | $0.5309 \pm 0.0197$ (3)  | Higher    |
| EdNet   | SeqVAE    | Rare Skill Path; Targeted synthetic only                       | AUPRC  | $0.4782 \pm 0.0078$ (3)  | Higher    |
| EdNet   | SeqVAE    | Rare Skill Path; Targeted synthetic only                       | AUROC  | $0.5251 \pm 0.0124$ (3)  | Higher    |

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| Dataset | Generator | Scope / comparison                                                | Metric                                      | Mean $\pm$ SD ( $n$ )               | Direction |
|---------|-----------|-------------------------------------------------------------------|---------------------------------------------|-------------------------------------|-----------|
| EdNet   | SeqVAE    | Rare Skill Path; Standard augmentation gain                       | AUPRC                                       | $0.0205 \pm 0.0099$ (3)             | Higher    |
| EdNet   | SeqVAE    | Rare Skill Path; Standard augmentation gain                       | AUROC                                       | $-0.0021 \pm 0.0025$ (3)            | Higher    |
| EdNet   | SeqVAE    | Rare Skill Path; Standard synthetic utility loss                  | AUPRC                                       | $0.0360 \pm 0.0154$ (3)             | Lower     |
| EdNet   | SeqVAE    | Rare Skill Path; Standard synthetic utility loss                  | AUROC                                       | $0.0583 \pm 0.0197$ (3)             | Lower     |
| EdNet   | SeqVAE    | Rare Skill Path; Targeted-vs-standard TSTR gain                   | AUPRC                                       | $-0.0061 \pm 0.0186$ (3)            | Higher    |
| EdNet   | SeqVAE    | Rare Skill Path; Targeted-vs-standard TSTR gain                   | AUROC                                       | $-0.0059 \pm 0.0185$ (3)            | Higher    |
| EdNet   | SeqVAE    | Rare Skill Path; Targeted augmentation gain                       | AUPRC                                       | $0.0146 \pm 0.0082$ (3)             | Higher    |
| EdNet   | SeqVAE    | Rare Skill Path; Targeted augmentation gain                       | AUROC                                       | $0.0028 \pm 0.0032$ (3)             | Higher    |
| EdNet   | SeqVAE    | Recovery; Real + standard synthetic                               | AUPRC                                       | $0.7254 \pm 0.0033$ (3)             | Higher    |
| EdNet   | SeqVAE    | Recovery; Real + standard synthetic                               | AUROC                                       | $0.7016 \pm 0.0034$ (3)             | Higher    |
| EdNet   | SeqVAE    | Recovery; Real + targeted synthetic                               | AUPRC                                       | $0.7296 \pm 0.0053$ (3)             | Higher    |
| EdNet   | SeqVAE    | Recovery; Real + targeted synthetic                               | AUROC                                       | $0.7058 \pm 0.0045$ (3)             | Higher    |
| EdNet   | SeqVAE    | Recovery; Real only (TRTR)                                        | AUPRC                                       | $0.7117 \pm 0.0000$ (3)             | Higher    |
| EdNet   | SeqVAE    | Recovery; Real only (TRTR)                                        | AUROC                                       | $0.6912 \pm 0.0000$ (3)             | Higher    |
| EdNet   | SeqVAE    | Recovery; Standard synthetic only (TSTR)                          | AUPRC                                       | $0.5820 \pm 0.0102$ (3)             | Higher    |
| EdNet   | SeqVAE    | Recovery; Standard synthetic only (TSTR)                          | AUROC                                       | $0.5853 \pm 0.0097$ (3)             | Higher    |
| EdNet   | SeqVAE    | Recovery; Targeted synthetic only                                 | AUPRC                                       | $0.5731 \pm 0.0206$ (3)             | Higher    |
| EdNet   | SeqVAE    | Recovery; Targeted synthetic only                                 | AUROC                                       | $0.5594 \pm 0.0360$ (3)             | Higher    |
| EdNet   | SeqVAE    | Recovery; Standard augmentation gain                              | AUPRC                                       | $0.0137 \pm 0.0033$ (3)             | Higher    |
| EdNet   | SeqVAE    | Recovery; Standard augmentation gain                              | AUROC                                       | $0.0103 \pm 0.0034$ (3)             | Higher    |
| EdNet   | SeqVAE    | Recovery; Standard synthetic utility loss                         | AUPRC                                       | $0.1297 \pm 0.0102$ (3)             | Lower     |
| EdNet   | SeqVAE    | Recovery; Standard synthetic utility loss                         | AUROC                                       | $0.1059 \pm 0.0097$ (3)             | Lower     |
| EdNet   | SeqVAE    | Recovery; Targeted-vs-standard TSTR gain                          | AUPRC                                       | $-0.0089 \pm 0.0167$ (3)            | Higher    |
| EdNet   | SeqVAE    | Recovery; Targeted-vs-standard TSTR gain                          | AUROC                                       | $-0.0259 \pm 0.0290$ (3)            | Higher    |
| EdNet   | SeqVAE    | Recovery; Targeted augmentation gain                              | AUPRC                                       | $0.0179 \pm 0.0053$ (3)             | Higher    |
| EdNet   | SeqVAE    | Recovery; Targeted augmentation gain                              | AUROC                                       | $0.0146 \pm 0.0045$ (3)             | Higher    |
| EdNet   | SeqVAE    | Late Correctness Decline                                          | Rare-group prevalence-error reduction       | $0.0029 \pm 0.0024$ (3)             | Higher    |
| EdNet   | SeqVAE    | Late Correctness Decline                                          | Rare-group trajectory-shape-error reduction | $0.0017 \pm 0.0129$ (3)             | Higher    |
| EdNet   | SeqVAE    | Persistent Low Correctness                                        | Rare-group prevalence-error reduction       | $-0.0006 \pm 0.0005$ (3)            | Higher    |
| EdNet   | SeqVAE    | Persistent Low Correctness                                        | Rare-group trajectory-shape-error reduction | $-0.0051 \pm 0.0036$ (2; 1 missing) | Higher    |
| EdNet   | SeqVAE    | Rapid Low-Accuracy Responding                                     | Rare-group prevalence-error reduction       | $0.0000 \pm 0.0002$ (3)             | Higher    |
| EdNet   | SeqVAE    | Rare Skill Path                                                   | Rare-group prevalence-error reduction       | $-0.0122 \pm 0.0054$ (3)            | Higher    |
| EdNet   | SeqVAE    | Rare Skill Path                                                   | Rare-group trajectory-shape-error reduction | $0.0123 \pm 0.0165$ (3)             | Higher    |
| EdNet   | SeqVAE    | Recovery                                                          | Rare-group prevalence-error reduction       | $-0.0007 \pm 0.0056$ (3)            | Higher    |
| EdNet   | SeqVAE    | Recovery                                                          | Rare-group trajectory-shape-error reduction | $0.0117 \pm 0.0084$ (3)             | Higher    |
| EdNet   | TimeGAN   | Real + standard synthetic                                         | AUPRC                                       | $0.7631 \pm 0.0003$ (3)             | Higher    |
| EdNet   | TimeGAN   | Real + standard synthetic                                         | AUROC                                       | $0.6675 \pm 0.0003$ (3)             | Higher    |
| EdNet   | TimeGAN   | Real + targeted synthetic                                         | AUPRC                                       | $0.7634 \pm 0.0001$ (3)             | Higher    |
| EdNet   | TimeGAN   | Real + targeted synthetic                                         | AUROC                                       | $0.6682 \pm 0.0002$ (3)             | Higher    |
| EdNet   | TimeGAN   | Real only (TRTR)                                                  | AUPRC                                       | $0.7687 \pm 0.0000$ (3)             | Higher    |
| EdNet   | TimeGAN   | Real only (TRTR)                                                  | AUROC                                       | $0.6722 \pm 0.0000$ (3)             | Higher    |
| EdNet   | TimeGAN   | Standard synthetic only (TSTR)                                    | AUPRC                                       | $0.6045 \pm 0.0084$ (3)             | Higher    |
| EdNet   | TimeGAN   | Standard synthetic only (TSTR)                                    | AUROC                                       | $0.4593 \pm 0.0184$ (3)             | Higher    |
| EdNet   | TimeGAN   | Targeted synthetic only                                           | AUPRC                                       | $0.6127 \pm 0.0034$ (3)             | Higher    |
| EdNet   | TimeGAN   | Targeted synthetic only                                           | AUROC                                       | $0.4633 \pm 0.0113$ (3)             | Higher    |
| EdNet   | TimeGAN   | Standard augmentation gain                                        | AUPRC                                       | $-0.0056 \pm 0.0003$ (3)            | Higher    |
| EdNet   | TimeGAN   | Standard augmentation gain                                        | AUROC                                       | $-0.0046 \pm 0.0003$ (3)            | Higher    |
| EdNet   | TimeGAN   | Standard synthetic utility loss                                   | AUPRC                                       | $0.1642 \pm 0.0084$ (3)             | Lower     |
| EdNet   | TimeGAN   | Standard synthetic utility loss                                   | AUROC                                       | $0.2128 \pm 0.0184$ (3)             | Lower     |
| EdNet   | TimeGAN   | Targeted-vs-standard TSTR gain                                    | AUPRC                                       | $0.0082 \pm 0.0092$ (3)             | Higher    |
| EdNet   | TimeGAN   | Targeted-vs-standard TSTR gain                                    | AUROC                                       | $0.0040 \pm 0.0200$ (3)             | Higher    |
| EdNet   | TimeGAN   | Targeted augmentation gain                                        | AUPRC                                       | $-0.0053 \pm 0.0001$ (3)            | Higher    |
| EdNet   | TimeGAN   | Targeted augmentation gain                                        | AUROC                                       | $-0.0040 \pm 0.0002$ (3)            | Higher    |
| EdNet   | TimeGAN   | Persistent Low-Correctness Prediction; Real + standard synthetic  | AUPRC                                       | $0.1334 \pm 0.0084$ (3)             | Higher    |
| EdNet   | TimeGAN   | Persistent Low-Correctness Prediction; Real + standard synthetic  | AUROC                                       | $0.6099 \pm 0.0023$ (3)             | Higher    |
| EdNet   | TimeGAN   | Persistent Low-Correctness Prediction; Real only (TRTR)           | AUPRC                                       | $0.1310 \pm 0.0000$ (3)             | Higher    |
| EdNet   | TimeGAN   | Persistent Low-Correctness Prediction; Real only (TRTR)           | AUROC                                       | $0.5859 \pm 0.0000$ (3)             | Higher    |
| EdNet   | TimeGAN   | Persistent Low-Correctness Prediction; Standard augmentation gain | AUPRC                                       | $0.0025 \pm 0.0084$ (3)             | Higher    |
| EdNet   | TimeGAN   | Persistent Low-Correctness Prediction; Standard augmentation gain | AUROC                                       | $0.0241 \pm 0.0023$ (3)             | Higher    |
| EdNet   | TimeGAN   | Recovery Prediction; Real + standard synthetic                    | AUPRC                                       | $0.2194 \pm 0.0000$ (3)             | Higher    |

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| Dataset | Generator | Scope / comparison                                          | Metric | Mean $\pm$ SD ( <i>n</i> ) | Direction |
|---------|-----------|-------------------------------------------------------------|--------|----------------------------|-----------|
| EdNet   | TimeGAN   | Recovery Prediction; Real + standard synthetic              | AUROC  | $0.4910 \pm 0.0000$ (3)    | Higher    |
| EdNet   | TimeGAN   | Recovery Prediction; Real only (TRTR)                       | AUPRC  | $0.2194 \pm 0.0000$ (3)    | Higher    |
| EdNet   | TimeGAN   | Recovery Prediction; Real only (TRTR)                       | AUROC  | $0.4910 \pm 0.0000$ (3)    | Higher    |
| EdNet   | TimeGAN   | Recovery Prediction; Standard augmentation gain             | AUPRC  | $0.0000 \pm 0.0000$ (3)    | Higher    |
| EdNet   | TimeGAN   | Recovery Prediction; Standard augmentation gain             | AUROC  | $0.0000 \pm 0.0000$ (3)    | Higher    |
| EdNet   | TimeGAN   | Late Correctness Decline; Real + standard synthetic         | AUPRC  | $0.6089 \pm 0.0016$ (3)    | Higher    |
| EdNet   | TimeGAN   | Late Correctness Decline; Real + standard synthetic         | AUROC  | $0.6742 \pm 0.0003$ (3)    | Higher    |
| EdNet   | TimeGAN   | Late Correctness Decline; Real + targeted synthetic         | AUPRC  | $0.6095 \pm 0.0036$ (3)    | Higher    |
| EdNet   | TimeGAN   | Late Correctness Decline; Real + targeted synthetic         | AUROC  | $0.6751 \pm 0.0009$ (3)    | Higher    |
| EdNet   | TimeGAN   | Late Correctness Decline; Real only (TRTR)                  | AUPRC  | $0.6426 \pm 0.0000$ (3)    | Higher    |
| EdNet   | TimeGAN   | Late Correctness Decline; Real only (TRTR)                  | AUROC  | $0.6902 \pm 0.0000$ (3)    | Higher    |
| EdNet   | TimeGAN   | Late Correctness Decline; Standard synthetic only (TSTR)    | AUPRC  | $0.4824 \pm 0.0294$ (3)    | Higher    |
| EdNet   | TimeGAN   | Late Correctness Decline; Standard synthetic only (TSTR)    | AUROC  | $0.5451 \pm 0.0086$ (3)    | Higher    |
| EdNet   | TimeGAN   | Late Correctness Decline; Targeted synthetic only           | AUPRC  | $0.3833 \pm 0.0107$ (3)    | Higher    |
| EdNet   | TimeGAN   | Late Correctness Decline; Targeted synthetic only           | AUROC  | $0.4067 \pm 0.0175$ (3)    | Higher    |
| EdNet   | TimeGAN   | Late Correctness Decline; Standard augmentation gain        | AUPRC  | $-0.0337 \pm 0.0016$ (3)   | Higher    |
| EdNet   | TimeGAN   | Late Correctness Decline; Standard augmentation gain        | AUROC  | $-0.0160 \pm 0.0003$ (3)   | Higher    |
| EdNet   | TimeGAN   | Late Correctness Decline; Standard synthetic utility loss   | AUPRC  | $0.1602 \pm 0.0294$ (3)    | Lower     |
| EdNet   | TimeGAN   | Late Correctness Decline; Standard synthetic utility loss   | AUROC  | $0.1451 \pm 0.0086$ (3)    | Lower     |
| EdNet   | TimeGAN   | Late Correctness Decline; Targeted-vs-standard TSTR gain    | AUPRC  | $-0.0991 \pm 0.0216$ (3)   | Higher    |
| EdNet   | TimeGAN   | Late Correctness Decline; Targeted-vs-standard TSTR gain    | AUROC  | $-0.1385 \pm 0.0139$ (3)   | Higher    |
| EdNet   | TimeGAN   | Late Correctness Decline; Targeted augmentation gain        | AUPRC  | $-0.0331 \pm 0.0036$ (3)   | Higher    |
| EdNet   | TimeGAN   | Late Correctness Decline; Targeted augmentation gain        | AUROC  | $-0.0151 \pm 0.0009$ (3)   | Higher    |
| EdNet   | TimeGAN   | Persistent Low Correctness; Real + standard synthetic       | AUPRC  | $0.1146 \pm 0.0014$ (3)    | Higher    |
| EdNet   | TimeGAN   | Persistent Low Correctness; Real + standard synthetic       | AUROC  | $0.5399 \pm 0.0020$ (3)    | Higher    |
| EdNet   | TimeGAN   | Persistent Low Correctness; Real + targeted synthetic       | AUPRC  | $0.1162 \pm 0.0031$ (3)    | Higher    |
| EdNet   | TimeGAN   | Persistent Low Correctness; Real + targeted synthetic       | AUROC  | $0.5522 \pm 0.0046$ (3)    | Higher    |
| EdNet   | TimeGAN   | Persistent Low Correctness; Real only (TRTR)                | AUPRC  | $0.1132 \pm 0.0000$ (3)    | Higher    |
| EdNet   | TimeGAN   | Persistent Low Correctness; Real only (TRTR)                | AUROC  | $0.5416 \pm 0.0000$ (3)    | Higher    |
| EdNet   | TimeGAN   | Persistent Low Correctness; Standard synthetic only (TSTR)  | AUPRC  | $0.1048 \pm 0.0134$ (3)    | Higher    |
| EdNet   | TimeGAN   | Persistent Low Correctness; Standard synthetic only (TSTR)  | AUROC  | $0.5324 \pm 0.0355$ (3)    | Higher    |
| EdNet   | TimeGAN   | Persistent Low Correctness; Targeted synthetic only         | AUPRC  | $0.0898 \pm 0.0053$ (3)    | Higher    |
| EdNet   | TimeGAN   | Persistent Low Correctness; Targeted synthetic only         | AUROC  | $0.4774 \pm 0.0172$ (3)    | Higher    |
| EdNet   | TimeGAN   | Persistent Low Correctness; Standard augmentation gain      | AUPRC  | $0.0014 \pm 0.0014$ (3)    | Higher    |
| EdNet   | TimeGAN   | Persistent Low Correctness; Standard augmentation gain      | AUROC  | $-0.0018 \pm 0.0020$ (3)   | Higher    |
| EdNet   | TimeGAN   | Persistent Low Correctness; Standard synthetic utility loss | AUPRC  | $0.0084 \pm 0.0134$ (3)    | Lower     |
| EdNet   | TimeGAN   | Persistent Low Correctness; Standard synthetic utility loss | AUROC  | $0.0092 \pm 0.0355$ (3)    | Lower     |
| EdNet   | TimeGAN   | Persistent Low Correctness; Targeted-vs-standard TSTR gain  | AUPRC  | $-0.0150 \pm 0.0181$ (3)   | Higher    |
| EdNet   | TimeGAN   | Persistent Low Correctness; Targeted-vs-standard TSTR gain  | AUROC  | $-0.0551 \pm 0.0492$ (3)   | Higher    |
| EdNet   | TimeGAN   | Persistent Low Correctness; Targeted augmentation gain      | AUPRC  | $0.0030 \pm 0.0031$ (3)    | Higher    |
| EdNet   | TimeGAN   | Persistent Low Correctness; Targeted augmentation gain      | AUROC  | $0.0105 \pm 0.0046$ (3)    | Higher    |
| EdNet   | TimeGAN   | Rapid Low-Accuracy Responding; Real + standard synthetic    | AUPRC  | $0.3042 \pm 0.0016$ (3)    | Higher    |
| EdNet   | TimeGAN   | Rapid Low-Accuracy Responding; Real + standard synthetic    | AUROC  | $0.5571 \pm 0.0025$ (3)    | Higher    |

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| Dataset | Generator | Scope / comparison                                             | Metric | Mean $\pm$ SD ( <i>n</i> ) | Direction |
|---------|-----------|----------------------------------------------------------------|--------|----------------------------|-----------|
| EdNet   | TimeGAN   | Rapid Low-Accuracy Responding; Real + targeted synthetic       | AUPRC  | 0.3053 $\pm$ 0.0014 (3)    | Higher    |
| EdNet   | TimeGAN   | Rapid Low-Accuracy Responding; Real + targeted synthetic       | AUROC  | 0.5602 $\pm$ 0.0016 (3)    | Higher    |
| EdNet   | TimeGAN   | Rapid Low-Accuracy Responding; Real only (TRTR)                | AUPRC  | 0.3092 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | TimeGAN   | Rapid Low-Accuracy Responding; Real only (TRTR)                | AUROC  | 0.5619 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | TimeGAN   | Rapid Low-Accuracy Responding; Standard synthetic only (TSTR)  | AUPRC  | 0.2823 $\pm$ 0.0059 (3)    | Higher    |
| EdNet   | TimeGAN   | Rapid Low-Accuracy Responding; Standard synthetic only (TSTR)  | AUROC  | 0.5352 $\pm$ 0.0102 (3)    | Higher    |
| EdNet   | TimeGAN   | Rapid Low-Accuracy Responding; Targeted synthetic only         | AUPRC  | 0.2627 $\pm$ 0.0043 (3)    | Higher    |
| EdNet   | TimeGAN   | Rapid Low-Accuracy Responding; Targeted synthetic only         | AUROC  | 0.4904 $\pm$ 0.0090 (3)    | Higher    |
| EdNet   | TimeGAN   | Rapid Low-Accuracy Responding; Standard augmentation gain      | AUPRC  | -0.0050 $\pm$ 0.0016 (3)   | Higher    |
| EdNet   | TimeGAN   | Rapid Low-Accuracy Responding; Standard augmentation gain      | AUROC  | -0.0048 $\pm$ 0.0025 (3)   | Higher    |
| EdNet   | TimeGAN   | Rapid Low-Accuracy Responding; Standard synthetic utility loss | AUPRC  | 0.0269 $\pm$ 0.0059 (3)    | Lower     |
| EdNet   | TimeGAN   | Rapid Low-Accuracy Responding; Standard synthetic utility loss | AUROC  | 0.0267 $\pm$ 0.0102 (3)    | Lower     |
| EdNet   | TimeGAN   | Rapid Low-Accuracy Responding; Targeted-vs-standard TSTR gain  | AUPRC  | -0.0196 $\pm$ 0.0101 (3)   | Higher    |
| EdNet   | TimeGAN   | Rapid Low-Accuracy Responding; Targeted-vs-standard TSTR gain  | AUROC  | -0.0448 $\pm$ 0.0190 (3)   | Higher    |
| EdNet   | TimeGAN   | Rapid Low-Accuracy Responding; Targeted augmentation gain      | AUPRC  | -0.0039 $\pm$ 0.0014 (3)   | Higher    |
| EdNet   | TimeGAN   | Rapid Low-Accuracy Responding; Targeted augmentation gain      | AUROC  | -0.0017 $\pm$ 0.0016 (3)   | Higher    |
| EdNet   | TimeGAN   | Rare Skill Path; Real + standard synthetic                     | AUPRC  | 0.5320 $\pm$ 0.0048 (3)    | Higher    |
| EdNet   | TimeGAN   | Rare Skill Path; Real + standard synthetic                     | AUROC  | 0.5826 $\pm$ 0.0029 (3)    | Higher    |
| EdNet   | TimeGAN   | Rare Skill Path; Real + targeted synthetic                     | AUPRC  | 0.5323 $\pm$ 0.0043 (3)    | Higher    |
| EdNet   | TimeGAN   | Rare Skill Path; Real + targeted synthetic                     | AUROC  | 0.5868 $\pm$ 0.0054 (3)    | Higher    |
| EdNet   | TimeGAN   | Rare Skill Path; Real only (TRTR)                              | AUPRC  | 0.5203 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | TimeGAN   | Rare Skill Path; Real only (TRTR)                              | AUROC  | 0.5892 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | TimeGAN   | Rare Skill Path; Standard synthetic only (TSTR)                | AUPRC  | 0.4560 $\pm$ 0.0178 (3)    | Higher    |
| EdNet   | TimeGAN   | Rare Skill Path; Standard synthetic only (TSTR)                | AUROC  | 0.4766 $\pm$ 0.0214 (3)    | Higher    |
| EdNet   | TimeGAN   | Rare Skill Path; Targeted synthetic only                       | AUPRC  | 0.4759 $\pm$ 0.0100 (3)    | Higher    |
| EdNet   | TimeGAN   | Rare Skill Path; Targeted synthetic only                       | AUROC  | 0.5215 $\pm$ 0.0142 (3)    | Higher    |
| EdNet   | TimeGAN   | Rare Skill Path; Standard augmentation gain                    | AUPRC  | 0.0117 $\pm$ 0.0048 (3)    | Higher    |
| EdNet   | TimeGAN   | Rare Skill Path; Standard augmentation gain                    | AUROC  | -0.0066 $\pm$ 0.0029 (3)   | Higher    |
| EdNet   | TimeGAN   | Rare Skill Path; Standard synthetic utility loss               | AUPRC  | 0.0643 $\pm$ 0.0178 (3)    | Lower     |
| EdNet   | TimeGAN   | Rare Skill Path; Standard synthetic utility loss               | AUROC  | 0.1126 $\pm$ 0.0214 (3)    | Lower     |
| EdNet   | TimeGAN   | Rare Skill Path; Targeted-vs-standard TSTR gain                | AUPRC  | 0.0198 $\pm$ 0.0156 (3)    | Higher    |
| EdNet   | TimeGAN   | Rare Skill Path; Targeted-vs-standard TSTR gain                | AUROC  | 0.0449 $\pm$ 0.0310 (3)    | Higher    |
| EdNet   | TimeGAN   | Rare Skill Path; Targeted augmentation gain                    | AUPRC  | 0.0120 $\pm$ 0.0043 (3)    | Higher    |
| EdNet   | TimeGAN   | Rare Skill Path; Targeted augmentation gain                    | AUROC  | -0.0025 $\pm$ 0.0054 (3)   | Higher    |
| EdNet   | TimeGAN   | Recovery; Real + standard synthetic                            | AUPRC  | 0.7074 $\pm$ 0.0012 (3)    | Higher    |
| EdNet   | TimeGAN   | Recovery; Real + standard synthetic                            | AUROC  | 0.6890 $\pm$ 0.0028 (3)    | Higher    |
| EdNet   | TimeGAN   | Recovery; Real + targeted synthetic                            | AUPRC  | 0.7146 $\pm$ 0.0007 (3)    | Higher    |
| EdNet   | TimeGAN   | Recovery; Real + targeted synthetic                            | AUROC  | 0.6964 $\pm$ 0.0004 (3)    | Higher    |
| EdNet   | TimeGAN   | Recovery; Real only (TRTR)                                     | AUPRC  | 0.7117 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | TimeGAN   | Recovery; Real only (TRTR)                                     | AUROC  | 0.6912 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | TimeGAN   | Recovery; Standard synthetic only (TSTR)                       | AUPRC  | 0.5143 $\pm$ 0.0159 (3)    | Higher    |
| EdNet   | TimeGAN   | Recovery; Standard synthetic only (TSTR)                       | AUROC  | 0.4371 $\pm$ 0.0233 (3)    | Higher    |
| EdNet   | TimeGAN   | Recovery; Targeted synthetic only                              | AUPRC  | 0.5762 $\pm$ 0.0282 (3)    | Higher    |
| EdNet   | TimeGAN   | Recovery; Targeted synthetic only                              | AUROC  | 0.5583 $\pm$ 0.0407 (3)    | Higher    |
| EdNet   | TimeGAN   | Recovery; Standard augmentation gain                           | AUPRC  | -0.0043 $\pm$ 0.0012 (3)   | Higher    |
| EdNet   | TimeGAN   | Recovery; Standard augmentation gain                           | AUROC  | -0.0022 $\pm$ 0.0028 (3)   | Higher    |
| EdNet   | TimeGAN   | Recovery; Standard synthetic utility loss                      | AUPRC  | 0.1974 $\pm$ 0.0159 (3)    | Lower     |
| EdNet   | TimeGAN   | Recovery; Standard synthetic utility loss                      | AUROC  | 0.2542 $\pm$ 0.0233 (3)    | Lower     |
| EdNet   | TimeGAN   | Recovery; Targeted-vs-standard TSTR gain                       | AUPRC  | 0.0619 $\pm$ 0.0232 (3)    | Higher    |

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| Dataset | Generator | Scope / comparison                                         | Metric                                      | Mean $\pm$ SD ( <i>n</i> ) | Direction |
|---------|-----------|------------------------------------------------------------|---------------------------------------------|----------------------------|-----------|
| EdNet   | TimeGAN   | Recovery; Targeted-vs-standard TSTR gain                   | AUROC                                       | 0.1212 $\pm$ 0.0313 (3)    | Higher    |
| EdNet   | TimeGAN   | Recovery; Targeted augmentation gain                       | AUPRC                                       | 0.0029 $\pm$ 0.0007 (3)    | Higher    |
| EdNet   | TimeGAN   | Recovery; Targeted augmentation gain                       | AUROC                                       | 0.0051 $\pm$ 0.0004 (3)    | Higher    |
| EdNet   | TimeGAN   | Late Correctness Decline                                   | Rare-group prevalence-error reduction       | 0.0000 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | TimeGAN   | Persistent Low Correctness                                 | Rare-group prevalence-error reduction       | 0.0000 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | TimeGAN   | Rapid Low-Accuracy Responding                              | Rare-group prevalence-error reduction       | 0.0000 $\pm$ 0.0000 (3)    | Higher    |
| EdNet   | TimeGAN   | Rare Skill Path                                            | Rare-group prevalence-error reduction       | -0.0172 $\pm$ 0.0070 (3)   | Higher    |
| EdNet   | TimeGAN   | Rare Skill Path                                            | Rare-group trajectory-shape-error reduction | -0.0004 $\pm$ 0.0001 (3)   | Higher    |
| EdNet   | TimeGAN   | Recovery                                                   | Rare-group prevalence-error reduction       | 0.0000 $\pm$ 0.0000 (3)    | Higher    |
| OULAD   | Markov    | Real + standard synthetic                                  | AUPRC                                       | 0.9179 $\pm$ 0.0001 (3)    | Higher    |
| OULAD   | Markov    | Real + standard synthetic                                  | AUROC                                       | 0.8802 $\pm$ 0.0001 (3)    | Higher    |
| OULAD   | Markov    | Real + targeted synthetic                                  | AUPRC                                       | 0.9181 $\pm$ 0.0001 (3)    | Higher    |
| OULAD   | Markov    | Real + targeted synthetic                                  | AUROC                                       | 0.8805 $\pm$ 0.0002 (3)    | Higher    |
| OULAD   | Markov    | Real only (TRTR)                                           | AUPRC                                       | 0.9225 $\pm$ 0.0000 (3)    | Higher    |
| OULAD   | Markov    | Real only (TRTR)                                           | AUROC                                       | 0.8872 $\pm$ 0.0000 (3)    | Higher    |
| OULAD   | Markov    | Standard synthetic only (TSTR)                             | AUPRC                                       | 0.8991 $\pm$ 0.0006 (3)    | Higher    |
| OULAD   | Markov    | Standard synthetic only (TSTR)                             | AUROC                                       | 0.8572 $\pm$ 0.0004 (3)    | Higher    |
| OULAD   | Markov    | Targeted synthetic only                                    | AUPRC                                       | 0.9009 $\pm$ 0.0003 (3)    | Higher    |
| OULAD   | Markov    | Targeted synthetic only                                    | AUROC                                       | 0.8589 $\pm$ 0.0002 (3)    | Higher    |
| OULAD   | Markov    | Standard augmentation gain                                 | AUPRC                                       | -0.0046 $\pm$ 0.0001 (3)   | Higher    |
| OULAD   | Markov    | Standard augmentation gain                                 | AUROC                                       | -0.0070 $\pm$ 0.0001 (3)   | Higher    |
| OULAD   | Markov    | Standard synthetic utility loss                            | AUPRC                                       | 0.0234 $\pm$ 0.0006 (3)    | Lower     |
| OULAD   | Markov    | Standard synthetic utility loss                            | AUROC                                       | 0.0301 $\pm$ 0.0004 (3)    | Lower     |
| OULAD   | Markov    | Targeted-vs-standard TSTR gain                             | AUPRC                                       | 0.0018 $\pm$ 0.0009 (3)    | Higher    |
| OULAD   | Markov    | Targeted-vs-standard TSTR gain                             | AUROC                                       | 0.0017 $\pm$ 0.0005 (3)    | Higher    |
| OULAD   | Markov    | Targeted augmentation gain                                 | AUPRC                                       | -0.0044 $\pm$ 0.0001 (3)   | Higher    |
| OULAD   | Markov    | Targeted augmentation gain                                 | AUROC                                       | -0.0068 $\pm$ 0.0002 (3)   | Higher    |
| OULAD   | Markov    | Course Failure Prediction; Real + standard synthetic       | AUPRC                                       | 0.5450 $\pm$ 0.0027 (3)    | Higher    |
| OULAD   | Markov    | Course Failure Prediction; Real + standard synthetic       | AUROC                                       | 0.7000 $\pm$ 0.0016 (3)    | Higher    |
| OULAD   | Markov    | Course Failure Prediction; Real only (TRTR)                | AUPRC                                       | 0.5467 $\pm$ 0.0000 (3)    | Higher    |
| OULAD   | Markov    | Course Failure Prediction; Real only (TRTR)                | AUROC                                       | 0.7034 $\pm$ 0.0000 (3)    | Higher    |
| OULAD   | Markov    | Course Failure Prediction; Standard synthetic only (TSTR)  | AUPRC                                       | 0.5261 $\pm$ 0.0044 (3)    | Higher    |
| OULAD   | Markov    | Course Failure Prediction; Standard synthetic only (TSTR)  | AUROC                                       | 0.6826 $\pm$ 0.0028 (3)    | Higher    |
| OULAD   | Markov    | Course Failure Prediction; Targeted synthetic only         | AUPRC                                       | 0.5234 $\pm$ 0.0040 (3)    | Higher    |
| OULAD   | Markov    | Course Failure Prediction; Targeted synthetic only         | AUROC                                       | 0.6815 $\pm$ 0.0043 (3)    | Higher    |
| OULAD   | Markov    | Course Failure Prediction; Standard augmentation gain      | AUPRC                                       | -0.0017 $\pm$ 0.0027 (3)   | Higher    |
| OULAD   | Markov    | Course Failure Prediction; Standard augmentation gain      | AUROC                                       | -0.0034 $\pm$ 0.0016 (3)   | Higher    |
| OULAD   | Markov    | Course Failure Prediction; Standard synthetic utility loss | AUPRC                                       | 0.0206 $\pm$ 0.0044 (3)    | Lower     |
| OULAD   | Markov    | Course Failure Prediction; Standard synthetic utility loss | AUROC                                       | 0.0208 $\pm$ 0.0028 (3)    | Lower     |
| OULAD   | Markov    | Dropout Prediction; Real + standard synthetic              | AUPRC                                       | 0.2963 $\pm$ 0.0010 (3)    | Higher    |
| OULAD   | Markov    | Dropout Prediction; Real + standard synthetic              | AUROC                                       | 0.6444 $\pm$ 0.0009 (3)    | Higher    |
| OULAD   | Markov    | Dropout Prediction; Real only (TRTR)                       | AUPRC                                       | 0.3074 $\pm$ 0.0000 (3)    | Higher    |
| OULAD   | Markov    | Dropout Prediction; Real only (TRTR)                       | AUROC                                       | 0.6526 $\pm$ 0.0000 (3)    | Higher    |
| OULAD   | Markov    | Dropout Prediction; Standard synthetic only (TSTR)         | AUPRC                                       | 0.2753 $\pm$ 0.0019 (3)    | Higher    |
| OULAD   | Markov    | Dropout Prediction; Standard synthetic only (TSTR)         | AUROC                                       | 0.6156 $\pm$ 0.0013 (3)    | Higher    |
| OULAD   | Markov    | Dropout Prediction; Targeted synthetic only                | AUPRC                                       | 0.2813 $\pm$ 0.0018 (3)    | Higher    |
| OULAD   | Markov    | Dropout Prediction; Targeted synthetic only                | AUROC                                       | 0.6225 $\pm$ 0.0006 (3)    | Higher    |
| OULAD   | Markov    | Dropout Prediction; Standard augmentation gain             | AUPRC                                       | -0.0111 $\pm$ 0.0010 (3)   | Higher    |
| OULAD   | Markov    | Dropout Prediction; Standard augmentation gain             | AUROC                                       | -0.0082 $\pm$ 0.0009 (3)   | Higher    |
| OULAD   | Markov    | Dropout Prediction; Standard synthetic utility loss        | AUPRC                                       | 0.0321 $\pm$ 0.0019 (3)    | Lower     |
| OULAD   | Markov    | Dropout Prediction; Standard synthetic utility loss        | AUROC                                       | 0.0370 $\pm$ 0.0013 (3)    | Lower     |
| OULAD   | Markov    | Late Disengagement; Real + standard synthetic              | AUPRC                                       | 0.8200 $\pm$ 0.0001 (3)    | Higher    |
| OULAD   | Markov    | Late Disengagement; Real + standard synthetic              | AUROC                                       | 0.8477 $\pm$ 0.0001 (3)    | Higher    |
| OULAD   | Markov    | Late Disengagement; Real + targeted synthetic              | AUPRC                                       | 0.8189 $\pm$ 0.0001 (3)    | Higher    |
| OULAD   | Markov    | Late Disengagement; Real + targeted synthetic              | AUROC                                       | 0.8467 $\pm$ 0.0002 (3)    | Higher    |

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| Dataset | Generator | Scope / comparison                                  | Metric | Mean $\pm$ SD ( <i>n</i> ) | Direction |
|---------|-----------|-----------------------------------------------------|--------|----------------------------|-----------|
| OULAD   | Markov    | Late Disengagement; Real only (TRTR)                | AUPRC  | $0.8294 \pm 0.0000$ (3)    | Higher    |
| OULAD   | Markov    | Late Disengagement; Real only (TRTR)                | AUROC  | $0.8542 \pm 0.0000$ (3)    | Higher    |
| OULAD   | Markov    | Late Disengagement; Standard synthetic only (TSTR)  | AUPRC  | $0.7986 \pm 0.0008$ (3)    | Higher    |
| OULAD   | Markov    | Late Disengagement; Standard synthetic only (TSTR)  | AUROC  | $0.8322 \pm 0.0002$ (3)    | Higher    |
| OULAD   | Markov    | Late Disengagement; Targeted synthetic only         | AUPRC  | $0.8004 \pm 0.0009$ (3)    | Higher    |
| OULAD   | Markov    | Late Disengagement; Targeted synthetic only         | AUROC  | $0.8319 \pm 0.0006$ (3)    | Higher    |
| OULAD   | Markov    | Late Disengagement; Standard augmentation gain      | AUPRC  | $-0.0094 \pm 0.0001$ (3)   | Higher    |
| OULAD   | Markov    | Late Disengagement; Standard augmentation gain      | AUROC  | $-0.0066 \pm 0.0001$ (3)   | Higher    |
| OULAD   | Markov    | Late Disengagement; Standard synthetic utility loss | AUPRC  | $0.0307 \pm 0.0008$ (3)    | Lower     |
| OULAD   | Markov    | Late Disengagement; Standard synthetic utility loss | AUROC  | $0.0220 \pm 0.0002$ (3)    | Lower     |
| OULAD   | Markov    | Late Disengagement; Targeted-vs-standard TSTR gain  | AUPRC  | $0.0018 \pm 0.0017$ (3)    | Higher    |
| OULAD   | Markov    | Late Disengagement; Targeted-vs-standard TSTR gain  | AUROC  | $-0.0003 \pm 0.0007$ (3)   | Higher    |
| OULAD   | Markov    | Late Disengagement; Targeted augmentation gain      | AUPRC  | $-0.0105 \pm 0.0001$ (3)   | Higher    |
| OULAD   | Markov    | Late Disengagement; Targeted augmentation gain      | AUROC  | $-0.0075 \pm 0.0002$ (3)   | Higher    |
| OULAD   | Markov    | Rare Activity Path; Real + standard synthetic       | AUPRC  | $0.9534 \pm 0.0002$ (3)    | Higher    |
| OULAD   | Markov    | Rare Activity Path; Real + standard synthetic       | AUROC  | $0.7968 \pm 0.0007$ (3)    | Higher    |
| OULAD   | Markov    | Rare Activity Path; Real + targeted synthetic       | AUPRC  | $0.9524 \pm 0.0001$ (3)    | Higher    |
| OULAD   | Markov    | Rare Activity Path; Real + targeted synthetic       | AUROC  | $0.7918 \pm 0.0007$ (3)    | Higher    |
| OULAD   | Markov    | Rare Activity Path; Real only (TRTR)                | AUPRC  | $0.9562 \pm 0.0000$ (3)    | Higher    |
| OULAD   | Markov    | Rare Activity Path; Real only (TRTR)                | AUROC  | $0.8097 \pm 0.0000$ (3)    | Higher    |
| OULAD   | Markov    | Rare Activity Path; Standard synthetic only (TSTR)  | AUPRC  | $0.9349 \pm 0.0014$ (3)    | Higher    |
| OULAD   | Markov    | Rare Activity Path; Standard synthetic only (TSTR)  | AUROC  | $0.7281 \pm 0.0044$ (3)    | Higher    |
| OULAD   | Markov    | Rare Activity Path; Targeted synthetic only         | AUPRC  | $0.9350 \pm 0.0008$ (3)    | Higher    |
| OULAD   | Markov    | Rare Activity Path; Targeted synthetic only         | AUROC  | $0.7279 \pm 0.0015$ (3)    | Higher    |
| OULAD   | Markov    | Rare Activity Path; Standard augmentation gain      | AUPRC  | $-0.0028 \pm 0.0002$ (3)   | Higher    |
| OULAD   | Markov    | Rare Activity Path; Standard augmentation gain      | AUROC  | $-0.0129 \pm 0.0007$ (3)   | Higher    |
| OULAD   | Markov    | Rare Activity Path; Standard synthetic utility loss | AUPRC  | $0.0213 \pm 0.0014$ (3)    | Lower     |
| OULAD   | Markov    | Rare Activity Path; Standard synthetic utility loss | AUROC  | $0.0816 \pm 0.0044$ (3)    | Lower     |
| OULAD   | Markov    | Rare Activity Path; Targeted-vs-standard TSTR gain  | AUPRC  | $0.0002 \pm 0.0020$ (3)    | Higher    |
| OULAD   | Markov    | Rare Activity Path; Targeted-vs-standard TSTR gain  | AUROC  | $-0.0002 \pm 0.0052$ (3)   | Higher    |
| OULAD   | Markov    | Rare Activity Path; Targeted augmentation gain      | AUPRC  | $-0.0037 \pm 0.0001$ (3)   | Higher    |
| OULAD   | Markov    | Rare Activity Path; Targeted augmentation gain      | AUROC  | $-0.0179 \pm 0.0007$ (3)   | Higher    |
| OULAD   | Markov    | Reengagement; Real + standard synthetic             | AUPRC  | $0.7599 \pm 0.0021$ (3)    | Higher    |
| OULAD   | Markov    | Reengagement; Real + standard synthetic             | AUROC  | $0.7462 \pm 0.0017$ (3)    | Higher    |
| OULAD   | Markov    | Reengagement; Real + targeted synthetic             | AUPRC  | $0.7669 \pm 0.0009$ (3)    | Higher    |
| OULAD   | Markov    | Reengagement; Real + targeted synthetic             | AUROC  | $0.7489 \pm 0.0004$ (3)    | Higher    |
| OULAD   | Markov    | Reengagement; Real only (TRTR)                      | AUPRC  | $0.7060 \pm 0.0000$ (3)    | Higher    |
| OULAD   | Markov    | Reengagement; Real only (TRTR)                      | AUROC  | $0.7062 \pm 0.0000$ (3)    | Higher    |
| OULAD   | Markov    | Reengagement; Standard synthetic only (TSTR)        | AUPRC  | $0.7503 \pm 0.0034$ (3)    | Higher    |
| OULAD   | Markov    | Reengagement; Standard synthetic only (TSTR)        | AUROC  | $0.7459 \pm 0.0025$ (3)    | Higher    |
| OULAD   | Markov    | Reengagement; Targeted synthetic only               | AUPRC  | $0.7553 \pm 0.0015$ (3)    | Higher    |
| OULAD   | Markov    | Reengagement; Targeted synthetic only               | AUROC  | $0.7463 \pm 0.0014$ (3)    | Higher    |
| OULAD   | Markov    | Reengagement; Standard augmentation gain            | AUPRC  | $0.0539 \pm 0.0021$ (3)    | Higher    |
| OULAD   | Markov    | Reengagement; Standard augmentation gain            | AUROC  | $0.0399 \pm 0.0017$ (3)    | Higher    |
| OULAD   | Markov    | Reengagement; Standard synthetic utility loss       | AUPRC  | $-0.0443 \pm 0.0034$ (3)   | Lower     |

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| Dataset | Generator | Scope / comparison                                         | Metric                                      | Mean $\pm$ SD ( <i>n</i> ) | Direction |
|---------|-----------|------------------------------------------------------------|---------------------------------------------|----------------------------|-----------|
| OULAD   | Markov    | Reengagement; Standard synthetic utility loss              | AUROC                                       | $-0.0397 \pm 0.0025$ (3)   | Lower     |
| OULAD   | Markov    | Reengagement; Targeted-vs-standard TSTR gain               | AUPRC                                       | $0.0050 \pm 0.0022$ (3)    | Higher    |
| OULAD   | Markov    | Reengagement; Targeted-vs-standard TSTR gain               | AUROC                                       | $0.0004 \pm 0.0016$ (3)    | Higher    |
| OULAD   | Markov    | Reengagement; Targeted augmentation gain                   | AUPRC                                       | $0.0610 \pm 0.0009$ (3)    | Higher    |
| OULAD   | Markov    | Reengagement; Targeted augmentation gain                   | AUROC                                       | $0.0426 \pm 0.0004$ (3)    | Higher    |
| OULAD   | Markov    | Late Disengagement                                         | Rare-group prevalence-error reduction       | $0.0119 \pm 0.0011$ (3)    | Higher    |
| OULAD   | Markov    | Late Disengagement                                         | Rare-group trajectory-shape-error reduction | $-0.0014 \pm 0.0056$ (3)   | Higher    |
| OULAD   | Markov    | Rare Activity Path                                         | Rare-group prevalence-error reduction       | $-0.0347 \pm 0.0030$ (3)   | Higher    |
| OULAD   | Markov    | Rare Activity Path                                         | Rare-group trajectory-shape-error reduction | $-0.0006 \pm 0.0050$ (3)   | Higher    |
| OULAD   | Markov    | Reengagement                                               | Rare-group prevalence-error reduction       | $0.0052 \pm 0.0007$ (3)    | Higher    |
| OULAD   | Markov    | Reengagement                                               | Rare-group trajectory-shape-error reduction | $-0.0015 \pm 0.0061$ (3)   | Higher    |
| OULAD   | SeqVAE    | Real + standard synthetic                                  | AUPRC                                       | $0.9179 \pm 0.0002$ (3)    | Higher    |
| OULAD   | SeqVAE    | Real + standard synthetic                                  | AUROC                                       | $0.8809 \pm 0.0003$ (3)    | Higher    |
| OULAD   | SeqVAE    | Real + targeted synthetic                                  | AUPRC                                       | $0.9181 \pm 0.0001$ (3)    | Higher    |
| OULAD   | SeqVAE    | Real + targeted synthetic                                  | AUROC                                       | $0.8810 \pm 0.0001$ (3)    | Higher    |
| OULAD   | SeqVAE    | Real only (TRTR)                                           | AUPRC                                       | $0.9225 \pm 0.0000$ (3)    | Higher    |
| OULAD   | SeqVAE    | Real only (TRTR)                                           | AUROC                                       | $0.8872 \pm 0.0000$ (3)    | Higher    |
| OULAD   | SeqVAE    | Standard synthetic only (TSTR)                             | AUPRC                                       | $0.8952 \pm 0.0006$ (3)    | Higher    |
| OULAD   | SeqVAE    | Standard synthetic only (TSTR)                             | AUROC                                       | $0.8529 \pm 0.0009$ (3)    | Higher    |
| OULAD   | SeqVAE    | Targeted synthetic only                                    | AUPRC                                       | $0.8966 \pm 0.0008$ (3)    | Higher    |
| OULAD   | SeqVAE    | Targeted synthetic only                                    | AUROC                                       | $0.8543 \pm 0.0008$ (3)    | Higher    |
| OULAD   | SeqVAE    | Standard augmentation gain                                 | AUPRC                                       | $-0.0046 \pm 0.0002$ (3)   | Higher    |
| OULAD   | SeqVAE    | Standard augmentation gain                                 | AUROC                                       | $-0.0063 \pm 0.0003$ (3)   | Higher    |
| OULAD   | SeqVAE    | Standard synthetic utility loss                            | AUPRC                                       | $0.0273 \pm 0.0006$ (3)    | Lower     |
| OULAD   | SeqVAE    | Standard synthetic utility loss                            | AUROC                                       | $0.0344 \pm 0.0009$ (3)    | Lower     |
| OULAD   | SeqVAE    | Targeted-vs-standard TSTR gain                             | AUPRC                                       | $0.0014 \pm 0.0012$ (3)    | Higher    |
| OULAD   | SeqVAE    | Targeted-vs-standard TSTR gain                             | AUROC                                       | $0.0014 \pm 0.0008$ (3)    | Higher    |
| OULAD   | SeqVAE    | Targeted augmentation gain                                 | AUPRC                                       | $-0.0044 \pm 0.0001$ (3)   | Higher    |
| OULAD   | SeqVAE    | Targeted augmentation gain                                 | AUROC                                       | $-0.0062 \pm 0.0001$ (3)   | Higher    |
| OULAD   | SeqVAE    | Course Failure Prediction; Real + standard synthetic       | AUPRC                                       | $0.5451 \pm 0.0020$ (3)    | Higher    |
| OULAD   | SeqVAE    | Course Failure Prediction; Real + standard synthetic       | AUROC                                       | $0.7019 \pm 0.0016$ (3)    | Higher    |
| OULAD   | SeqVAE    | Course Failure Prediction; Real only (TRTR)                | AUPRC                                       | $0.5467 \pm 0.0000$ (3)    | Higher    |
| OULAD   | SeqVAE    | Course Failure Prediction; Real only (TRTR)                | AUROC                                       | $0.7034 \pm 0.0000$ (3)    | Higher    |
| OULAD   | SeqVAE    | Course Failure Prediction; Standard synthetic only (TSTR)  | AUPRC                                       | $0.5277 \pm 0.0029$ (3)    | Higher    |
| OULAD   | SeqVAE    | Course Failure Prediction; Standard synthetic only (TSTR)  | AUROC                                       | $0.6876 \pm 0.0026$ (3)    | Higher    |
| OULAD   | SeqVAE    | Course Failure Prediction; Targeted synthetic only         | AUPRC                                       | $0.5302 \pm 0.0024$ (3)    | Higher    |
| OULAD   | SeqVAE    | Course Failure Prediction; Targeted synthetic only         | AUROC                                       | $0.6873 \pm 0.0051$ (3)    | Higher    |
| OULAD   | SeqVAE    | Course Failure Prediction; Standard augmentation gain      | AUPRC                                       | $-0.0015 \pm 0.0020$ (3)   | Higher    |
| OULAD   | SeqVAE    | Course Failure Prediction; Standard augmentation gain      | AUROC                                       | $-0.0015 \pm 0.0016$ (3)   | Higher    |
| OULAD   | SeqVAE    | Course Failure Prediction; Standard synthetic utility loss | AUPRC                                       | $0.0190 \pm 0.0029$ (3)    | Lower     |
| OULAD   | SeqVAE    | Course Failure Prediction; Standard synthetic utility loss | AUROC                                       | $0.0158 \pm 0.0026$ (3)    | Lower     |
| OULAD   | SeqVAE    | Dropout Prediction; Real + standard synthetic              | AUPRC                                       | $0.3013 \pm 0.0015$ (3)    | Higher    |
| OULAD   | SeqVAE    | Dropout Prediction; Real + standard synthetic              | AUROC                                       | $0.6512 \pm 0.0025$ (3)    | Higher    |
| OULAD   | SeqVAE    | Dropout Prediction; Real only (TRTR)                       | AUPRC                                       | $0.3074 \pm 0.0000$ (3)    | Higher    |
| OULAD   | SeqVAE    | Dropout Prediction; Real only (TRTR)                       | AUROC                                       | $0.6526 \pm 0.0000$ (3)    | Higher    |
| OULAD   | SeqVAE    | Dropout Prediction; Standard synthetic only (TSTR)         | AUPRC                                       | $0.2802 \pm 0.0034$ (3)    | Higher    |
| OULAD   | SeqVAE    | Dropout Prediction; Standard synthetic only (TSTR)         | AUROC                                       | $0.6271 \pm 0.0068$ (3)    | Higher    |
| OULAD   | SeqVAE    | Dropout Prediction; Targeted synthetic only                | AUPRC                                       | $0.2802 \pm 0.0037$ (3)    | Higher    |
| OULAD   | SeqVAE    | Dropout Prediction; Targeted synthetic only                | AUROC                                       | $0.6276 \pm 0.0011$ (3)    | Higher    |
| OULAD   | SeqVAE    | Dropout Prediction; Standard augmentation gain             | AUPRC                                       | $-0.0061 \pm 0.0015$ (3)   | Higher    |
| OULAD   | SeqVAE    | Dropout Prediction; Standard augmentation gain             | AUROC                                       | $-0.0014 \pm 0.0025$ (3)   | Higher    |
| OULAD   | SeqVAE    | Dropout Prediction; Standard synthetic utility loss        | AUPRC                                       | $0.0272 \pm 0.0034$ (3)    | Lower     |
| OULAD   | SeqVAE    | Dropout Prediction; Standard synthetic utility loss        | AUROC                                       | $0.0254 \pm 0.0068$ (3)    | Lower     |

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| Dataset | Generator | Scope / comparison                                  | Metric | Mean $\pm$ SD ( <i>n</i> ) | Direction |
|---------|-----------|-----------------------------------------------------|--------|----------------------------|-----------|
| OULAD   | SeqVAE    | Late Disengagement; Real + standard synthetic       | AUPRC  | $0.8233 \pm 0.0006$ (3)    | Higher    |
| OULAD   | SeqVAE    | Late Disengagement; Real + standard synthetic       | AUROC  | $0.8491 \pm 0.0004$ (3)    | Higher    |
| OULAD   | SeqVAE    | Late Disengagement; Real + targeted synthetic       | AUPRC  | $0.8228 \pm 0.0006$ (3)    | Higher    |
| OULAD   | SeqVAE    | Late Disengagement; Real + targeted synthetic       | AUROC  | $0.8494 \pm 0.0002$ (3)    | Higher    |
| OULAD   | SeqVAE    | Late Disengagement; Real only (TRTR)                | AUPRC  | $0.8294 \pm 0.0000$ (3)    | Higher    |
| OULAD   | SeqVAE    | Late Disengagement; Real only (TRTR)                | AUROC  | $0.8542 \pm 0.0000$ (3)    | Higher    |
| OULAD   | SeqVAE    | Late Disengagement; Standard synthetic only (TSTR)  | AUPRC  | $0.8037 \pm 0.0017$ (3)    | Higher    |
| OULAD   | SeqVAE    | Late Disengagement; Standard synthetic only (TSTR)  | AUROC  | $0.8289 \pm 0.0014$ (3)    | Higher    |
| OULAD   | SeqVAE    | Late Disengagement; Targeted synthetic only         | AUPRC  | $0.8034 \pm 0.0011$ (3)    | Higher    |
| OULAD   | SeqVAE    | Late Disengagement; Targeted synthetic only         | AUROC  | $0.8292 \pm 0.0005$ (3)    | Higher    |
| OULAD   | SeqVAE    | Late Disengagement; Standard augmentation gain      | AUPRC  | $-0.0061 \pm 0.0006$ (3)   | Higher    |
| OULAD   | SeqVAE    | Late Disengagement; Standard augmentation gain      | AUROC  | $-0.0051 \pm 0.0004$ (3)   | Higher    |
| OULAD   | SeqVAE    | Late Disengagement; Standard synthetic utility loss | AUPRC  | $0.0256 \pm 0.0017$ (3)    | Lower     |
| OULAD   | SeqVAE    | Late Disengagement; Standard synthetic utility loss | AUROC  | $0.0254 \pm 0.0014$ (3)    | Lower     |
| OULAD   | SeqVAE    | Late Disengagement; Targeted-vs-standard TSTR gain  | AUPRC  | $-0.0004 \pm 0.0025$ (3)   | Higher    |
| OULAD   | SeqVAE    | Late Disengagement; Targeted-vs-standard TSTR gain  | AUROC  | $0.0003 \pm 0.0008$ (3)    | Higher    |
| OULAD   | SeqVAE    | Late Disengagement; Targeted augmentation gain      | AUPRC  | $-0.0066 \pm 0.0006$ (3)   | Higher    |
| OULAD   | SeqVAE    | Late Disengagement; Targeted augmentation gain      | AUROC  | $-0.0048 \pm 0.0002$ (3)   | Higher    |
| OULAD   | SeqVAE    | Rare Activity Path; Real + standard synthetic       | AUPRC  | $0.9533 \pm 0.0002$ (3)    | Higher    |
| OULAD   | SeqVAE    | Rare Activity Path; Real + standard synthetic       | AUROC  | $0.7998 \pm 0.0008$ (3)    | Higher    |
| OULAD   | SeqVAE    | Rare Activity Path; Real + targeted synthetic       | AUPRC  | $0.9531 \pm 0.0003$ (3)    | Higher    |
| OULAD   | SeqVAE    | Rare Activity Path; Real + targeted synthetic       | AUROC  | $0.7990 \pm 0.0010$ (3)    | Higher    |
| OULAD   | SeqVAE    | Rare Activity Path; Real only (TRTR)                | AUPRC  | $0.9562 \pm 0.0000$ (3)    | Higher    |
| OULAD   | SeqVAE    | Rare Activity Path; Real only (TRTR)                | AUROC  | $0.8097 \pm 0.0000$ (3)    | Higher    |
| OULAD   | SeqVAE    | Rare Activity Path; Standard synthetic only (TSTR)  | AUPRC  | $0.9397 \pm 0.0004$ (3)    | Higher    |
| OULAD   | SeqVAE    | Rare Activity Path; Standard synthetic only (TSTR)  | AUROC  | $0.7454 \pm 0.0008$ (3)    | Higher    |
| OULAD   | SeqVAE    | Rare Activity Path; Targeted synthetic only         | AUPRC  | $0.9403 \pm 0.0014$ (3)    | Higher    |
| OULAD   | SeqVAE    | Rare Activity Path; Targeted synthetic only         | AUROC  | $0.7460 \pm 0.0048$ (3)    | Higher    |
| OULAD   | SeqVAE    | Rare Activity Path; Standard augmentation gain      | AUPRC  | $-0.0029 \pm 0.0002$ (3)   | Higher    |
| OULAD   | SeqVAE    | Rare Activity Path; Standard augmentation gain      | AUROC  | $-0.0099 \pm 0.0008$ (3)   | Higher    |
| OULAD   | SeqVAE    | Rare Activity Path; Standard synthetic utility loss | AUPRC  | $0.0165 \pm 0.0004$ (3)    | Lower     |
| OULAD   | SeqVAE    | Rare Activity Path; Standard synthetic utility loss | AUROC  | $0.0643 \pm 0.0008$ (3)    | Lower     |
| OULAD   | SeqVAE    | Rare Activity Path; Targeted-vs-standard TSTR gain  | AUPRC  | $0.0006 \pm 0.0012$ (3)    | Higher    |
| OULAD   | SeqVAE    | Rare Activity Path; Targeted-vs-standard TSTR gain  | AUROC  | $0.0005 \pm 0.0040$ (3)    | Higher    |
| OULAD   | SeqVAE    | Rare Activity Path; Targeted augmentation gain      | AUPRC  | $-0.0030 \pm 0.0003$ (3)   | Higher    |
| OULAD   | SeqVAE    | Rare Activity Path; Targeted augmentation gain      | AUROC  | $-0.0107 \pm 0.0010$ (3)   | Higher    |
| OULAD   | SeqVAE    | Reengagement; Real + standard synthetic             | AUPRC  | $0.7246 \pm 0.0040$ (3)    | Higher    |
| OULAD   | SeqVAE    | Reengagement; Real + standard synthetic             | AUROC  | $0.7154 \pm 0.0028$ (3)    | Higher    |
| OULAD   | SeqVAE    | Reengagement; Real + targeted synthetic             | AUPRC  | $0.7327 \pm 0.0055$ (3)    | Higher    |
| OULAD   | SeqVAE    | Reengagement; Real + targeted synthetic             | AUROC  | $0.7208 \pm 0.0030$ (3)    | Higher    |
| OULAD   | SeqVAE    | Reengagement; Real only (TRTR)                      | AUPRC  | $0.7060 \pm 0.0000$ (3)    | Higher    |
| OULAD   | SeqVAE    | Reengagement; Real only (TRTR)                      | AUROC  | $0.7062 \pm 0.0000$ (3)    | Higher    |
| OULAD   | SeqVAE    | Reengagement; Standard synthetic only (TSTR)        | AUPRC  | $0.6964 \pm 0.0025$ (3)    | Higher    |
| OULAD   | SeqVAE    | Reengagement; Standard synthetic only (TSTR)        | AUROC  | $0.6982 \pm 0.0031$ (3)    | Higher    |
| OULAD   | SeqVAE    | Reengagement; Targeted synthetic only               | AUPRC  | $0.6997 \pm 0.0053$ (3)    | Higher    |

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| Dataset | Generator | Scope / comparison                                         | Metric                                      | Mean $\pm$ SD ( $n$ )    | Direction |
|---------|-----------|------------------------------------------------------------|---------------------------------------------|--------------------------|-----------|
| OULAD   | SeqVAE    | Reengagement; Targeted synthetic only                      | AUROC                                       | $0.7008 \pm 0.0068$ (3)  | Higher    |
| OULAD   | SeqVAE    | Reengagement; Standard augmentation gain                   | AUPRC                                       | $0.0186 \pm 0.0040$ (3)  | Higher    |
| OULAD   | SeqVAE    | Reengagement; Standard augmentation gain                   | AUROC                                       | $0.0091 \pm 0.0028$ (3)  | Higher    |
| OULAD   | SeqVAE    | Reengagement; Standard synthetic utility loss              | AUPRC                                       | $0.0096 \pm 0.0025$ (3)  | Lower     |
| OULAD   | SeqVAE    | Reengagement; Standard synthetic utility loss              | AUROC                                       | $0.0080 \pm 0.0031$ (3)  | Lower     |
| OULAD   | SeqVAE    | Reengagement; Targeted-vs-standard TSTR gain               | AUPRC                                       | $0.0033 \pm 0.0033$ (3)  | Higher    |
| OULAD   | SeqVAE    | Reengagement; Targeted-vs-standard TSTR gain               | AUROC                                       | $0.0026 \pm 0.0038$ (3)  | Higher    |
| OULAD   | SeqVAE    | Reengagement; Targeted augmentation gain                   | AUPRC                                       | $0.0267 \pm 0.0055$ (3)  | Higher    |
| OULAD   | SeqVAE    | Reengagement; Targeted augmentation gain                   | AUROC                                       | $0.0146 \pm 0.0030$ (3)  | Higher    |
| OULAD   | SeqVAE    | Late Disengagement                                         | Rare-group prevalence-error reduction       | $-0.0030 \pm 0.0046$ (3) | Higher    |
| OULAD   | SeqVAE    | Late Disengagement                                         | Rare-group trajectory-shape-error reduction | $-0.0018 \pm 0.0029$ (3) | Higher    |
| OULAD   | SeqVAE    | Rare Activity Path                                         | Rare-group prevalence-error reduction       | $0.0166 \pm 0.0034$ (3)  | Higher    |
| OULAD   | SeqVAE    | Rare Activity Path                                         | Rare-group trajectory-shape-error reduction | $0.0013 \pm 0.0068$ (3)  | Higher    |
| OULAD   | SeqVAE    | Reengagement                                               | Rare-group prevalence-error reduction       | $-0.0001 \pm 0.0017$ (3) | Higher    |
| OULAD   | SeqVAE    | Reengagement                                               | Rare-group trajectory-shape-error reduction | $-0.0002 \pm 0.0102$ (3) | Higher    |
| OULAD   | TimeGAN   | Real + standard synthetic                                  | AUPRC                                       | $0.9166 \pm 0.0034$ (3)  | Higher    |
| OULAD   | TimeGAN   | Real + standard synthetic                                  | AUROC                                       | $0.8790 \pm 0.0041$ (3)  | Higher    |
| OULAD   | TimeGAN   | Real + targeted synthetic                                  | AUPRC                                       | $0.9168 \pm 0.0035$ (3)  | Higher    |
| OULAD   | TimeGAN   | Real + targeted synthetic                                  | AUROC                                       | $0.8793 \pm 0.0043$ (3)  | Higher    |
| OULAD   | TimeGAN   | Real only (TRTR)                                           | AUPRC                                       | $0.9225 \pm 0.0000$ (3)  | Higher    |
| OULAD   | TimeGAN   | Real only (TRTR)                                           | AUROC                                       | $0.8872 \pm 0.0000$ (3)  | Higher    |
| OULAD   | TimeGAN   | Standard synthetic only (TSTR)                             | AUPRC                                       | $0.8201 \pm 0.0924$ (3)  | Higher    |
| OULAD   | TimeGAN   | Standard synthetic only (TSTR)                             | AUROC                                       | $0.7503 \pm 0.1291$ (3)  | Higher    |
| OULAD   | TimeGAN   | Targeted synthetic only                                    | AUPRC                                       | $0.7857 \pm 0.1353$ (3)  | Higher    |
| OULAD   | TimeGAN   | Targeted synthetic only                                    | AUROC                                       | $0.7182 \pm 0.1712$ (3)  | Higher    |
| OULAD   | TimeGAN   | Standard augmentation gain                                 | AUPRC                                       | $-0.0059 \pm 0.0034$ (3) | Higher    |
| OULAD   | TimeGAN   | Standard augmentation gain                                 | AUROC                                       | $-0.0082 \pm 0.0041$ (3) | Higher    |
| OULAD   | TimeGAN   | Standard synthetic utility loss                            | AUPRC                                       | $0.1023 \pm 0.0924$ (3)  | Lower     |
| OULAD   | TimeGAN   | Standard synthetic utility loss                            | AUROC                                       | $0.1369 \pm 0.1291$ (3)  | Lower     |
| OULAD   | TimeGAN   | Targeted-vs-standard TSTR gain                             | AUPRC                                       | $-0.0344 \pm 0.0429$ (3) | Higher    |
| OULAD   | TimeGAN   | Targeted-vs-standard TSTR gain                             | AUROC                                       | $-0.0321 \pm 0.0421$ (3) | Higher    |
| OULAD   | TimeGAN   | Targeted augmentation gain                                 | AUPRC                                       | $-0.0057 \pm 0.0035$ (3) | Higher    |
| OULAD   | TimeGAN   | Targeted augmentation gain                                 | AUROC                                       | $-0.0080 \pm 0.0043$ (3) | Higher    |
| OULAD   | TimeGAN   | Course Failure Prediction; Real + standard synthetic       | AUPRC                                       | $0.5287 \pm 0.0211$ (3)  | Higher    |
| OULAD   | TimeGAN   | Course Failure Prediction; Real + standard synthetic       | AUROC                                       | $0.6968 \pm 0.0067$ (3)  | Higher    |
| OULAD   | TimeGAN   | Course Failure Prediction; Real only (TRTR)                | AUPRC                                       | $0.5467 \pm 0.0000$ (3)  | Higher    |
| OULAD   | TimeGAN   | Course Failure Prediction; Real only (TRTR)                | AUROC                                       | $0.7034 \pm 0.0000$ (3)  | Higher    |
| OULAD   | TimeGAN   | Course Failure Prediction; Standard synthetic only (TSTR)  | AUPRC                                       | $0.4411 \pm 0.0581$ (3)  | Higher    |
| OULAD   | TimeGAN   | Course Failure Prediction; Standard synthetic only (TSTR)  | AUROC                                       | $0.6276 \pm 0.0645$ (3)  | Higher    |
| OULAD   | TimeGAN   | Course Failure Prediction; Targeted synthetic only         | AUPRC                                       | $0.4503 \pm 0.0558$ (3)  | Higher    |
| OULAD   | TimeGAN   | Course Failure Prediction; Targeted synthetic only         | AUROC                                       | $0.6210 \pm 0.0800$ (3)  | Higher    |
| OULAD   | TimeGAN   | Course Failure Prediction; Standard augmentation gain      | AUPRC                                       | $-0.0179 \pm 0.0211$ (3) | Higher    |
| OULAD   | TimeGAN   | Course Failure Prediction; Standard augmentation gain      | AUROC                                       | $-0.0066 \pm 0.0067$ (3) | Higher    |
| OULAD   | TimeGAN   | Course Failure Prediction; Standard synthetic utility loss | AUPRC                                       | $0.1055 \pm 0.0581$ (3)  | Lower     |
| OULAD   | TimeGAN   | Course Failure Prediction; Standard synthetic utility loss | AUROC                                       | $0.0758 \pm 0.0645$ (3)  | Lower     |
| OULAD   | TimeGAN   | Dropout Prediction; Real + standard synthetic              | AUPRC                                       | $0.2983 \pm 0.0042$ (3)  | Higher    |
| OULAD   | TimeGAN   | Dropout Prediction; Real + standard synthetic              | AUROC                                       | $0.6404 \pm 0.0090$ (3)  | Higher    |
| OULAD   | TimeGAN   | Dropout Prediction; Real only (TRTR)                       | AUPRC                                       | $0.3074 \pm 0.0000$ (3)  | Higher    |
| OULAD   | TimeGAN   | Dropout Prediction; Real only (TRTR)                       | AUROC                                       | $0.6526 \pm 0.0000$ (3)  | Higher    |
| OULAD   | TimeGAN   | Dropout Prediction; Standard synthetic only (TSTR)         | AUPRC                                       | $0.2758 \pm 0.0109$ (3)  | Higher    |
| OULAD   | TimeGAN   | Dropout Prediction; Standard synthetic only (TSTR)         | AUROC                                       | $0.6092 \pm 0.0023$ (3)  | Higher    |
| OULAD   | TimeGAN   | Dropout Prediction; Targeted synthetic only                | AUPRC                                       | $0.2687 \pm 0.0100$ (3)  | Higher    |
| OULAD   | TimeGAN   | Dropout Prediction; Targeted synthetic only                | AUROC                                       | $0.5934 \pm 0.0051$ (3)  | Higher    |

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| Dataset | Generator | Scope / comparison                                  | Metric | Mean $\pm$ SD ( <i>n</i> ) | Direction |
|---------|-----------|-----------------------------------------------------|--------|----------------------------|-----------|
| OULAD   | TimeGAN   | Dropout Prediction; Standard augmentation gain      | AUPRC  | $-0.0091 \pm 0.0042$ (3)   | Higher    |
| OULAD   | TimeGAN   | Dropout Prediction; Standard augmentation gain      | AUROC  | $-0.0121 \pm 0.0090$ (3)   | Higher    |
| OULAD   | TimeGAN   | Dropout Prediction; Standard synthetic utility loss | AUPRC  | $0.0316 \pm 0.0109$ (3)    | Lower     |
| OULAD   | TimeGAN   | Dropout Prediction; Standard synthetic utility loss | AUROC  | $0.0433 \pm 0.0023$ (3)    | Lower     |
| OULAD   | TimeGAN   | Late Disengagement; Real + standard synthetic       | AUPRC  | $0.8254 \pm 0.0010$ (3)    | Higher    |
| OULAD   | TimeGAN   | Late Disengagement; Real + standard synthetic       | AUROC  | $0.8492 \pm 0.0014$ (3)    | Higher    |
| OULAD   | TimeGAN   | Late Disengagement; Real + targeted synthetic       | AUPRC  | $0.8258 \pm 0.0013$ (3)    | Higher    |
| OULAD   | TimeGAN   | Late Disengagement; Real + targeted synthetic       | AUROC  | $0.8499 \pm 0.0013$ (3)    | Higher    |
| OULAD   | TimeGAN   | Late Disengagement; Real only (TRTR)                | AUPRC  | $0.8294 \pm 0.0000$ (3)    | Higher    |
| OULAD   | TimeGAN   | Late Disengagement; Real only (TRTR)                | AUROC  | $0.8542 \pm 0.0000$ (3)    | Higher    |
| OULAD   | TimeGAN   | Late Disengagement; Standard synthetic only (TSTR)  | AUPRC  | $0.7247 \pm 0.1029$ (3)    | Higher    |
| OULAD   | TimeGAN   | Late Disengagement; Standard synthetic only (TSTR)  | AUROC  | $0.7491 \pm 0.1010$ (3)    | Higher    |
| OULAD   | TimeGAN   | Late Disengagement; Targeted synthetic only         | AUPRC  | $0.6644 \pm 0.1858$ (3)    | Higher    |
| OULAD   | TimeGAN   | Late Disengagement; Targeted synthetic only         | AUROC  | $0.6869 \pm 0.1912$ (3)    | Higher    |
| OULAD   | TimeGAN   | Late Disengagement; Standard augmentation gain      | AUPRC  | $-0.0040 \pm 0.0010$ (3)   | Higher    |
| OULAD   | TimeGAN   | Late Disengagement; Standard augmentation gain      | AUROC  | $-0.0051 \pm 0.0014$ (3)   | Higher    |
| OULAD   | TimeGAN   | Late Disengagement; Standard synthetic utility loss | AUPRC  | $0.1046 \pm 0.1029$ (3)    | Lower     |
| OULAD   | TimeGAN   | Late Disengagement; Standard synthetic utility loss | AUROC  | $0.1051 \pm 0.1010$ (3)    | Lower     |
| OULAD   | TimeGAN   | Late Disengagement; Targeted-vs-standard TSTR gain  | AUPRC  | $-0.0603 \pm 0.0829$ (3)   | Higher    |
| OULAD   | TimeGAN   | Late Disengagement; Targeted-vs-standard TSTR gain  | AUROC  | $-0.0623 \pm 0.0903$ (3)   | Higher    |
| OULAD   | TimeGAN   | Late Disengagement; Targeted augmentation gain      | AUPRC  | $-0.0035 \pm 0.0013$ (3)   | Higher    |
| OULAD   | TimeGAN   | Late Disengagement; Targeted augmentation gain      | AUROC  | $-0.0044 \pm 0.0013$ (3)   | Higher    |
| OULAD   | TimeGAN   | Rare Activity Path; Real + standard synthetic       | AUPRC  | $0.9533 \pm 0.0015$ (3)    | Higher    |
| OULAD   | TimeGAN   | Rare Activity Path; Real + standard synthetic       | AUROC  | $0.7968 \pm 0.0074$ (3)    | Higher    |
| OULAD   | TimeGAN   | Rare Activity Path; Real + targeted synthetic       | AUPRC  | $0.9531 \pm 0.0017$ (3)    | Higher    |
| OULAD   | TimeGAN   | Rare Activity Path; Real + targeted synthetic       | AUROC  | $0.7959 \pm 0.0083$ (3)    | Higher    |
| OULAD   | TimeGAN   | Rare Activity Path; Real only (TRTR)                | AUPRC  | $0.9562 \pm 0.0000$ (3)    | Higher    |
| OULAD   | TimeGAN   | Rare Activity Path; Real only (TRTR)                | AUROC  | $0.8097 \pm 0.0000$ (3)    | Higher    |
| OULAD   | TimeGAN   | Rare Activity Path; Standard synthetic only (TSTR)  | AUPRC  | $0.9169 \pm 0.0222$ (3)    | Higher    |
| OULAD   | TimeGAN   | Rare Activity Path; Standard synthetic only (TSTR)  | AUROC  | $0.6805 \pm 0.0635$ (3)    | Higher    |
| OULAD   | TimeGAN   | Rare Activity Path; Targeted synthetic only         | AUPRC  | $0.8905 \pm 0.0570$ (3)    | Higher    |
| OULAD   | TimeGAN   | Rare Activity Path; Targeted synthetic only         | AUROC  | $0.6100 \pm 0.1554$ (3)    | Higher    |
| OULAD   | TimeGAN   | Rare Activity Path; Standard augmentation gain      | AUPRC  | $-0.0029 \pm 0.0015$ (3)   | Higher    |
| OULAD   | TimeGAN   | Rare Activity Path; Standard augmentation gain      | AUROC  | $-0.0129 \pm 0.0074$ (3)   | Higher    |
| OULAD   | TimeGAN   | Rare Activity Path; Standard synthetic utility loss | AUPRC  | $0.0393 \pm 0.0222$ (3)    | Lower     |
| OULAD   | TimeGAN   | Rare Activity Path; Standard synthetic utility loss | AUROC  | $0.1292 \pm 0.0635$ (3)    | Lower     |
| OULAD   | TimeGAN   | Rare Activity Path; Targeted-vs-standard TSTR gain  | AUPRC  | $-0.0264 \pm 0.0349$ (3)   | Higher    |
| OULAD   | TimeGAN   | Rare Activity Path; Targeted-vs-standard TSTR gain  | AUROC  | $-0.0706 \pm 0.0922$ (3)   | Higher    |
| OULAD   | TimeGAN   | Rare Activity Path; Targeted augmentation gain      | AUPRC  | $-0.0031 \pm 0.0017$ (3)   | Higher    |
| OULAD   | TimeGAN   | Rare Activity Path; Targeted augmentation gain      | AUROC  | $-0.0138 \pm 0.0083$ (3)   | Higher    |
| OULAD   | TimeGAN   | Reengagement; Real + standard synthetic             | AUPRC  | $0.7524 \pm 0.0077$ (3)    | Higher    |
| OULAD   | TimeGAN   | Reengagement; Real + standard synthetic             | AUROC  | $0.7544 \pm 0.0034$ (3)    | Higher    |
| OULAD   | TimeGAN   | Reengagement; Real + targeted synthetic             | AUPRC  | $0.7536 \pm 0.0099$ (3)    | Higher    |
| OULAD   | TimeGAN   | Reengagement; Real + targeted synthetic             | AUROC  | $0.7580 \pm 0.0025$ (3)    | Higher    |

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| Dataset | Generator | Scope / comparison                            | Metric                                      | Mean $\pm$ SD ( $n$ )               | Direction |
|---------|-----------|-----------------------------------------------|---------------------------------------------|-------------------------------------|-----------|
| OULAD   | TimeGAN   | Reengagement; Real only (TRTR)                | AUPRC                                       | $0.7060 \pm 0.0000$ (3)             | Higher    |
| OULAD   | TimeGAN   | Reengagement; Real only (TRTR)                | AUROC                                       | $0.7062 \pm 0.0000$ (3)             | Higher    |
| OULAD   | TimeGAN   | Reengagement; Standard synthetic only (TSTR)  | AUPRC                                       | $0.6463 \pm 0.1331$ (3)             | Higher    |
| OULAD   | TimeGAN   | Reengagement; Standard synthetic only (TSTR)  | AUROC                                       | $0.6268 \pm 0.1687$ (3)             | Higher    |
| OULAD   | TimeGAN   | Reengagement; Targeted synthetic only         | AUPRC                                       | $0.6866 \pm 0.0826$ (3)             | Higher    |
| OULAD   | TimeGAN   | Reengagement; Targeted synthetic only         | AUROC                                       | $0.6953 \pm 0.0869$ (3)             | Higher    |
| OULAD   | TimeGAN   | Reengagement; Standard augmentation gain      | AUPRC                                       | $0.0464 \pm 0.0077$ (3)             | Higher    |
| OULAD   | TimeGAN   | Reengagement; Standard augmentation gain      | AUROC                                       | $0.0482 \pm 0.0034$ (3)             | Higher    |
| OULAD   | TimeGAN   | Reengagement; Standard synthetic utility loss | AUPRC                                       | $0.0597 \pm 0.1331$ (3)             | Lower     |
| OULAD   | TimeGAN   | Reengagement; Standard synthetic utility loss | AUROC                                       | $0.0795 \pm 0.1687$ (3)             | Lower     |
| OULAD   | TimeGAN   | Reengagement; Targeted-vs-standard TSTR gain  | AUPRC                                       | $0.0403 \pm 0.0513$ (3)             | Higher    |
| OULAD   | TimeGAN   | Reengagement; Targeted-vs-standard TSTR gain  | AUROC                                       | $0.0685 \pm 0.0826$ (3)             | Higher    |
| OULAD   | TimeGAN   | Reengagement; Targeted augmentation gain      | AUPRC                                       | $0.0477 \pm 0.0099$ (3)             | Higher    |
| OULAD   | TimeGAN   | Reengagement; Targeted augmentation gain      | AUROC                                       | $0.0517 \pm 0.0025$ (3)             | Higher    |
| OULAD   | TimeGAN   | Late Disengagement                            | Rare-group prevalence-error reduction       | $0.0086 \pm 0.0076$ (3)             | Higher    |
| OULAD   | TimeGAN   | Late Disengagement                            | Rare-group trajectory-shape-error reduction | $-0.0064 \pm 0.0054$ (2; 1 missing) | Higher    |
| OULAD   | TimeGAN   | Rare Activity Path                            | Rare-group prevalence-error reduction       | $-0.0000 \pm 0.0002$ (3)            | Higher    |
| OULAD   | TimeGAN   | Rare Activity Path                            | Rare-group trajectory-shape-error reduction | $0.0053 \pm 0.0155$ (2; 1 missing)  | Higher    |
| OULAD   | TimeGAN   | Reengagement                                  | Rare-group prevalence-error reduction       | $-0.0016 \pm 0.0022$ (3)            | Higher    |
| OULAD   | TimeGAN   | Reengagement                                  | Rare-group trajectory-shape-error reduction | $-0.0046 \pm 0.0052$ (2; 1 missing) | Higher    |

Table S13: Complete publication results for disclosure risk. Every numeric metric-summary entry is shown; SD is the population SD across the fixed seeds and  $n$  is the number of estimable seeds.

| Dataset     | Generator | Scope / comparison           | Metric                                | Mean $\pm$ SD ( $n$ )    | Direction |
|-------------|-----------|------------------------------|---------------------------------------|--------------------------|-----------|
| ASSISTments | Markov    | Standard synthetic           | Exact-duplicate rate                  | $0.0000 \pm 0.0000$ (3)  | Lower     |
| ASSISTments | Markov    | Standard synthetic           | Mean nearest-training distance        | $0.9532 \pm 0.0009$ (3)  | Higher    |
| ASSISTments | Markov    | Standard synthetic           | Membership-inference advantage        | $0.0119 \pm 0.0088$ (3)  | Lower     |
| ASSISTments | Markov    | Standard synthetic           | Near-duplicate rate                   | $0.0000 \pm 0.0000$ (3)  | Lower     |
| ASSISTments | Markov    | Targeted synthetic           | Exact-duplicate rate                  | $0.0000 \pm 0.0000$ (3)  | Lower     |
| ASSISTments | Markov    | Targeted synthetic           | Mean nearest-training distance        | $0.9517 \pm 0.0009$ (3)  | Higher    |
| ASSISTments | Markov    | Targeted synthetic           | Membership-inference advantage        | $0.0528 \pm 0.0194$ (3)  | Lower     |
| ASSISTments | Markov    | Targeted synthetic           | Near-duplicate rate                   | $0.0000 \pm 0.0000$ (3)  | Lower     |
| ASSISTments | Markov    | Targeted-minus-standard risk | Exact-duplicate risk change           | $0.0000 \pm 0.0000$ (3)  | Lower     |
| ASSISTments | Markov    | Targeted-minus-standard risk | Nearest-neighbor exposure risk change | $0.0015 \pm 0.0016$ (3)  | Lower     |
| ASSISTments | Markov    | Targeted-minus-standard risk | Membership-inference risk change      | $0.0409 \pm 0.0142$ (3)  | Lower     |
| ASSISTments | Markov    | Targeted-minus-standard risk | Near-duplicate risk change            | $0.0000 \pm 0.0000$ (3)  | Lower     |
| ASSISTments | SeqVAE    | Standard synthetic           | Exact-duplicate rate                  | $0.0000 \pm 0.0000$ (3)  | Lower     |
| ASSISTments | SeqVAE    | Standard synthetic           | Mean nearest-training distance        | $0.9657 \pm 0.0015$ (3)  | Higher    |
| ASSISTments | SeqVAE    | Standard synthetic           | Membership-inference advantage        | $0.0235 \pm 0.0095$ (3)  | Lower     |
| ASSISTments | SeqVAE    | Standard synthetic           | Near-duplicate rate                   | $0.0000 \pm 0.0000$ (3)  | Lower     |
| ASSISTments | SeqVAE    | Targeted synthetic           | Exact-duplicate rate                  | $0.0000 \pm 0.0000$ (3)  | Lower     |
| ASSISTments | SeqVAE    | Targeted synthetic           | Mean nearest-training distance        | $0.9677 \pm 0.0020$ (3)  | Higher    |
| ASSISTments | SeqVAE    | Targeted synthetic           | Membership-inference advantage        | $0.0412 \pm 0.0253$ (3)  | Lower     |
| ASSISTments | SeqVAE    | Targeted synthetic           | Near-duplicate rate                   | $0.0000 \pm 0.0000$ (3)  | Lower     |
| ASSISTments | SeqVAE    | Targeted-minus-standard risk | Exact-duplicate risk change           | $0.0000 \pm 0.0000$ (3)  | Lower     |
| ASSISTments | SeqVAE    | Targeted-minus-standard risk | Nearest-neighbor exposure risk change | $-0.0020 \pm 0.0005$ (3) | Lower     |
| ASSISTments | SeqVAE    | Targeted-minus-standard risk | Membership-inference risk change      | $0.0178 \pm 0.0276$ (3)  | Lower     |
| ASSISTments | SeqVAE    | Targeted-minus-standard risk | Near-duplicate risk change            | $0.0000 \pm 0.0000$ (3)  | Lower     |
| ASSISTments | TimeGAN   | Standard synthetic           | Exact-duplicate rate                  | $0.0000 \pm 0.0000$ (3)  | Lower     |
| ASSISTments | TimeGAN   | Standard synthetic           | Mean nearest-training distance        | $0.8614 \pm 0.0412$ (3)  | Higher    |
| ASSISTments | TimeGAN   | Standard synthetic           | Membership-inference advantage        | $0.0011 \pm 0.0009$ (3)  | Lower     |
| ASSISTments | TimeGAN   | Standard synthetic           | Near-duplicate rate                   | $0.0000 \pm 0.0000$ (3)  | Lower     |
| ASSISTments | TimeGAN   | Targeted synthetic           | Exact-duplicate rate                  | $0.0000 \pm 0.0000$ (3)  | Lower     |
| ASSISTments | TimeGAN   | Targeted synthetic           | Mean nearest-training distance        | $0.8676 \pm 0.0294$ (3)  | Higher    |
| ASSISTments | TimeGAN   | Targeted synthetic           | Membership-inference advantage        | $0.0221 \pm 0.0173$ (3)  | Lower     |
| ASSISTments | TimeGAN   | Targeted synthetic           | Near-duplicate rate                   | $0.0000 \pm 0.0000$ (3)  | Lower     |
| ASSISTments | TimeGAN   | Targeted-minus-standard risk | Exact-duplicate risk change           | $0.0000 \pm 0.0000$ (3)  | Lower     |
| ASSISTments | TimeGAN   | Targeted-minus-standard risk | Nearest-neighbor exposure risk change | $-0.0062 \pm 0.0124$ (3) | Lower     |
| ASSISTments | TimeGAN   | Targeted-minus-standard risk | Membership-inference risk change      | $0.0210 \pm 0.0169$ (3)  | Lower     |
| ASSISTments | TimeGAN   | Targeted-minus-standard risk | Near-duplicate risk change            | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet       | Markov    | Standard synthetic           | Exact-duplicate rate                  | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet       | Markov    | Standard synthetic           | Mean nearest-training distance        | $0.9937 \pm 0.0008$ (3)  | Higher    |
| EdNet       | Markov    | Standard synthetic           | Membership-inference advantage        | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet       | Markov    | Standard synthetic           | Near-duplicate rate                   | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet       | Markov    | Targeted synthetic           | Exact-duplicate rate                  | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet       | Markov    | Targeted synthetic           | Mean nearest-training distance        | $0.9923 \pm 0.0009$ (3)  | Higher    |

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| Dataset | Generator | Scope / comparison           | Metric                                     | Mean $\pm$ SD ( $n$ )    | Direction |
|---------|-----------|------------------------------|--------------------------------------------|--------------------------|-----------|
| EdNet   | Markov    | Targeted synthetic           | Membership-inference advantage             | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet   | Markov    | Targeted synthetic           | Near-duplicate rate                        | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet   | Markov    | Targeted-minus-standard risk | Exact-duplicate risk change                | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet   | Markov    | Targeted-minus-standard risk | Nearest-neighbor exposure risk change      | $0.0015 \pm 0.0008$ (3)  | Lower     |
| EdNet   | Markov    | Targeted-minus-standard risk | Membership-inference risk change           | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet   | Markov    | Targeted-minus-standard risk | Near-duplicate risk change                 | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet   | SeqVAE    | Standard synthetic           | Exact-duplicate rate                       | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet   | SeqVAE    | Standard synthetic           | Mean nearest-training distance             | $0.9869 \pm 0.0017$ (3)  | Higher    |
| EdNet   | SeqVAE    | Standard synthetic           | Membership-inference advantage             | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet   | SeqVAE    | Standard synthetic           | Near-duplicate rate                        | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet   | SeqVAE    | Targeted synthetic           | Exact-duplicate rate                       | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet   | SeqVAE    | Targeted synthetic           | Mean nearest-training distance             | $0.9852 \pm 0.0013$ (3)  | Higher    |
| EdNet   | SeqVAE    | Targeted synthetic           | Membership-inference advantage             | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet   | SeqVAE    | Targeted synthetic           | Near-duplicate rate                        | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet   | SeqVAE    | Targeted-minus-standard risk | Exact-duplicate risk change                | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet   | SeqVAE    | Targeted-minus-standard risk | Nearest-neighbor exposure risk change      | $0.0017 \pm 0.0008$ (3)  | Lower     |
| EdNet   | SeqVAE    | Targeted-minus-standard risk | Membership-inference risk change           | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet   | SeqVAE    | Targeted-minus-standard risk | Near-duplicate risk change                 | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet   | TimeGAN   | Standard synthetic           | Exact-duplicate rate                       | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet   | TimeGAN   | Standard synthetic           | Mean nearest-training distance             | $0.9933 \pm 0.0007$ (3)  | Higher    |
| EdNet   | TimeGAN   | Standard synthetic           | Membership-inference advantage             | $0.0049 \pm 0.0069$ (3)  | Lower     |
| EdNet   | TimeGAN   | Standard synthetic           | Near-duplicate rate                        | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet   | TimeGAN   | Targeted synthetic           | Exact-duplicate rate                       | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet   | TimeGAN   | Targeted synthetic           | Mean nearest-training distance             | $0.9942 \pm 0.0001$ (3)  | Higher    |
| EdNet   | TimeGAN   | Targeted synthetic           | Membership-inference advantage             | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet   | TimeGAN   | Targeted synthetic           | Near-duplicate rate                        | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet   | TimeGAN   | Targeted-minus-standard risk | Exact-duplicate risk change                | $0.0000 \pm 0.0000$ (3)  | Lower     |
| EdNet   | TimeGAN   | Targeted-minus-standard risk | Nearest-neighbor exposure risk change      | $-0.0009 \pm 0.0007$ (3) | Lower     |
| EdNet   | TimeGAN   | Targeted-minus-standard risk | Membership-inference risk change           | $-0.0049 \pm 0.0069$ (3) | Lower     |
| EdNet   | TimeGAN   | Targeted-minus-standard risk | Near-duplicate risk change                 | $0.0000 \pm 0.0000$ (3)  | Lower     |
| OULAD   | Markov    | Standard synthetic           | Behavior-projection exact-duplicate rate   | $0.0290 \pm 0.0009$ (3)  | Lower     |
| OULAD   | Markov    | Targeted synthetic           | Behavior-projection exact-duplicate rate   | $0.0315 \pm 0.0005$ (3)  | Lower     |
| OULAD   | Markov    | Targeted-minus-standard risk | Behavior-projection exact-copy risk change | $0.0025 \pm 0.0011$ (3)  | Lower     |
| OULAD   | SeqVAE    | Standard synthetic           | Behavior-projection exact-duplicate rate   | $0.0031 \pm 0.0009$ (3)  | Lower     |
| OULAD   | SeqVAE    | Targeted synthetic           | Behavior-projection exact-duplicate rate   | $0.0027 \pm 0.0003$ (3)  | Lower     |
| OULAD   | SeqVAE    | Targeted-minus-standard risk | Behavior-projection exact-copy risk change | $-0.0004 \pm 0.0012$ (3) | Lower     |
| OULAD   | TimeGAN   | Standard synthetic           | Behavior-projection exact-duplicate rate   | $0.1557 \pm 0.0906$ (3)  | Lower     |
| OULAD   | TimeGAN   | Targeted synthetic           | Behavior-projection exact-duplicate rate   | $0.1587 \pm 0.0973$ (3)  | Lower     |
| OULAD   | TimeGAN   | Targeted-minus-standard risk | Behavior-projection exact-copy risk change | $0.0030 \pm 0.0071$ (3)  | Lower     |

## G Signed Rare-Pathway Prevalence

Absolute prevalence error supports generator comparison but does not identify whether a pathway was underproduced or overproduced. Table S14 therefore reports the real, standard-synthetic, and targeted-synthetic learner prevalences and both signed differences for every publication pathway.

Table S14: Signed rare-pathway prevalence results. Prevalences are learner percentages and differences are synthetic minus real in percentage points. Positive differences mean overrepresentation; negative differences mean underrepresentation. Values are mean  $\pm$  population SD across the three fixed generator seeds.

| Dataset     | Generator | Pathway                        | Real (%)        | Standard (%)      | synth. Targeted (%) | synth. Standard diff. (pp) | signed Targeted signed diff. (pp) |
|-------------|-----------|--------------------------------|-----------------|-------------------|---------------------|----------------------------|-----------------------------------|
| ASSISTments | Markov    | High Hint Use                  | 4.20 $\pm$ 0.00 | 1.27 $\pm$ 0.15   | 5.79 $\pm$ 0.30     | -2.93 $\pm$ 0.15           | +1.59 $\pm$ 0.30                  |
| ASSISTments | Markov    | Late Correctness Decline       | 2.65 $\pm$ 0.00 | 3.15 $\pm$ 0.13   | 4.66 $\pm$ 0.28     | +0.50 $\pm$ 0.13           | +2.00 $\pm$ 0.28                  |
| ASSISTments | Markov    | Persistent Low Correctness     | 2.45 $\pm$ 0.00 | 0.48 $\pm$ 0.17   | 2.23 $\pm$ 0.55     | -1.97 $\pm$ 0.17           | -0.22 $\pm$ 0.55                  |
| ASSISTments | Markov    | Rapid Low-Accuracy Re-sponding | 2.57 $\pm$ 0.00 | 1.17 $\pm$ 0.14   | 4.12 $\pm$ 0.34     | -1.40 $\pm$ 0.14           | +1.56 $\pm$ 0.34                  |
| ASSISTments | Markov    | Rare Skill Path                | 4.87 $\pm$ 0.00 | 2.11 $\pm$ 0.76   | 2.77 $\pm$ 0.68     | -2.76 $\pm$ 0.76           | -2.10 $\pm$ 0.68                  |
| ASSISTments | Markov    | Recovery                       | 3.32 $\pm$ 0.00 | 3.42 $\pm$ 0.31   | 4.31 $\pm$ 0.46     | +0.10 $\pm$ 0.31           | +0.98 $\pm$ 0.46                  |
| ASSISTments | SeqVAE    | High Hint Use                  | 4.20 $\pm$ 0.00 | 0.30 $\pm$ 0.27   | 0.51 $\pm$ 0.24     | -3.90 $\pm$ 0.27           | -3.69 $\pm$ 0.24                  |
| ASSISTments | SeqVAE    | Late Correctness Decline       | 2.65 $\pm$ 0.00 | 1.81 $\pm$ 0.70   | 2.43 $\pm$ 0.56     | -0.85 $\pm$ 0.70           | -0.22 $\pm$ 0.56                  |
| ASSISTments | SeqVAE    | Persistent Low Correctness     | 2.45 $\pm$ 0.00 | 0.08 $\pm$ 0.09   | 0.09 $\pm$ 0.10     | -2.37 $\pm$ 0.09           | -2.36 $\pm$ 0.10                  |
| ASSISTments | SeqVAE    | Rapid Low-Accuracy Re-sponding | 2.57 $\pm$ 0.00 | 0.11 $\pm$ 0.06   | 0.12 $\pm$ 0.14     | -2.46 $\pm$ 0.06           | -2.45 $\pm$ 0.14                  |
| ASSISTments | SeqVAE    | Rare Skill Path                | 4.87 $\pm$ 0.00 | 98.52 $\pm$ 0.73  | 98.98 $\pm$ 0.62    | +93.65 $\pm$ 0.73          | +94.11 $\pm$ 0.62                 |
| ASSISTments | SeqVAE    | Recovery                       | 3.32 $\pm$ 0.00 | 2.33 $\pm$ 0.97   | 2.52 $\pm$ 0.63     | -0.99 $\pm$ 0.97           | -0.81 $\pm$ 0.63                  |
| ASSISTments | TimeGAN   | High Hint Use                  | 4.20 $\pm$ 0.00 | 0.02 $\pm$ 0.03   | 0.01 $\pm$ 0.01     | -4.18 $\pm$ 0.03           | -4.19 $\pm$ 0.01                  |
| ASSISTments | TimeGAN   | Late Correctness Decline       | 2.65 $\pm$ 0.00 | 0.28 $\pm$ 0.40   | 0.05 $\pm$ 0.07     | -2.37 $\pm$ 0.40           | -2.61 $\pm$ 0.07                  |
| ASSISTments | TimeGAN   | Persistent Low Correctness     | 2.45 $\pm$ 0.00 | 0.00 $\pm$ 0.00   | 0.00 $\pm$ 0.00     | -2.45 $\pm$ 0.00           | -2.45 $\pm$ 0.00                  |
| ASSISTments | TimeGAN   | Rapid Low-Accuracy Re-sponding | 2.57 $\pm$ 0.00 | 0.00 $\pm$ 0.00   | 0.00 $\pm$ 0.00     | -2.57 $\pm$ 0.00           | -2.57 $\pm$ 0.00                  |
| ASSISTments | TimeGAN   | Rare Skill Path                | 4.87 $\pm$ 0.00 | 14.90 $\pm$ 20.30 | 8.44 $\pm$ 11.46    | +10.03 $\pm$ 20.30         | +3.57 $\pm$ 11.46                 |
| ASSISTments | TimeGAN   | Recovery                       | 3.32 $\pm$ 0.00 | 0.36 $\pm$ 0.51   | 0.08 $\pm$ 0.11     | -2.96 $\pm$ 0.51           | -3.25 $\pm$ 0.11                  |
| EdNet       | Markov    | Late Correctness Decline       | 7.61 $\pm$ 0.00 | 2.07 $\pm$ 0.14   | 3.24 $\pm$ 0.08     | -5.54 $\pm$ 0.14           | -4.37 $\pm$ 0.08                  |
| EdNet       | Markov    | Persistent Low Correctness     | 4.76 $\pm$ 0.00 | 0.35 $\pm$ 0.10   | 0.78 $\pm$ 0.09     | -4.41 $\pm$ 0.10           | -3.98 $\pm$ 0.09                  |
| EdNet       | Markov    | Rapid Low-Accuracy Re-sponding | 3.92 $\pm$ 0.00 | 0.39 $\pm$ 0.10   | 7.63 $\pm$ 0.38     | -3.53 $\pm$ 0.10           | +3.71 $\pm$ 0.38                  |
| EdNet       | Markov    | Rare Skill Path                | 5.01 $\pm$ 0.00 | 1.76 $\pm$ 0.60   | 2.07 $\pm$ 0.17     | -3.25 $\pm$ 0.60           | -2.94 $\pm$ 0.17                  |
| EdNet       | Markov    | Recovery                       | 4.35 $\pm$ 0.00 | 2.13 $\pm$ 0.07   | 2.04 $\pm$ 0.20     | -2.22 $\pm$ 0.07           | -2.31 $\pm$ 0.20                  |
| EdNet       | SeqVAE    | Late Correctness Decline       | 7.61 $\pm$ 0.00 | 1.41 $\pm$ 0.35   | 1.70 $\pm$ 0.17     | -6.20 $\pm$ 0.35           | -5.91 $\pm$ 0.17                  |
| EdNet       | SeqVAE    | Persistent Low Correctness     | 4.76 $\pm$ 0.00 | 0.17 $\pm$ 0.08   | 0.11 $\pm$ 0.03     | -4.59 $\pm$ 0.08           | -4.65 $\pm$ 0.03                  |
| EdNet       | SeqVAE    | Rapid Low-Accuracy Re-sponding | 3.92 $\pm$ 0.00 | 0.02 $\pm$ 0.02   | 0.02 $\pm$ 0.02     | -3.90 $\pm$ 0.02           | -3.90 $\pm$ 0.02                  |
| EdNet       | SeqVAE    | Rare Skill Path                | 5.01 $\pm$ 0.00 | 90.29 $\pm$ 0.99  | 91.51 $\pm$ 0.48    | +85.28 $\pm$ 0.99          | +86.50 $\pm$ 0.48                 |
| EdNet       | SeqVAE    | Recovery                       | 4.35 $\pm$ 0.00 | 1.96 $\pm$ 0.40   | 1.90 $\pm$ 0.16     | -2.38 $\pm$ 0.40           | -2.45 $\pm$ 0.16                  |
| EdNet       | TimeGAN   | Late Correctness Decline       | 7.61 $\pm$ 0.00 | 0.00 $\pm$ 0.00   | 0.00 $\pm$ 0.00     | -7.61 $\pm$ 0.00           | -7.61 $\pm$ 0.00                  |
| EdNet       | TimeGAN   | Persistent Low Correctness     | 4.76 $\pm$ 0.00 | 0.00 $\pm$ 0.00   | 0.00 $\pm$ 0.00     | -4.76 $\pm$ 0.00           | -4.76 $\pm$ 0.00                  |
| EdNet       | TimeGAN   | Rapid Low-Accuracy Re-sponding | 3.92 $\pm$ 0.00 | 0.00 $\pm$ 0.00   | 0.00 $\pm$ 0.00     | -3.92 $\pm$ 0.00           | -3.92 $\pm$ 0.00                  |
| EdNet       | TimeGAN   | Rare Skill Path                | 5.01 $\pm$ 0.00 | 88.06 $\pm$ 1.19  | 89.78 $\pm$ 0.49    | +83.05 $\pm$ 1.19          | +84.77 $\pm$ 0.49                 |
| EdNet       | TimeGAN   | Recovery                       | 4.35 $\pm$ 0.00 | 0.00 $\pm$ 0.00   | 0.00 $\pm$ 0.00     | -4.35 $\pm$ 0.00           | -4.35 $\pm$ 0.00                  |
| OULAD       | Markov    | Late Disengagement             | 6.74 $\pm$ 0.00 | 9.30 $\pm$ 0.14   | 8.11 $\pm$ 0.05     | +2.56 $\pm$ 0.14           | +1.38 $\pm$ 0.05                  |
| OULAD       | Markov    | Rare Activity Path             | 4.46 $\pm$ 0.00 | 1.73 $\pm$ 0.07   | 10.67 $\pm$ 0.24    | -2.73 $\pm$ 0.07           | +6.20 $\pm$ 0.24                  |
| OULAD       | Markov    | Reengagement                   | 0.22 $\pm$ 0.00 | 3.43 $\pm$ 0.09   | 2.91 $\pm$ 0.10     | +3.21 $\pm$ 0.09           | +2.68 $\pm$ 0.10                  |
| OULAD       | SeqVAE    | Late Disengagement             | 6.74 $\pm$ 0.00 | 4.85 $\pm$ 0.33   | 4.55 $\pm$ 0.25     | -1.88 $\pm$ 0.33           | -2.18 $\pm$ 0.25                  |
| OULAD       | SeqVAE    | Rare Activity Path             | 4.46 $\pm$ 0.00 | 2.38 $\pm$ 0.10   | 4.04 $\pm$ 0.28     | -2.09 $\pm$ 0.10           | -0.43 $\pm$ 0.28                  |
| OULAD       | SeqVAE    | Reengagement                   | 0.22 $\pm$ 0.00 | 2.41 $\pm$ 0.12   | 2.42 $\pm$ 0.12     | +2.19 $\pm$ 0.12           | +2.20 $\pm$ 0.12                  |
| OULAD       | TimeGAN   | Late Disengagement             | 6.74 $\pm$ 0.00 | 7.82 $\pm$ 5.56   | 6.96 $\pm$ 4.92     | +1.09 $\pm$ 5.56           | +0.23 $\pm$ 4.92                  |
| OULAD       | TimeGAN   | Rare Activity Path             | 4.46 $\pm$ 0.00 | 0.21 $\pm$ 0.15   | 0.20 $\pm$ 0.15     | -4.26 $\pm$ 0.15           | -4.26 $\pm$ 0.15                  |
| OULAD       | TimeGAN   | Reengagement                   | 0.22 $\pm$ 0.00 | 2.92 $\pm$ 2.07   | 3.08 $\pm$ 2.18     | +2.70 $\pm$ 2.07           | +2.86 $\pm$ 2.18                  |

# H Complete Publication Status Results

Table S15 reports learner-level outcome-model collapse. Table S16 summarizes shape estimability by dataset and generator. The machine-readable supplement-generated/rare-shape-estimability.csv retains all 126 seed-level pathway rows, including exact support counts and fully estimable combinations. Table S17 then records the dataset-level contract for every metric in the dictionary. The companion files supplement-generated/complete-publication-statuses.csv and supplement-generated/metric-applicability.csv retain the same status and applicability information in machine-readable form.

Table S15: Learner-level outcome-task statuses. Every source status entry is represented: overall task status and any non-estimable arm are shown together rather than replacing undefined metrics with zero.

| Dataset     | Generator | Task                                  | Overall task status and seeds                                                  | Non-estimable arms and seeds                                                                                                                                                           |
|-------------|-----------|---------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASSISTments | Markov    | Persistent Low-Correctness Prediction | completed (3): 20260703, 20260704, 20260705                                    | None                                                                                                                                                                                   |
| ASSISTments | Markov    | Recovery Prediction                   | completed (3): 20260703, 20260704, 20260705                                    | None                                                                                                                                                                                   |
| ASSISTments | SeqVAE    | Persistent Low-Correctness Prediction | completed (2): 20260703, 20260705; completed with model collapse (1): 20260704 | Standard synthetic only (TSTR): not estimable model collapse (1): 20260704                                                                                                             |
| ASSISTments | SeqVAE    | Recovery Prediction                   | completed (3): 20260703, 20260704, 20260705                                    | None                                                                                                                                                                                   |
| ASSISTments | TimeGAN   | Persistent Low-Correctness Prediction | completed with model collapse (3): 20260703, 20260704, 20260705                | Standard synthetic only (TSTR): not estimable model collapse (3): 20260703, 20260704, 20260705 Targeted synthetic only: not estimable model collapse (3): 20260703, 20260704, 20260705 |
| ASSISTments | TimeGAN   | Recovery Prediction                   | completed (1): 20260704; completed with model collapse (2): 20260703, 20260705 | Standard synthetic only (TSTR): not estimable model collapse (2): 20260703, 20260705 Targeted synthetic only: not estimable model collapse (2): 20260703, 20260705                     |
| EdNet       | Markov    | Persistent Low-Correctness Prediction | completed (3): 20260703, 20260704, 20260705                                    | None                                                                                                                                                                                   |
| EdNet       | Markov    | Recovery Prediction                   | completed (3): 20260703, 20260704, 20260705                                    | None                                                                                                                                                                                   |
| EdNet       | SeqVAE    | Persistent Low-Correctness Prediction | completed (3): 20260703, 20260704, 20260705                                    | None                                                                                                                                                                                   |
| EdNet       | SeqVAE    | Recovery Prediction                   | completed (3): 20260703, 20260704, 20260705                                    | None                                                                                                                                                                                   |
| EdNet       | TimeGAN   | Persistent Low-Correctness Prediction | completed with model collapse (3): 20260703, 20260704, 20260705                | Standard synthetic only (TSTR): not estimable model collapse (3): 20260703, 20260704, 20260705 Targeted synthetic only: not estimable model collapse (3): 20260703, 20260704, 20260705 |
| EdNet       | TimeGAN   | Recovery Prediction                   | completed with model collapse (3): 20260703, 20260704, 20260705                | Standard synthetic only (TSTR): not estimable model collapse (3): 20260703, 20260704, 20260705 Targeted synthetic only: not estimable model collapse (3): 20260703, 20260704, 20260705 |
| OULAD       | Markov    | Course Failure Prediction             | completed (3): 20260703, 20260704, 20260705                                    | None                                                                                                                                                                                   |
| OULAD       | Markov    | Dropout Prediction                    | completed (3): 20260703, 20260704, 20260705                                    | None                                                                                                                                                                                   |
| OULAD       | SeqVAE    | Course Failure Prediction             | completed (3): 20260703, 20260704, 20260705                                    | None                                                                                                                                                                                   |
| OULAD       | SeqVAE    | Dropout Prediction                    | completed (3): 20260703, 20260704, 20260705                                    | None                                                                                                                                                                                   |
| OULAD       | TimeGAN   | Course Failure Prediction             | completed (3): 20260703, 20260704, 20260705                                    | None                                                                                                                                                                                   |
| OULAD       | TimeGAN   | Dropout Prediction                    | completed (3): 20260703, 20260704, 20260705                                    | None                                                                                                                                                                                   |

Table S16: Rare-pathway conditional-shape support summarized by dataset and generator. Each fraction is the number of estimable pathway-seed cells over all eligible cells; each arm requires at least 10 qualifying real and synthetic learners. “Affected pathways” counts pathways with at least one non-estimable standard, targeted, or paired cell. The accompanying rare-shape-estimability.csv retains all 126 seed-level records, exact statuses, and real/synthetic learner counts.

| Dataset     | Generator | Eligible cells | Standard shape | Targeted shape | Paired comparison | Affected pathways |
|-------------|-----------|----------------|----------------|----------------|-------------------|-------------------|
| ASSISTments | Markov    | 18             | 17/18          | 18/18          | 17/18             | 1/6               |
| ASSISTments | SeqVAE    | 18             | 10/18          | 12/18          | 10/18             | 3/6               |
| ASSISTments | TimeGAN   | 18             | 4/18           | 2/18           | 2/18              | 6/6               |
| EdNet       | Markov    | 15             | 15/15          | 15/15          | 15/15             | 0/5               |
| EdNet       | SeqVAE    | 15             | 11/15          | 11/15          | 11/15             | 2/5               |
| EdNet       | TimeGAN   | 15             | 3/15           | 3/15           | 3/15              | 4/5               |
| OULAD       | Markov    | 9              | 9/9            | 9/9            | 9/9               | 0/3               |
| OULAD       | SeqVAE    | 9              | 9/9            | 9/9            | 9/9               | 0/3               |
| OULAD       | TimeGAN   | 9              | 6/9            | 6/9            | 6/9               | 3/3               |

Table S17: Complete dataset-level applicability grid for the publication metrics. “Reported” means a numeric row is emitted when estimable; support- and model-dependent cases retain explicit statuses. N/A denotes a missing or constant required signal, whereas excluded quantities exist diagnostically but are not generator-comparable under the declared postprocessing contract.

| Construct                      | Metric                                               | ASSISTments                           | EdNet                                 | OULAD                                         |
|--------------------------------|------------------------------------------------------|---------------------------------------|---------------------------------------|-----------------------------------------------|
| Population composition         | Behavior-rate error                                  | Reported                              | Reported                              | Reported                                      |
| Population composition         | Event-type frequency divergence                      | Reported                              | Reported                              | Reported                                      |
| Population composition         | Average feature-distribution divergence              | Reported                              | Reported                              | Reported                                      |
| Population composition         | Real-versus-synthetic detectability AU-ROC           | Reported                              | Reported                              | Excluded: postprocessing confounds comparison |
| Learning process               | Behavior-transition divergence                       | Reported                              | Reported                              | Reported                                      |
| Learning process               | Three-event sequence divergence                      | Reported                              | Reported                              | Reported                                      |
| Learning process               | Trajectory-shape error                               | Reported                              | Reported                              | Reported                                      |
| Learning process               | Behavior-persistence error                           | Reported                              | Reported                              | Reported                                      |
| Learning process               | Response/inactivity-gap distribution divergence      | N/A: no declared gap signal           | Reported                              | Reported                                      |
| Learning process               | Behavior-hint lag error                              | Reported                              | N/A: hint is constant zero            | N/A: no hint signal                           |
| Learning process               | Feature-dependence error                             | Reported                              | Reported                              | Reported                                      |
| Learner-pathway representation | Subgroup behavior error                              | Reported                              | Reported                              | Reported                                      |
| Learner-pathway representation | Subgroup-size error                                  | Reported                              | Reported                              | Reported                                      |
| Learner-pathway representation | Subgroup hint-use error                              | Reported                              | N/A: hint is constant zero            | N/A: no hint signal                           |
| Learner-pathway representation | Rare-group prevalence error                          | Reported                              | Reported                              | Reported                                      |
| Learner-pathway representation | Signed rare-group prevalence difference              | Reported                              | Reported                              | Reported                                      |
| Learner-pathway representation | Rare-group trajectory-shape error                    | Reported when support permits         | Reported when support permits         | Reported when support permits                 |
| Analytic usefulness            | AUPRC                                                | Reported                              | Reported                              | Reported                                      |
| Analytic usefulness            | AUROC                                                | Reported                              | Reported                              | Reported                                      |
| Analytic usefulness            | Standard synthetic utility loss                      | Reported                              | Reported                              | Reported                                      |
| Analytic usefulness            | Standard augmentation gain                           | Reported                              | Reported                              | Reported                                      |
| Analytic usefulness            | Targeted-vs-standard TSTR gain                       | Reported                              | Reported                              | Reported                                      |
| Analytic usefulness            | Targeted augmentation gain                           | Reported                              | Reported                              | Reported                                      |
| Analytic usefulness            | Pathway-level predictive usefulness                  | Reported when support permits         | Reported when support permits         | Reported when support permits                 |
| Analytic usefulness            | Rare-group prevalence-match change                   | Reported                              | Reported                              | Reported                                      |
| Analytic usefulness            | Rare-group shape-error reduction                     | Reported when support permits         | Reported when support permits         | Reported when support permits                 |
| Analytic usefulness            | Learner-level outcome-task AUPRC/AUROC and contrasts | Reported or status if model collapses | Reported or status if model collapses | Reported or status if model collapses         |
| Disclosure risk                | Exact-duplicate rate                                 | Reported on full modeled trajectory   | Reported on full modeled trajectory   | Excluded: postprocessing confounds comparison |
| Disclosure risk                | Near-duplicate rate                                  | Reported on full modeled trajectory   | Reported on full modeled trajectory   | Excluded: postprocessing confounds comparison |
| Disclosure risk                | Mean nearest-training distance                       | Reported on full modeled trajectory   | Reported on full modeled trajectory   | Excluded: postprocessing confounds comparison |
| Disclosure risk                | Membership-inference advantage                       | Reported on full modeled trajectory   | Reported on full modeled trajectory   | Excluded: postprocessing confounds comparison |
| Disclosure risk                | Behavior-projection exact-duplicate rate             | N/A: full-trajectory audit used       | N/A: full-trajectory audit used       | Reported on behavior projection only          |
| Disclosure risk                | Targeted-minus-standard privacy-risk change          | Reported on full modeled trajectory   | Reported on full modeled trajectory   | Reported on behavior projection only          |

# I Supporting Disclosure-Audit Figure

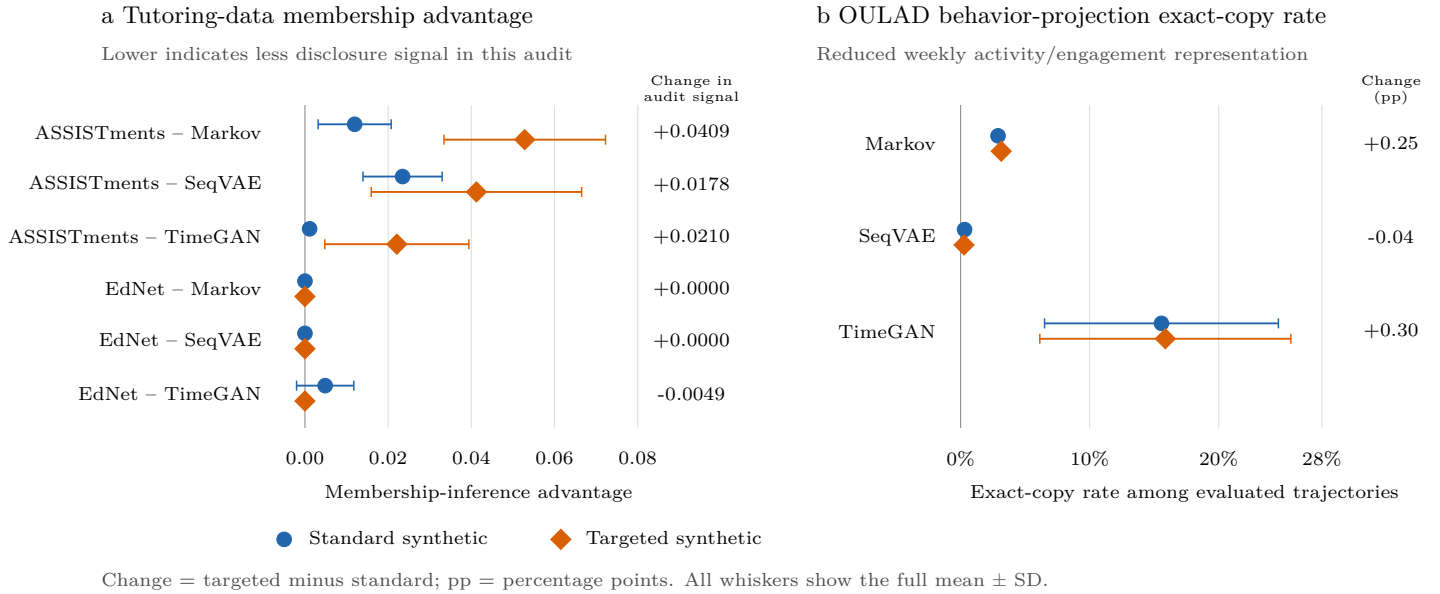


Figure S1: Supporting dataset-level disclosure-risk audit. Panel (a) reports membership-inference advantage for full normalized tutoring trajectories over the declared audit columns. Panel (b) reports exact copying for the OULAD dominant-activity/engagement projection; full-record membership inference is not generator-comparable there. The change in audit signal is targeted minus standard; positive values indicate a larger tested indicator after targeting. Points are means and whiskers show the full mean  $\pm$  population SD across three seeds; a descriptive SD whisker may extend below zero. The audit does not estimate disclosure risk specifically for learners on uncommon pathways. These are empirical memorization indicators, not formal privacy guarantees.

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